Perturbative Hamiltonian truncation of the long-range Ising model

Finite-cutoff coefficients C_2 to C_8, window corrections, extrapolations in $d = 1, 2, 3$, and how to reproduce them

Companion note to the public repository, folder `HT_Pert/`

Draft

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1 Summary

This note documents every perturbative Hamiltonian-truncation (HT) number quoted in the paper *The long-range Ising CFT: perturbative and nonperturbative approaches* and in its Supplemental Material, and how to reproduce each from the folder `HT_Pert/` of the public repository. The truncation at finite coupling is documented separately, in the note *Hamiltonian truncation of the long-range Ising model at finite coupling* (folder `HT/`).

The method expands the truncated Hamiltonian of the ϕ^4-deformed generalized free field in $\epsilon = 2s - d$ at fixed cutoff. Two exact relations fix the couplings, the gap $\Delta_\varepsilon = \sum_n C_n(\Lambda)\epsilon^n$ follows order by order, and the coefficients are extrapolated in the cutoff (Section 2). Two observations recur. First, in every dimension, truncating the total energy or level of a state leaves it a vacuum window that does not close as the cutoff grows. A window correction with no free parameter removes the resulting bias. In $d = 1$, with it, post hoc fits of the exact sums give C_4 and C_5 to 2×10^{-5} or better, and the solver data give C_4 to C_7 to 1×10^{-3} or better. In $d = 2$ it works through ϵ^4, and through ϵ^5 with two further rules that are not derived in full. In $d = 3$ it works through ϵ^4, with ϵ^5 provisional. It removes the ϵ^2 violation of the descendant spacings in all three dimensions (in

$d = 3$ for the one state tested), and the ϵ^3 violation in the single-state sectors of $d = 1$; in $d = 2$ it removes at least 95% of the ϵ^3 violation of single-quantum descendants at $L = 14$ and 16. Second, convergence is slow: the corrections fall like $\Lambda^{-1/2}$ times powers of $\log \Lambda$ in $d = 1$ and like $1/L$ times powers of $\ln L$ in $d = 2$ and $d = 3$, so the tail model carries a large part of every limit and dominates the error. Section 2.2 gives the reference values (Table 1), the deviation $\delta = |C_n^{\mathrm{HT}}/C_n - 1|$ and the status words used below.

One dimension (Section 3). C_2 and C_3 come from exact sums over intermediate states to $\Lambda = 2000$, with a fit rule fixed in advance: $\delta = 4 \times 10^{-12}$ and 2×10^{-8}. The same sums extend C_4 to $\Lambda = 2000$ and C_5 to 500, and fits frozen before the largest runs predict them to 10^{-9}. The frozen C_4 fits land 1×10^{-6} from the exact value, with the basis and the median chosen post hoc. Fixing the $\log^3$ tail of C_5 by the exponentiated C_4 tail gives C_5 to 2×10^{-5}, post hoc. With the complete window, the solver data to $\Lambda = 118$ give C_6 to 8×10^{-4} and C_7 to 1×10^{-3}, with fit windows and tail models chosen after the data were seen. The same data give a C_4 range that lies below the exact value and excludes it (largest deviation 2×10^{-4}) and a C_5 range that contains it (largest deviation 3×10^{-4}). C_8 lies 7×10^{-4} from the reference, inside its uncertainty, when the tail models are calibrated on C_6 and C_7.

Two dimensions (Section 4). The complete window, built from the level left to each counterterm, removes the $\ln L$ divergence of C_4. C_2 to C_4 agree with the continuum values to better than 10^{-7}, and C_5 to 5×10^{-5}, if the C_5 data carry no residual $\ln L$ term. The data cannot exclude one; with the bound judged from the median of the fits it would move C_5 by up to about 0.1, and the spread of single fits would allow -0.7 to $+0.5$. C_6 to C_8 are not reproduced.

Three dimensions (Section 5). With the complete window of $d = 2$ carried over unchanged, exact sums over intermediate states (C_2 to $L = 8000$, C_3 and C_4 to $L = 600$) reproduce C_2 (15 digits), C_3 ($\delta = 1 \times 10^{-7}$) and C_4 ($\delta = 5 \times 10^{-6}$). C_5, from $L \le 40$, gives -6.0 ± 0.3 against -5.850 (provisional and calibrated on $C_4^{(3)}$; a term in $L^{-1/2}$ that $L \le 40$ cannot exclude would lower it by 0.5 to 1). The vacuum window alone diverges at order ϵ^4. C_6 to C_8 are not reproduced.

Section 8 maps every number to its code, records, command and tolerance.

2 Model, cutoffs and perturbation theory

2.1 Model and observable

The long-range Ising critical point in d dimensions is described by the generalized free field (GFF) ϕ with nonlocal kinetic term $\phi(-\partial^2)^{s/2}\phi$, deformed by a ϕ^4 interaction [1, 2, 3]. With $\epsilon = 2s - d$ the field dimension is

$$\Delta_\sigma = \frac{d-s}{2} = \frac{d-\epsilon}{4}, \tag{1}$$

exactly, since no local counterterm renormalizes the nonlocal kinetic term [3]. The nonlocal equation of motion gives the shadow relation $\Delta_{\phi^3} = d - \Delta_\sigma$ [3]. These two exact statements are the only physics inputs of the truncation: they fix the two couplings of the Hamiltonian order by order. The observable is the thermal operator $\varepsilon = [\phi^2]$,

$$\Delta_\varepsilon(\epsilon) = \sum_{n \ge 0} C_n \epsilon^n, \qquad C_0 = \frac{d}{2}, \qquad C_1 = -\frac{1}{6}. \tag{2}$$

We write $C_n(\Lambda)$ or $C_n(L)$ for a coefficient computed at finite cutoff, and C_n, or $C_n^{(d)}$ where the dimension must be named, for the continuum value.

2.2 Reference values and conventions

Table 1 lists the continuum coefficients that the HT results are compared with. They come from the ϵ expansion of the GFF with Feynman graphs. The note *Thermal dimensions in the long-range Ising model*, in the folder `GFF/` of the public repository, documents them, and the paper *The long-range Ising CFT: perturbative and nonperturbative approaches* summarizes them. In this note they are used only for comparison, except where a fit is explicitly labeled as calibrated on them.

	$d=1$	$d=2$	$d=3$
C_2	$\frac{1}{9}(\pi+4\log 2)=0.657131263981$	$\frac{4}{9}\log 2=0.308065413582$	$\frac{1}{9}(2+4\log 2-\pi)=0.181221785405554$
C_3	-3.719856322902	-0.874850012653	-0.353581520433
C_4	45.42488010356	5.048868431	1.228063545
C_5	-731.13947662454	-38.833645896	-5.850351421
C_6	14331.68703275805	362.366 ± 0.004	33.5754 ± 0.0005
C_7	-322550 ± 41	-3876.1 ± 2.1	-220.35 ± 0.13
C_8	$(8.060 \pm 0.017) \times 10^6$	45990 ± 33	1602.6 ± 3.9

Table 1: Continuum coefficients $C_n^{(d)}$ used as references, from the GFF note (folder `GFF/`). C_2 to C_5 are closed forms, and $C_6^{(1)}$ is a finite representation in special-function values (an independent sum over all six-loop graphs gives 14331.6870328(5)). The other entries are Monte Carlo estimates; their $\pm$ is the empirical allowance of the GFF note, about four to five sampling standard errors, not a standard deviation.

Conventions. The deviation of an HT value from the reference is $\delta = |C_n^{\mathrm{HT}}/C_n - 1|$; for a range, δ is the largest deviation of its ends. Quoted HT uncertainties are spreads over fit models, fit windows or prescriptions, not standard errors; values are rounded with the PDG rule. Each result carries the status word of the analysis that produced it. The words are *final* (the analysis is complete; it may still contain post hoc choices), *blind* (fit rules fixed before the data they predict existed), *post hoc* (a choice made after the data, or the continuum value, were seen), *calibrated* (tail models selected by their reproduction of known continuum coefficients), *provisional*, or *not reproduced*.

2.3 Radial quantization and the Hamiltonian

In radial quantization the GFF is a set of free oscillators. In $d=1$ the "sphere" is the pair of points $\sigma = \pm 1$ and

$$\phi_\sigma = \sum_{n\geq 0} u_n\, \sigma^n \big(a_n + a_n^\dagger\big), \qquad u_n^2 = \frac{(2\Delta_\sigma)_n}{n!} = \frac{\left(\frac{1-\epsilon}{2}\right)_n}{n!}, \qquad E(a_n^\dagger) = \Delta_\sigma + n. \tag{3}$$

With the level $\ell = \sum_n n N_n$ and the particle number $P = \sum_n N_n$,

$$H = H_0 + \epsilon F_1 + r(\epsilon)\, M + g(\epsilon)\, V, \qquad H_0 = \ell + \tfrac{P}{4}, \qquad F_1 = -\tfrac{P}{4}, \tag{4}$$

where $M = \langle{:}\phi^2{:}\rangle$ and $V = \frac{1}{12}\langle{:}\phi^4{:}\rangle$ are averages over the sphere, normal ordered. Their matrix elements are used as they stand in the truncated space: operators are never commuted past each

other, because in a truncated space a commutator identity no longer holds. The sphere average keeps only transitions that change ℓ by an even amount, so the parity of ℓ is conserved; the reference states below have $\ell = 0$, and odd-ℓ states such as $\partial\phi$ enter only the descendant tests. The field parity $(-1)^P$ is conserved as well. ϵ enters only through $u_n(\epsilon)$ and F_1; the Taylor coefficients of u_n are kept exactly to the working order. The $d = 2$ and $d = 3$ mode functions and weights are given in Sections 4 and 5.

2.4 Cutoffs

$d = 1$**: frozen energy cutoff.** The Fock space keeps the states whose energy at $\epsilon = 0$ satisfies

$$E^0 = \ell + \tfrac{P}{4} \le \Lambda \qquad \Longleftrightarrow \qquad 4\ell + P \le 4\Lambda, \tag{5}$$

and the thresholds are not moved with ϵ. Energies at $\epsilon = 0$ are multiples of $\frac{1}{4}$.

$d = 2, 3$**: level cutoff.** The zero mode (the mode with the lowest energy, Δ_σ) is kept untruncated; the other modes form the bath, whose total level is cut at L: $\ell \le L$ in $d = 2$, $N_b \le L$ in $d = 3$ (Sections 4, 5).

2.5 Tuning and the coefficients

The reference states are the zero-mode states $|p\rangle = (a_0^\dagger)^p|0\rangle/\sqrt{p!}$ with $\ell = 0$: $p = 0$ is the vacuum, $p = 1$ and $p = 3$ are ϕ and ϕ^3, $p = 2$ is ϕ^2, and $p = 6$ (ϕ^6) gives the gap channel used for the tail exponents. The couplings $r = \sum_{k\ge1} r_k\epsilon^k$ and $g = \sum_{k\ge1} g_k\epsilon^k$ are fixed order by order by

$$E_1 - E_0 = \Delta_\sigma, \qquad E_3 - E_0 = d - \Delta_\sigma, \qquad \Delta_\varepsilon = E_2 - E_0 = \sum_n C_n(\Lambda)\epsilon^n. \tag{6}$$

At $\epsilon = 0$, $\langle p|M|p\rangle - \langle 0|M|0\rangle = 2p$ and $\langle p|V|p\rangle - \langle 0|V|0\rangle = p(p-1)/2$, so the solve is triangular. At first order $r_1 = 0$ and $g_1 = \frac{1}{3}$, so $C_0 = \frac{d}{2}$ and $C_1 = -\frac{1}{6}$ at every cutoff. With $\mathcal{E}_p$ the order-ϵ^n energy of $|p\rangle$ without the r_n, g_n terms,

$$C_n = \mathcal{E}_2 - \mathcal{E}_1 - \tfrac{1}{3}\big(\mathcal{E}_3 - \mathcal{E}_0\big) = -\tfrac{1}{3}\,\Delta_p^3\mathcal{E}_p \qquad (n \ge 2)\,: \tag{7}$$

shifts linear or quadratic in p are absorbed by r and g.

Evaluation. The ϵ series are computed by formal power-series algebra at fixed cutoff, never by sampling ϵ. At low orders a Rayleigh–Schrödinger recursion on truncated power series in ϵ gives the wave functions. At high orders a trial state correct through ϵ^k (order k) gives the energy through ϵ^{2k+1} by its Rayleigh quotient, and the eighth order adds one scalar term (Section 3.1). At order k no intermediate state of $|p\rangle$ has more than $p + 4k$ particles, or more than $4k$ particles in modes $n \ge 1$, so these particle caps are exact at the order computed. Independently, the order-n Rayleigh–Schrödinger expressions can be summed exactly over the intermediate states of the truncated Hamiltonian, grouped by the levels of the intermediate states between consecutive vertices, without building a basis. This gives C_2 to C_4 to $\Lambda = 2000$ and C_5 to $\Lambda = 500$ in $d = 1$, C_4, C_5 to large L in $d = 2$, and C_2 to C_4 to $L = 600$ and C_5 to $L = 40$ in $d = 3$.

2.6 Large-cutoff tails

At second order the $d = 1$ truncation has a closed form: at every cutoff

$$C_2(\Lambda) = \sum_{\substack{n \text{ even} \\ 0 \le n \le \Lambda - 2}} \bar{u}_n^2 \Big[\frac{2}{81(n - \frac{1}{2})} + \frac{2}{9(n + \frac{1}{2})} + \frac{2}{81(n+1)} + [n > 0] \frac{2}{9n} \Big], \qquad \bar{u}_n^2 = u_n^2 \big|_{\epsilon = 0} = \frac{(1/2)_n}{n!}, \tag{8}$$

which every solver run checks, and its tail is

$$C_2(\Lambda) - C_2 \simeq - \frac{40}{81\sqrt{\pi}} \, \Lambda^{-1/2} = -0.2786121 \, \Lambda^{-1/2}. \tag{9}$$

Cutoff effects scale as $\Lambda^{-(\Delta_{\text{irr}} - 1)}$ with Δ_{irr} the dimension of the leading irrelevant operator, ϕ^6. Its gap $\Delta_6 = E_6 - E_0$, computed in the same truncation, defines

$$\omega(\epsilon) = \Delta_6(\epsilon) - 1 = \tfrac{1}{2} + \tfrac{7}{2}\epsilon + \sum_{k \ge 2} \omega_k \epsilon^k, \tag{10}$$

with $\omega_1 = -\frac{6}{4} + 15 g_1 = \frac{7}{2}$ from the one-loop rule $[\epsilon] \Delta_{\phi^n} = -\frac{n}{4} + \binom{n}{2} g_1$, the same rule that gives $C_1 = -\frac{1}{6}$ for $n = 2$. The tail model of the extrapolations is

$$\sum_n \big[C_n(\Lambda) - C_n \big] \epsilon^n = \Lambda^{-1/2} \Big(\sum_{n \ge 2} a_n \epsilon^n \Big) \, e^{-[\omega(\epsilon) - \frac{1}{2}] \log \Lambda} + \rho(\epsilon, \Lambda), \tag{11}$$

with the remainder ρ built from the terms $\Lambda^{-p} \log^j \Lambda$ of one of six bases fixed in advance:

basis	terms (p, j) of ρ
A	$(1,0)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$, $(2,0)$
B	A and $(2,1)$
C	$(1,0)$, $(1,1)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$
D	C and $(2,0)$
E	D and $(2,1)$
F	$(1,0)$, $(\frac{3}{2},0)$, $(\frac{3}{2},1)$, $(\frac{3}{2},2)$, $(2,0)$

The model is fitted order by order, with C_2 and C_3 fixed to their continuum values and C_4 to C_8 fitted in sequence. Identifying the gap with the leading irrelevant operator is an assumption for this regulator.

Top logarithm. If the $\Lambda^{-1/2}$ tail comes from a single channel whose amplitude starts at ϵ^2, Eq. (11) fixes the coefficient of the highest power of $\log \Lambda$ at each order:

$$C_n(\Lambda) - C_n \ni c_n \, \Lambda^{-1/2} \log^{n-2} \Lambda, \qquad c_n = a_2 \frac{(-\omega_1)^{n-2}}{(n-2)!}, \qquad a_2 = -\frac{40}{81\sqrt{\pi}}, \tag{12}$$

independently of $\omega_{k \ge 2}$, of $a_{k \ge 3}$ and of channels that start at ϵ^3 or later. The exact sums test it: c_3 agrees to 5×10^{-6} and c_4 to 0.3% (Section 3.4).

2.7 The vacuum window

Truncating the total energy of a state makes the cutoff effectively nonlocal and state dependent [5]. In ϕ^4 theory in $2 + 1$ dimensions this spoils the cancellation of disconnected vacuum bubbles in Hamiltonian perturbation theory and gives new UV divergences, which Ref. [6] addressed with

perturbative state-dependent counterterms that add back the missing states. A basis state of energy E_i^0 is dressed only by vacuum bubbles with $E_b \le \Lambda - E_i^0$, while the vacuum keeps all bubbles with $E_b \le \Lambda$, so the state misses

$$E_b \in \big(\Lambda - E_i^0, \Lambda\big], \qquad N_{\mathrm{w},i} = \#\big\{\text{odd } k \in (\Lambda - E_i^0, \Lambda]\big\}. \tag{13}$$

At $\epsilon = 0$ every four-particle shell of odd energy k carries $-g^2/6$, so the deficit is proportional to the number N_{w} of missing shells, which is of order one at every cutoff and does not decay as $\Lambda \to \infty$. The descendant spacings $v_n(\mathcal{O}) = E(\partial^n\mathcal{O}) - E(\mathcal{O}) - n$, which vanish in a conformal theory, measure it: raw HT gives $v_n = \frac{N_{\mathrm{w}}}{54}\epsilon^2 + O(\epsilon^3)$.

Complete window, $d = 1$. We add to every basis state m the diagonal shift

$$D_m^{\mathrm{conn}} = E_c(\Lambda) - E_c\big(\lfloor \Lambda - E_m^0 \rfloor_{1/2}\big), \tag{14}$$

with $\lfloor\cdot\rfloor_{1/2}$ the half-integer floor and $E_c(x)$ the vacuum energy at cutoff x of the truncated Hamiltonian plus the same correction, and retune r and g. Since $D_{s+b}^{\mathrm{conn}} = D_s^{\mathrm{conn}} + D_b^{[\Lambda - E_s]}$ for a disconnected bubble b, each window is restored exactly once; the literal vacuum-energy difference $E_{\mathrm{vac}}(\Lambda) - E_{\mathrm{vac}}(\Lambda - E_m^0)$ would count the bubble-bubble class twice. There is no free parameter. At second order in g, D^{conn} reduces to the isolated window $g^2 D$ of the second-order vacuum bubbles, the counterpart of Patch I of Ref. [6]: a state-dependent counterterm that adds back the same missing four-particle states, with the unperturbed energy of the level being computed in its energy denominator, where $g^2 D$ has the bubble energy alone. Its Patch II, called a temporary fix there, restores towers of iterated bubbles up to five bubbles with every bubble cut at the full cutoff; D^{conn} contains such towers through E_c, cut as in the truncated vacuum.

Level cutoff, $d = 2, 3$. With bath level $\ell \le L$ the vacuum window is $D_L(\ell) = E_v(L) - E_v(L - \ell)$ (called the consistent window in the $d = 2$ production code). In $d = 2$ divergent sub-bubbles inside a state also see only the level left by the other quanta; the complete window W that cures this is defined in Section 4 and derived in Appendix A. The same complete window applies unchanged in $d = 3$ (Section 5).

2.8 Notation across dimensions

Table 2 collects the symbols that differ between the three dimensions.

3 Results in one dimension

3.1 Computations and data

Table 3 lists the computations behind the $d = 1$ numbers. All of them use the Hamiltonian (4), the cutoff (5) and the tuning (6); they differ in the order reached, in how the ϵ series is evaluated, and in the window. The eighth order needs, besides third-order trial states, the scalar $\langle b_4, R_p b_4\rangle$, with b_4 the fourth-order source vector of the reference state $|p\rangle$ and R_p its reduced resolvent. Production runs used SDSC Expanse and the symmetry cluster of the Perimeter Institute.

	$d = 1$	$d = 2$	$d = 3$
cutoff	energy Λ, Eq. (5)	bath level L, $\ell \le L$	bath level L, $N_b \le L$
level of a state	ℓ	ℓ (operator $\hat{\ell}$)	N_b (operator $\hat{N}_b$); ℓ is the angular momentum
vacuum window	second-order part $g^2 D$ (isolated window)	consistent window $D_L(\ell)$	vacuum window $D_L(N_b)$
complete window	$D^{\rm conn}$, Eq. (14)	$W = D^{\rm zm} + D^M$; at ϵ^5 the rules W_c, W_{cs}	W of $d = 2$; W_c, W_{cs} at ϵ^5
descendant spacing	$v_n(\mathcal{O})$	R_n (coefficient of ϵ^n)	as in $d = 2$
leading tail	$\Lambda^{-1/2} \log^k \Lambda$	$L^{-1} \ln^k L$	$L^{-1} \ln^k L$

Table 2: Notation that differs between dimensions. The "raw" truncation has no window in every dimension. $D^{\rm conn}$ has the self-consistent form of the $d = 2$ consistent window; "complete" names, in each dimension, the window that restores the tested descendant spacings and continuum coefficients there.

3.2 Raw coefficients and slow convergence

The tails are long. At $\Lambda = 118$ the raw C_4 reaches only 67% of its limit and the raw C_7 only 44%, and even at $\Lambda = 500$ the exact $C_5^{\rm conn}(500) = -519.770442010804$ is still far from $C_5 = -731.14$. The tail model of Section 2.6 therefore carries a large part of every limit. Every extrapolation of the raw truncation, with any of the bases A–F, gives a C_4 about 0.19% below the exact value; the window explains this offset.

3.3 The window in one dimension

With $D^{\rm conn}$ added, the shift of the coefficients does not decay:

$$C_4^{\rm conn}(\Lambda) - C_4^{\rm raw}(\Lambda) \simeq 0.0803 + \frac{\log \Lambda}{17496}, \tag{15}$$

where $0.0803 = K + \eta_4$, with K the finite part of the isolated $g^2 D$ window and $\eta_4 = 0.0186910$ the limit of the remaining fourth-order shift. At $\Lambda = 118$, $\Delta C_4 = 0.080$ and $\Delta C_5 = 0.71$. The exact sums follow ΔC_5 to $\Lambda = 500$, where it is 0.717651334579: the two $\log \Lambda$ terms of its pieces nearly cancel, and ΔC_5 tends to a constant near 0.72. Beyond ϵ^5 the complete window is computed numerically to $\Lambda = 72$; there ΔC_6, ΔC_7, ΔC_8 are +1.1717361, -1.828185, -8.5641, respectively.

Descendant spacings. The spacings v_n test the window without any continuum input (Table 4). States are labeled by their particle number P and level ℓ; a sector is a pair (P, ℓ). The raw truncation gives $N_{\rm w}/54$ at ϵ^2 and about $0.146\, N_{\rm w}$ at ϵ^3. With $D^{\rm conn}$ the ϵ^2 violation vanishes, and the ϵ^3 one is within $0.003\, N_{\rm w}$ in the sectors with a single state; the degenerate block $P = 2$, $\ell = 2$ gives $+0.004$ to $+0.006\, N_{\rm w}$ over the fit models, and whether it vanishes is not settled. At ϵ^4, $D^{\rm conn}$ removes 96–100% of the violation left by $g^2 D$ alone. At ϵ^5 the violation is consistent with zero within about $\pm(0.1\text{–}0.2)\, N_{\rm w}$, against $+2.3$ to $+2.8\, N_{\rm w}$ raw and -1.3 to $-1.6\, N_{\rm w}$ with $g^2 D$ alone; the $\partial\phi$ residual is not converged.

Cutoff placement. If the cutoff sits at $\Lambda = 2m + \frac{1}{4}$ or $2m + \frac{1}{2}$, with m an integer, the window of ϕ is empty, and raw HT satisfies $E(\partial\phi) - E(\phi) = 1$ exactly through ϵ^3 (to 10^{-14}). At $\Lambda = 2m + \frac{1}{4}$ the

computation	reach	method, where
HT solver, thermal	C_0–C_7, every even $\Lambda \leq 118$	Rayleigh quotient of third-order trial states in the Fock basis (up to 3.5×10^9 states at $\Lambda = 118$); Expanse, 111 new cutoffs; `code/lri_hpc_Hamitonian`
HT solver, ϕ^6 gap	Δ_6 to ϵ^5 for $\Lambda \leq 160$, to ϵ^3 for $\Lambda \leq 832$	same solver; exponents ω_k of Eq. (10)
eighth-order code	C_8, every even $\Lambda \leq 92$	third-order trial states plus $\langle b_4, R_p b_4 \rangle$; raw and isolated g^2D window; symmetry; `code/lri_c8_hpc`
window engines	ΔC_4, ΔC_5 exact, $\Lambda \leq 152$	two independent engines plus closed-form sums over the states at the top of the window; `code/lri_hpc_Hamitonian/analysis`, `code/lri_eh_hpc`
complete-window solver	C_0–C_8 with D^{conn}, every even $\Lambda \leq 72$	numerical, eighth order; beyond 72 a continuation validated blind (Section 3.4); symmetry; `code/lri_d1_full_coefficient_flow`
exact sums	C_2–C_4 to $\Lambda = 2000$, C_5 to $\Lambda = 500$, raw and D^{conn}	Rayleigh–Schrödinger summed over intermediate states; symmetry; `code/lri_1d_c4_exact`, `code/lri_1d_c5_exact`

Table 3: The $d = 1$ computations. Where they overlap they agree to 2×10^{-11} or better (Section 6). The eighth-order code also reproduces the saved records of its original version to 2.1×10^{-8}.

order	raw	g^2D alone	with D^{conn}
$[\epsilon^2]v_n$	$N_{\text{w}}/54$	0	0
$[\epsilon^3]v_n$	$\approx 0.146\, N_{\text{w}}$	$-N_{\text{w}}/27$	$\lvert\cdot\rvert \leq 0.003\, N_{\text{w}}$ (single-state sectors)
$[\epsilon^4]v_n$		$\approx -0.46\, N_{\text{w}}$	96–100% of the g^2D value removed
$[\epsilon^5]v_n$	$+2.3$ to $+2.8\, N_{\text{w}}$	-1.3 to $-1.6\, N_{\text{w}}$	0 within $\pm(0.1$–$0.2)\, N_{\text{w}}$

Table 4: Violations of the descendant spacings in $d = 1$, per missing shell N_{w} (Eq. (13)). Values from ϵ^3 on are extrapolated in Λ; ranges are spreads over fit models. The raw value at ϵ^4 was not extrapolated. The g^2D column and the ϵ^5 raw value are from `analysis/DIAGNOSIS.md` of `lri_hpc_Hamitonian`.

ϵ^4 spacing of $\partial\phi$ measures the part η_4 of the C_4 window that g^2D leaves out, without extrapolating C_4: with g^2D alone it should equal

$$[\epsilon^4]\, v_1(\phi) = 3\eta_4 = 0.0561. \tag{16}$$

The fits give $+0.0495$ (median), 12% below this prediction, which lies just above the range of the fits. With D^{conn} a residual of about a tenth of $3\eta_4$, with the opposite sign, remains. The test thus confirms η_4 at the level of 12%.

3.4 Extrapolations of C_2 to C_8

Table 5 collects the $d = 1$ results.

C_2 and C_3. The exact $C_2(\Lambda)$, $C_3(\Lambda)$ at every even $\Lambda \leq 2000$ are fitted by a rule fixed before any fit was run: free intercept; for C_2 the bases $\Lambda^{-j/2}$, $j \leq 6$ and $j \leq 8$; for C_3 the bases $\{\Lambda^{-1/2}, \Lambda^{-1/2}\log\Lambda, \Lambda^{-1}, \Lambda^{-1}\log\Lambda, \Lambda^{-3/2}, \Lambda^{-3/2}\log\Lambda, \Lambda^{-2}\}$ and the same with $\Lambda^{-2}\log\Lambda$, $\Lambda^{-5/2}$ added; lower cutoffs 100, 200, 400; the median of the six fits, with half their range. This gives $C_2 = 0.657131263984(10)$ ($\delta = 4 \times 10^{-12}$) and $C_3 = -3.7198563(10)$ ($\delta = 2 \times 10^{-8}$). The same rule

on the solver range $\Lambda \le 118$ gives 0.657131272(35) and $-3.71984(8)$. Earlier drafts of the paper quoted, instead, the closest of twenty solver fits to the exact values; that selection used the exact values and is no longer used.

C_4 from the solver. With the complete window, the bases A–F at lower cutoffs 40–70 give 45.415–45.419, all below the exact value. Two further extrapolations extend the range upward (`lri_1d_dconn_refit`). The first is an ensemble of joint fits of C_4 to C_8 that share their tail parameters, kept when their residuals reach the noise floor of the data. The second is a model-independent extrapolation that fits a grid of tail sets and selects among them with rules calibrated on held-out cutoffs. The union of the three is 45.415–45.420. It lies below the exact C_4 and excludes it; its largest deviation is 2×10^{-4}. The fit windows were chosen after the data were seen.

C_4 from the exact sums. With $C_4^{\mathrm{conn}}(\Lambda)$ to $\Lambda = 2000$, fits of the form $A + \sum c_{jk} \Lambda^{-j/2} \log^k \Lambda$ were frozen from the $\Lambda \le 1000$ result before the $\Lambda = 2000$ run existed, for several bases and lower cutoffs. In the basis L2 ($\Lambda^{-1/2}$ and Λ^{-1} with $\log^{0,1,2}$, $\Lambda^{-3/2}$ with $\log^{0,1}$, and Λ^{-2}) they predict $C_4^{\mathrm{conn}}(2000)$ to 4.0×10^{-10}, while fits without logarithms miss by up to 10^{-2}. Which frozen fits to quote was decided after their intercepts had been compared with the exact C_4: basis L2, lower cutoffs 100, 200, 300, median and half-range. This gives $C_4 = 45.4249(9)$, 1×10^{-6} from the exact value; the refit on all data gives 45.4249235266 at lower cutoff 200. The quoted uncertainty is the half-range of these three fits only; fits with larger lower cutoffs have condition numbers above 10^{12} and scatter by up to 3×10^{-3}.

C_5 from the solver. The bases A–F miss C_5 by -1.53 to -0.51, i.e. by up to 2×10^{-3} relative. Freeing the leading tail amplitudes (the coefficients of $\Lambda^{-1/2}$ and $\Lambda^{-1/2} \log \Lambda$ at orders 4 and 5) after the data were seen moves the misses to -0.22 to $+0.06$, and the ensemble of joint fits extends the range upward. The union is -731.36 to -730.97; it contains the exact C_5, and its largest deviation is 3×10^{-4}. The fits that reach the exact C_5 need an effective ω_3 of 380–430, not the gap value 556; Section 7 discusses this.

C_5 from the exact sums. The exact $C_5^{\mathrm{conn}}(\Lambda)$ to $\Lambda = 500$ carries $\Lambda^{-1/2} \log^k \Lambda$ up to $k = 3$. Frozen fits predict the unseen cutoffs to 3×10^{-9} (records frozen at $\Lambda \le 400$) and 1.5×10^{-8} ($\Lambda \le 320$). Their free intercepts, however, determine C_5 only to a few tenths: without the $\log^3$ term (basis L2) the intercept -730.8460638788 is biased; with it (basis M3, which adds $\Lambda^{-1/2} \log^3 \Lambda$ to L2) the intercept is -731.0822822059 at the lowest cutoff and unstable above. With the intercept fixed to the exact C_5, only the bases with $\log^3$ fit at the noise floor, and the fitted $\log^3$ coefficient is $1.945 - -1.994$. Eq. (12) predicts it from the C_4 data, $c_5 = -\frac{\omega_1}{3} c_4 = 1.9965$. The relation (12) holds at ϵ^3 to 5×10^{-6} and at ϵ^4 to 0.3% (Section 7); the step from c_4 to c_5 itself is not tested independently. Fixing c_5 at this value and fitting the rest with a free intercept gives $-731.1467, -731.1397, -731.1359$ for the three best-conditioned fits. A separately written fit of the same constrained model gives $C_5 = -731.13(10)$, the quoted determination. Its uncertainty is the spread over tail bases and fit windows, and the deviation from the exact value, 2×10^{-5}, lies well inside it. This is post hoc: the constraint was adopted after free-intercept fits had left C_5 open to about ± 0.3, with the exact value known, and the selection thresholds of the first constrained fit were set after a run had printed the exact C_5.

C_6, C_7 and C_8. Beyond $\Lambda = 72$ an ensemble of fits to the computed shifts continues ΔC_6, ΔC_7 to $\Lambda = 118$ and ΔC_8 to $\Lambda = 92$. The same rule, frozen on $\Lambda \le 44$ before the data at $46 \le \Lambda \le 72$

existed, predicted those data within a tenth of its own uncertainty. The window changes the order-6 to order-8 extrapolations little; the spread of tail models dominates. The bases A–F, the ensemble of joint fits (16th to 84th percentiles) and the model-independent extrapolation give $C_6 \in 14\,320$–$14\,342$ (at most 8×10^{-4} from C_6) and $C_7 \in -322\,890$ to $-322\,230$ (at most 1×10^{-3}; the range includes the uncertainty of the continuation). For C_8 the eighth-order data to $\Lambda = 92$ with the continued window are used, and the tail models are selected by their reproduction of C_6 and C_7: the selected estimators span $(8.061$–$8.067) \times 10^6$, with central value 8.0652×10^6, 7×10^{-4} from the reference and inside its uncertainty (2×10^{-3}). Unselected tail models span $(8.045$–$8.073) \times 10^6$. This C_8 is calibrated on the diagrammatic C_6 and C_7 and is not an independent determination.

order	HT	δ	source, status
C_2	0.657131263984(10)	4×10^{-12}	exact sums, $\Lambda \le 2000$; rule fixed in advance
C_3	−3.7198563(10)	2×10^{-8}	same
C_4	45.4249(9)	1×10^{-6}	exact sums to $\Lambda = 1000$, fits frozen before $\Lambda = 2000$; choice of fits post hoc
	45.415–45.420	2×10^{-4}	solver, $\Lambda \le 118$, D^{conn}; fit windows post hoc; excludes C_4
C_5	−731.13(10)	2×10^{-5}	exact sums to $\Lambda = 500$, $\log^3$ tail fixed by Eq. (12); post hoc
	−731.36 to −730.97	3×10^{-4}	solver; leading tail freed post hoc (bases A–F alone: 2×10^{-3})
C_6	14 320–14 342	8×10^{-4}	solver plus complete window to 72; post hoc; spread of tail models
C_7	−322 890 to −322 230	1×10^{-3}	same
C_8	8.0652×10^6	7×10^{-4}	$\Lambda \le 92$; calibrated on C_6, C_7

Table 5: $d = 1$ results; δ against Table 1. Ranges and uncertainties are spreads over tail models and fit windows, not standard errors.

3.5 A test protocol fixed before most of the data

A test protocol for the raw C_4 and C_5 was written after the first new thermal cutoff ($\Lambda = 84$) had been seen and before the rest. It set numeric thresholds for three criteria, evaluated in the bases A, B and C: stability under changes of the fit window, prediction of held-out cutoffs, and insensitivity to the tail model. Outcome: the held-out predictions pass for both orders; window stability fails for C_4 in one basis and for C_5 in two of the three; tail insensitivity passes for C_4 and fails for C_5. The protocol's verdict for this mixed outcome is "unresolved", and no later analysis has revised it. Everything above beyond C_3, apart from the blind checks of the exact sums, is post hoc.

4 Results in two dimensions

In $d = 2$ the truncation is a level cutoff $\ell \le L$ and the coefficients are computed in three ways: a production solver to eighth order for $L \le 34$, exact-rational and Cauchy-integral solvers at small L, and sums over intermediate states that give $C_4(L)$ to $L = 2000$ and $C_5(L)$ to $L = 300$. Through ϵ^3 the truncation converges to the continuum. At ϵ^4 every window in use diverges like $\ln L$; a complete window, derived from the level budget of divergent sub-bubbles, removes the logarithm and gives C_4 to 5×10^{-8}. At ϵ^5 it needs two further pieces and then gives C_5 to 5×10^{-5}, if the data carry

no residual $\ln L$ term (Sec. 4.5). C_6 to C_8 are not reproduced. Table 6 collects the results, and Appendices A to C give the derivations and the supporting data.

	HT, $L \to \infty$	continuum	$\lvert C_n^{\mathrm{HT}}/C_n - 1\rvert$	status
C_2	0.30806541358	$\frac{4}{9}\log 2$	35 digits	final
C_3	$-0.8748500(1)$	-0.8748500127	1×10^{-8}	final
C_4, complete window W	5.0488682(10)	5.0488684314	5×10^{-8}	final
C_5, scheme W_{cs} (Sec. 4.5)	$-38.835(13)$	-38.8336459	5×10^{-5}	final; no residual $\ln L$ assumed
C_6 to C_8	not reproduced	362.366(4), ...		not reproduced

Table 6: $d = 2$ perturbative results. "Final" refers to the quoted value in the sense of Sec. 2.2: the analysis is complete, with post hoc choices. The windows behind C_4 and C_5 were derived after their effects had been seen, and the rules W_c and W_{cs} behind C_5 are provisional (Table 31). Uncertainties are spreads over tail models and fit windows, not standard errors; that of C_4 is the half spread of nine fits, rounded up (Sec. 4.4). The C_5 uncertainty assumes that the data carry no residual $\ln L$ term. They cannot exclude one: with the bound judged from the median of the fits it would move C_5 by up to about 0.1, and the spread of single fits would allow -0.7 to $+0.5$ (Sec. 4.5). Continuum values from Table 1; in $d = 2$, $C_3 = \frac{4}{27}\log^2 2 - \frac{\Gamma(1/4)^8}{324\pi^4}$ and $C_4 = \frac{1}{648}\big[16P_4(2) - 64\log^3 2 - 32\log 2\, Q_2 - 359\zeta(3)\big]$, with $Q_2 = \frac{\Gamma(1/4)^8}{24\pi^4} = 12.77138119864696$ and the primitive four-loop wheel value $P_4(2) = 250.48731558186729$ (GFF note, section "Finite multivariable representations for the nonprimitive masters").

4.1 Model and level cutoff

On the unit circle the field has one oscillator A_{ab} for each $a, b \geq 0$, describing $\partial^a \bar{\partial}^b \phi$, with energy $\Delta_\sigma + a + b$, spin $a - b$ and weight

$$\phi(\theta) = \sum_{a,b} \sqrt{w_{ab}}\, \big(e^{\mathrm{i}(a-b)\theta} A_{ab}^\dagger + e^{-\mathrm{i}(a-b)\theta} A_{ab}\big), \qquad w_{ab} = \frac{(\Delta_\sigma)_a (\Delta_\sigma)_b}{a!\, b!}, \qquad \Delta_\sigma = \tfrac{1}{2} - \tfrac{\epsilon}{4}. \tag{17}$$

A Fock state has level $\ell = \sum (a+b)\, n_{ab}$ and particle number $P = \sum n_{ab}$, and its free energy is $\ell + P\Delta_\sigma = H_0 + \epsilon F_1$, with $H_0 = \ell + \frac{P}{2}$ and $F_1 = -\frac{P}{4}$ (the $d = 1$ Hamiltonian, Eq. (4), has $H_0 = \ell + \frac{P}{4}$). With $M = \langle{:}\phi^2{:}\rangle_\theta$ and $V = \frac{1}{12}\langle{:}\phi^4{:}\rangle_\theta$,

$$H(\epsilon) = H_0 + \epsilon F_1 + r(\epsilon) M + g(\epsilon) V + D, \tag{18}$$

where D is the window. We keep the states with $\ell \leq L$ (even), spin zero and even under $a \leftrightarrow b$; the zero mode $(0, 0)$ costs no level and is not truncated; the other modes form the bath. r and g are tuned at cutoff L, and $C_n(L)$ follows from the reference states $|p\rangle$, $p = 0, \ldots, 3$, as in Sec. 2, with $C_0 = 1$, $C_1 = -\frac{1}{6}$, $r_1 = 0$, $g_1 = \frac{1}{3}$ at every L. Through ϵ^8 no intermediate state has more than $p + 16$ particles, so this cap is exact. We write $Z = A_{00} + A_{00}^\dagger$, $\chi = \phi - Z$ for the non-zero modes, $M' = M - {:}Z^2{:}$, and χ_{aa} for the one-quantum state of mode (a, a).

Windows. The raw truncation has $D = 0$. The consistent window is the vacuum window $D_L(\ell) = E_v(L) - E_v(L - \ell)$ with self-consistent vacua $E_v(\lambda)$ at sub-cutoffs $\lambda \leq L$, each computed with the same window. The literal window is the same difference with the vacua $E_v(\lambda)$ of the raw truncation; it differs from the consistent window from ϵ^6 on. The zero-mode window replaces E_v by the self-consistent energy $E_z(\lambda)$ of the channel with the same number z of zero-mode quanta. All of them are diagonal.

Terms. The zero-mode channel $|z\rangle = (A^\dagger_{00})^z|0\rangle/\sqrt{z!}$ has free energy $E^0_z = \frac{z}{2}$; $|z,\chi\rangle$ is a state with z zero-mode quanta and bath part χ of level ℓ_χ, and $|k\rangle$ is an intermediate state of free energy E^0_k. In a time-ordered product of vertices, a *slice* is the intermediate state between two neighboring vertices. A sub-process (a bubble, sunset or mass insertion) absorbs or emits some χ quanta, its *legs*; the other quanta present are *spectators*, *hard* if their level is comparable with L and *soft* otherwise. A term of M' with one creator and one annihilator is a *hop*; one with two creators or two annihilators is a *pair* term. The *exchange sunset* of χ_{aa} is the ϵ^2 ordering in which the new χ_{aa} is emitted, together with a bubble, before the old one is absorbed, so that two (a,a) quanta are present at once. An *auxiliary channel* is the eigenvalue branch $E_z(\lambda)$, with right eigenvector $\psi_z(\lambda)$, of the Hamiltonian at a sub-cutoff $\lambda \le L$ through $|z\rangle$; the windows use it as the reference for zero-mode clusters inside larger states. A *ket-referenced (Bloch)* effective interaction uses the energy denominators of the state it acts on, so it is not symmetric in z and z'. Newton's forward-difference form in p, $E_p = E_0 + p\,e_1 + \binom{p}{2}e_2 + \dots$, defines the *Newton coefficients* $e_1 = E_1 - E_0$ and $e_2 = E_2 - 2E_1 + E_0$. A *tower* is a fit model: a constant plus terms $\ln^j L/L^k$ (Appendix A.3). A *tagged* line is an external quantum replaced by a copy that the cutoff and the windows see with an assigned level, its *charge* (Appendix C.2).

4.2 Solvers

The production solver `code/lri_d2_coeff_hpc/` computes C_0 to C_8, r_n, g_n and every reference energy for $L \le 34$ in the raw and consistent schemes, in float64 truncated power series in ϵ ($L = 16$ to 30 on the symmetry cluster, 32 and 34 on SDSC Expanse); the literal window is implemented and checked at small L. Two clean-room solvers, written from the model specification without seeing the production code or each other, check it: exact Rayleigh–Schrödinger perturbation theory in rational arithmetic ($L \le 18$) and Cauchy integrals of the non-perturbative truncated Hamiltonian at complex ϵ with 256-bit arithmetic ($L \le 8$). The production solver agrees with the exact one to 4×10^{-13} (largest scaled difference over all C_n, r_n, g_n, energies and vacua, $n \le 8$), and the Cauchy solver with the exact one to 5×10^{-46} (Sec. 6).

The large-L results come from separate codes in `code/lri_2d_coeff_independent_check/`. With V and M factorized into zero-mode and χ parts, a K-vertex chain is a sum over χ-line patterns of a Wick count, an exact zero-mode path sum and a χ level sum; at $\epsilon = 0$ the weights factorize holomorphically, $\sum_{ab} w_{ab} y^a \bar{y}^b = (1-y)^{-1/2}(1-\bar{y})^{-1/2}$, and the cutoff only bounds the largest slice level, so one run gives every even L. These codes reproduce the exact rationals and the production solver: $C_2(L)$ and $C_3(L)$ to 4×10^{-14} (absolute), $C_4(L)$ to 1.1×10^{-13} and $C_5(L)$ to 2.2×10^{-13} (relative). The C_5 engine was later optimized; per task it agrees with the original to 7.3×10^{-16}, and the assembled $C_5(L)$ reproduces the earlier data at $L \le 200$ to 7.6×10^{-13}.

The zero-mode scheme has its own exact solver (`zeromode_exact/`, $L \le 18$; Appendix A.3). Every other window scheme at $L \le 16$, and every descendant spacing, comes from two further exact solvers in the same package: `complete_exact/` (scheme W and its variants as explicit operators, exact through ϵ^8 for $L \le 16$) and its copy `residual_log/ce_wc/` with rule W_c and the external-line windows (Appendix B.2).

4.3 Second and third order

At ϵ^2 only $-\frac{P}{4} + g_1 V$ acts and

$$C_2(L) = \frac{1}{81}\Big[17 + \sum_{a=1}^{L/2} w_{aa}\Big(\frac{2}{2a-1} + \frac{18}{2a} + \frac{18}{2a+1} + \frac{2}{2a+2}\Big)\Big], \qquad C_2(L) - C_2 \simeq -\frac{40}{81\pi}\,\frac{1}{L}, \quad (19)$$

with w_{aa} at $\epsilon = 0$. Summed to $L = 10^5$ with its asymptotic tail it equals $\frac{4}{9}\log 2 = 0.30806541358\ldots$ to all 35 digits evaluated (numerical agreement, not a proof). $C_3(L)$ is exact to $L = 600$; its tail has a $\ln L/L$ term with coefficient $0.70734 \approx \frac{9}{2} \cdot \frac{40}{81\pi}$ (the factor $\frac{9}{2}$ identified numerically; Appendix A.2), and towers in $1/L^k \ln^j L$ give $C_3 = -0.8748500(1)$ against -0.8748500127. The schemes do not differ through ϵ^3.

4.4 Fourth order: the level budget and the complete window

Divergence of the raw, consistent and zero-mode windows. Exact $C_4(L)$ to $L = 2000$ in the raw, consistent and zero-mode schemes (Fig. 1a) has the large-L form $A + \beta \ln L$ (plus $-\kappa L$ in raw, $\kappa = 2.2073 \times 10^{-4}$):

scheme	A	fitted β	derived β
raw	5.0021	-0.026381	$-\frac{725}{8748\pi} = -0.0263803$
consistent	5.0026(5)	$-0.026563(42)$	$-\frac{365}{4374\pi} = -0.0265622$
zero-mode	5.0213(5)	$-0.010480(42)$	$-\frac{8}{243\pi} = -0.0104793$

For the consistent and zero-mode schemes, A and the fitted β are medians of twelve fits, with half their spread in brackets; the twelve include variants with the $\ln^2 L/L$ coefficient fixed to $-5/\pi$ (with T_{22} these fit 2.5 to 50 times worse, with $T_{22} + \ln^3 L/L$ about as well; `c4_exact/fits/fit_summary.txt`). The raw row follows from the consistent fit and the exact window shift. The consistent scheme thus diverges like $-0.027 \ln L$. Predictions for $L = 1200$ to 2000, recorded from fits to $L \le 1000$ before those data existed, miss by at most 8.3×10^{-8} with the logarithm; at $L = 2000$ the models without it miss by at least 4.1×10^{-7}.

Cause. The cutoff bounds the total level of an intermediate state, so a divergent sub-bubble inside a state with other quanta of level ℓ keeps only $L - \ell$, while r, g and the vacuum are tuned on level-0 states with the full L. In $d = 2$ the mass (sunset) sub-bubble diverges linearly, $r_2 \simeq \frac{2L}{27\pi}$ (exact slope 0.0235785 at $L = 2000$, $\frac{2}{27\pi} = 0.0235785$), so the deficit is proportional to ℓ and does not decay with L. It is an operator $\sim\{\hat{\ell}, M\}$; through the zero-mode amplitude it is cubic in p, so no choice of r and g absorbs it. Splitting the deficit by window class gives the derived coefficients above: the consistent window restores the vacuum class, the diagonal zero-mode window $\frac{3}{5}$ of the zero-mode sunset class, and the rest is the part of the sunset that changes the zero-mode number by ± 2 ($\frac{8}{243\pi}$; Appendix A.4).

Complete window. Each counterterm acting in a state is evaluated at the budget left by the quanta that do not take part. At the order that enters C_4, $D = D^{\rm zm} + D^M$ with

$$D^{\rm zm}|z,\chi\rangle = \sum_{z'} \left[h^{(2)}_{z'z}(L) - h^{(2)}_{z'z}(L-\ell_\chi)\right]|z',\chi\rangle, \qquad h^{(2)}_{z'z}(\lambda) = g_1^2 \sum_{k:\,\ell_k \le \lambda} \frac{\langle z'|V|k\rangle\langle k|V|z\rangle}{E^0_z - E^0_k}, \tag{20}$$

$$D^M = -\frac{g_1^2}{3\pi}\{\hat{\ell}, M'\} = -\frac{1}{2}\frac{dr_2}{dL}\{\hat{\ell}, M'\}, \tag{21}$$

where $\ell_\chi \ge 2$ is the level of the χ part and $h^{(2)}$ is the second-order effective interaction of the zero mode, a matrix in z in its ket-referenced (Bloch) form; its diagonal contains the consistent and zero-mode windows. The mass window D^M makes a mass insertion between states of levels ℓ_s, $\ell_{s'}$ use $r(L - \frac{1}{2}(\ell_s + \ell_{s'}))$. $D^{\rm zm}$ follows from the linked-cluster cancellation of two components that share only the zero mode; D^M is derived at leading order in $1/L$. Appendix B.1 gives the all-order form of D^M, with the Newton coefficient e_1 in place of r; at ϵ^2 the two forms agree up to $\mathcal{O}(1/L)$. The C_4 data to $L = 2000$ and the brute-force checks of Appendix A.5 use Eq. (21); the exact solvers `complete_exact` and `ce_wc` and the ϵ^5 data use the all-order form. Neither has a fitted parameter.

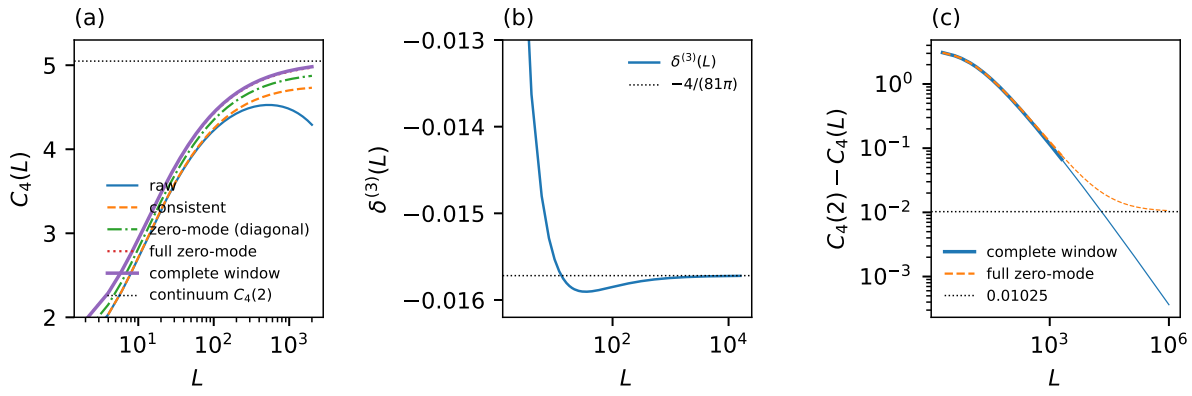


Figure 1: (a) Exact $C_4(L)$ in $d = 2$ for $L \leq 2000$ in five windows: raw, consistent, diagonal zero-mode, full zero-mode (diagonal and off-diagonal) and complete; only the complete window converges to $C_4^{(2)}$ (dotted). (b) The ϵ^3 external-line coefficient $\delta^{(3)}(L)$ of Eq. (24) for $L \leq 16000$, with its limit $-4/(81\pi)$ (dotted); it is not monotone. (c) Distance to the continuum value, $C_4^{(2)} - C_4(L)$, for the complete window (solid) and the full zero-mode window (dashed); thin lines continue each with its tail fit T_{22} ($300 \leq L \leq 2000$, Appendix A.5). The complete window falls like $\ln^2 L/L$; the full zero-mode window levels off at 0.01025 (dotted), the part that the mass window D^M supplies. Script `make_fig_d2.py`.

The window codes agree with brute-force exact perturbation theory with D as an explicit operator at $L \leq 12$ (Appendix A.5).

Result. With the complete window $C_4(L)$ has no $\ln L$ term (Fig. 1a). The full zero-mode window alone tends to 5.0386139; the mass window D^M adds the missing 0.0102543. Fits of $C_4^W(L)$ for $L \leq 2000$ with tail towers in $\ln^j L/L^k$ give

$$C_4 = 5.0488682(10) \qquad \text{against} \qquad C_4^{(2)} = 5.0488684314, \tag{22}$$

with a half spread of 6×10^{-7} over the nine stable no-log fits (towers T_{22}, $T_{22} + \ln^3 L/L$ and T_{23}), rounded up to the quoted 10^{-6}. No-log fits outside this set move the limit by up to 2×10^{-5} (one over-fitted tower excepted). A free $\ln L$ term $\beta \ln L$ in the stable towers comes out at $|\beta| \leq 3.5 \times 10^{-6}$ (T_{33} gives -8.1×10^{-6}) and moves the limit by up to 4.3×10^{-5} (Appendix A.5); the complete window has no $\ln L$ by construction (Table 12). C_2 to C_4 thus agree with the continuum to better than 10^{-7}.

4.5 Fifth order

All-order window and double counting. At all orders $D^{\rm zm}$ uses the Bloch matrix of the auxiliary channels $|z\rangle$ at sub-cutoff λ, $h(\lambda) = U\,\mathrm{diag}(E_z(\lambda))\,U^{-1}$ with $U_{z'z} = \langle z'|\psi_z(\lambda)\rangle$, and the mass window is $D^M = \frac{1}{4}\{e_1(L) - e_1(L - \hat{\ell}), M'\}$ with the Newton coefficient $e_1(\lambda) = E_1(\lambda) - E_0(\lambda)$; degenerate partners of $|z\rangle$ ($z \geq 5$) are projected out. If the auxiliary objects carry their own windows self-consistently, $C_5^W(L)$ diverges linearly, with coefficient -0.0019 ± 0.0004 (the uncertainty of the linear term of raw C_5; Table 16). The cause is a double counting: a sunset nested in an auxiliary channel is restored once by the outer D^M and once more inside $h(L - \ell)$. Rule W_c removes it: an auxiliary channel at sub-cutoff $\lambda < L$ represents a zero-mode cluster inside a state of the cutoff-L theory, so

its windows are referred to the outer cutoff, $[h(L) - h(\lambda - \ell_\chi)]$ and $\frac{1}{4}\{e_1(L) - e_1(\lambda - \hat{\ell}), M'\}$. It leaves C_4 unchanged and removes the linear term (Appendices B.1 to B.5).

External-line window. In W_c the ϵ^3 spacing of χ_{aa} (Sec. 4.6) tends to a constant, $R_3 \approx 0.063\, a\, w_{aa}$ (0.062 to $0.063\, a\, w_{aa}$ at $L = 16$ for $a = 1, 2, 3$, changing slowly with L). An exact test with a tagged external line (Appendix C.2) locates it: if the external quantum is invisible to the cutoff and the windows (budget-free), $R_3 \to 0$; if it is charged its level, R_3 reproduces its value in W_c. D^M charges both legs of a mass sub-process with their average level in every slice, which is exact for the ϵ^2 sunset but not for the mixed orderings at ϵ^3. Scheme W_{cs} adds at ϵ^3

$$D^{(3)} \mathrel{+}= \frac{\delta^{(3)}(L)}{2}\, X_0, \qquad \delta^{(3)}(L) = \tfrac{1}{4}\big[e_1^{(3)}(L-2) - e_1^{\mathrm{tag},(3)}(L;2,2)\big], \tag{23}$$

where X is M' with each term multiplied by the summed level of its χ legs (plus for hops, minus for pair creation and annihilation), X_0 its ϵ^0 part, and $e_1^{\mathrm{tag}}(\lambda;2,2)$ is $E - E_0$ of the ϕ channel $|1\rangle$ at cutoff λ whose external zero-mode quantum is tagged with charges $(2,2)$: it has level 0 in H_0, but the cutoff and the windows count it as level 2 before it is absorbed and after it is re-emitted (Appendix C.3). $\delta^{(3)}(L)$ is an exact rational at each L, one number per cutoff; nothing is fitted (Appendix C.3 derives the operator form). Chain sums give it for every even $L \le 16000$ (equal to the rationals at $L \le 16$ to 4×10^{-16}). It is not monotone, with a minimum -0.0159046 at $L = 34$ (Fig. 1b), and

$$\delta^{(3)}(L) = -\frac{4}{81\pi} - \frac{1}{54\pi}\,\frac{\ln L}{L} + \mathcal{O}(1/L), \tag{24}$$

where the limit $\delta_\infty = -\frac{4}{81\pi}$ (to 7×10^{-11} by free-limit fits) and the $\ln L/L$ coefficient are identified numerically, not derived. With this identification the shift $4a\, w_{aa}\delta_\infty = -\frac{16}{81\pi} a\, w_{aa}$ of $R_3(\chi_{aa})$ that the window produces is two thirds of the ϵ^2 exchange-sunset defect $\frac{8}{27\pi} a\, w_{aa}$ of R_2, and $\frac{16}{81\pi} = 0.0629$ is close to the W_c remainder above. The shift is therefore expected to cancel that remainder as $L \to \infty$; this is not established, since R_3 is known only to $L = 16$. W_{cs} leaves C_0 to C_4, $r_{\le 3}$ and $g_{\le 3}$ unchanged and shifts C_5 by $\delta^{(3)}(L)S(L)$, with $S(L)$ the C_5 response to $\frac{1}{2}X_0$; S has no $\ln L$ and $S_\infty \equiv S(\infty) = 0.367187$, so the shift tends to $\delta_\infty S_\infty = -0.005772$.

Data and tail (Appendix B.6). $C_5^{W_{cs}}(L)$ is raw C_5 to $L = 300$ plus the exact W_c shift and $\delta^{(3)}S$ (both known to $L = 1000$). The tails of C_2 to C_4 carry $\ln^{n-2} L/L$, and hold-out tests select $\ln^3 L/L$ for C_5. The fits use

$$C_5(L) = A + \sum_{j=0}^{3} a_j \frac{\ln^j L}{L} + \sum_{j=0}^{k_2} a'_j \frac{\ln^j L}{L^2}, \qquad k_2 \in \{0,1,2\}, \tag{25}$$

with no $\ln L$ term, on every even L from $L_{\min} \in \{30, 40, 60, 80, 100\}$ to 300. Fitted to $L \le 200$, these forms predict $L = 202$ to 300 to 4.3×10^{-6}, against 9.4×10^{-6} with a free $\ln L$; fitted to $L \le 240$, both predict $L = 242$ to 300 to about 6×10^{-7} (Appendix B.6). The median and half range of the 15 fits give

$$C_5 = -38.835(13) \qquad \text{against} \qquad C_5^{(2)} = -38.8336459, \tag{26}$$

with the unrounded median -38.83544, 0.0018 below the continuum value. Towers with one more power of $\ln L$ at $1/L$ ($k_1 = 4$), which the hold-out test does not exclude, move the median by only $+0.002$ but widen the half range of the fits from 0.013 to 0.050 (Appendix B.6); the quoted uncertainty does not include this. Without the external-line window, $C_5^{W_c} = -38.830(13)$ (median -38.8297), higher by about $-\delta_\infty S_\infty$ as expected. Both agree with the continuum value within the spread of the fits, so C_5 does not distinguish W_c from W_{cs}; the case for the external-line window rests on the spacing R_3 (Sec. 4.6).

Residual logarithm. A term $b \ln L$ that the data cannot see would move the limit. Over the 15 fits of Eq. (25), the median of the free coefficient drifts from -0.011 at $L_{\max} = 200$ to $+0.004$ at 300; a second set of twelve fits ($L_{\min} = 30$ to 80) drifts from -0.0134 to $+0.0054$ (Appendix B.6). On C_4^W, which has no $\ln L$, the 15-fit estimator returns $+0.0005$. On a synthetic series with a C_5-like tail and a known b, both sets read about 0.004 low; corrected for this bias, the median is about $+0.008$ (15 fits; $+0.009$ with 12). We take $|b| \lesssim 0.01$ as a judgement from this median, its drift and its bias; it is not a confidence bound. The individual fits of the 15-fit set range from -0.05 to $+0.07$, but the extreme ones have nearly as many parameters as points; taken as a bound they would allow shifts of C_5 between about -0.7 and $+0.5$. In the 15 fits, fixing $b = \mp 0.01$ moves the limit by about ± 0.1, and $b = +0.008$ would move it by about -0.08; bounding $|b|$ well below 0.01 would need $L \gtrsim 400$. The external-line class itself gives no $\ln L$: the $\ln L$ terms of S cancel within the hop and within the pair sector separately.

Status. The complete window has no fitted parameter, removes the $\ln L$ and is verified through ϵ^4. At ϵ^5 the rule W_c is needed to remove a linear divergence, and the external-line window to reduce R_3 (by at least 95% at $L = 14$ and 16). With both, C_5 agrees with the continuum. Both rules are provisional: they are not derived in full, and the window is not proven complete at this order. Like all windows here, they were derived after their effect had been seen and with the continuum values known.

4.6 Descendant spacings

A descendant lies an integer above its primary at every order. With

$$R_n(L) = \big[E(\chi_{aa}) - E(\phi) - 2a\big]_{\epsilon^n}, \tag{27}$$

the ϵ^n coefficient of the spacing of $\chi_{aa} = \partial^a \bar{\partial}^a \phi$ computed with the same r, g and window, and the same for the descendants $(\partial\bar{\partial})^a \phi^2$ of ϕ^2, the exact small-L solvers `complete_exact` and `ce_wc` (Sec. 4.2) give every R_n, $n \le 8$, as rationals for $L \le 16$ in each scheme. These tests need no continuum input; Appendix C gives the data.

- ϵ^2: in the consistent and zero-mode windows the exchange sunset of χ_{aa} has only $L - 4a$, and $R_2 \to \frac{8a\, w_{aa}}{27\pi}$ (0.0243 for χ_{11} at $L = 16$). D^M shifts it by exactly $-\frac{8a\, w_{aa}}{27\pi}$, so in the complete window R_2 decays like $1/L$ ($0.0114/L$ for χ_{11}), also for the ϕ^2 descendants.

- ϵ^3, single-quantum descendants: $R_3(\chi_{11})$ is 0.0156 in W_c and -1.6×10^{-4} in W_{cs} at $L = 16$. At $L = 14$ and 16 the window leaves at most 3% of the W_c value for $a = 1$ to 3, so at least 95% of the defect is removed at $L = 14$ and 16. Free-limit fits of χ_{11} leave a few percent, and the window is expected to cancel the defect as $L \to \infty$ (above); R_3 itself is not computed beyond $L = 16$.

- ϵ^3, descendants of ϕ^2: at $L = 16$, $R_3 = -0.057$ (level 2) and -0.059 (level 4) in W_{cs}; the level-4 value still moves away from zero. Not settled.

- ϵ^4: in W_c and W_{cs}, $R_4(\chi_{11})$ grows, with a local slope of about 0.028 per unit of $\ln L$ at $L = 8$ to 16, while with budget-free legs it goes to zero: the same leg defect one order higher. A partial χ-line window ($W_{cs4\chi}$, calibrated on χ_{11}, with the ϵ^4 difference between the χ-line and zero-mode self-energies as a spectator term) closes 95% and 90% of the gaps of χ_{22} and χ_{33} to their budget-free lines at $L = 14$, but the χ_{22} gap is not shrinking, the ϕ^2 descendants still grow, and $R_5(\chi_{aa})$ grows. The window coefficient grows roughly like $\ln L$ at $L \le 16$, which suggests that C_6 in W_c carries about $+0.010 \ln L$ (a prediction, not a measurement). The ϵ^4 window is not yet known.

4.7 Sixth to eighth order

C_6 to C_8 are not reproduced. At $L = 16$, $C_6 \approx 165.4$ in W_c, W_{cs} and $W_{cs4\chi}$ (Table 30), 54% below the continuum value 362.366(4), and $C_6(L)$ still rises by about 7.5 per step of 2 in L. The external-line windows shift C_6 by about 10^{-4} of its value there, while W and W_c differ by about 1% (Appendix B.7). The production tables to $L = 34$ (raw and consistent) are shipped as finite-cutoff data, but those windows diverge from ϵ^4 on, and the external-line windows from ϵ^4 on are not known.

5 Results in three dimensions

Model and cutoff. In $d = 3$ the generalized free field is quantized on S^2 (Sec. 2, Ref. [4]). A mode (n, ℓ, m) has energy $\Delta_\sigma + 2n + \ell$, with ℓ the angular momentum; the zero mode $n = \ell = 0$ is the local oscillator, and all other modes form the bath. We keep the states with bath level

$$N_b = \sum_{(n,\ell,m)\neq(0,0,0)} (2n + \ell)\, n_{n\ell m} \leq L \tag{28}$$

and do not truncate the zero mode. The perturbation is $H = H_0 + r\, M + g\, V$ with $M = \int d\Omega\, {:}\phi^2{:}/(4\pi)$ and $V = \int d\Omega\, {:}\phi^4{:}/(48\pi)$. r and g are tuned order by order to $E_1 - E_0 = \Delta_\sigma$ and $E_3 - E_0 = 3 - \Delta_\sigma$, and

$$\Delta_\varepsilon(\epsilon; L) = E_2 - E_0 = \tfrac{3}{2} - \tfrac{\epsilon}{6} + \sum_{n\geq 2} C_n(L)\, \epsilon^n, \qquad \Delta_\sigma = \frac{3-\epsilon}{4}. \tag{29}$$

The continuum values $C_n^{(3)}$ are those of Table 1.

Two codes. The production package (`code/lri_3d_hpc/`) builds the truncated Hamiltonian explicitly in the SO(3)-scalar, inversion-even sector and solves the Rayleigh–Schrödinger recursion through ϵ^8 in x87 long double. It reached $L = 16$, 18 and 20 (196 585 scalar bath states at $L = 20$, from 182 727 113 before the projection; about 2550 service units on SDSC Expanse, charge units of about one per core-hour, with large-memory jobs charged more). It implements two prescriptions: raw, and the vacuum window $D_L(N_b) = E_v(L) - E_v(L - N_b)$.

The exact-sum package (`code/lri_3d_c4_exact/`) evaluates the same coefficients as sums over intermediate states, with exact partial-wave reductions on S^2 and exact zero-mode projections, at every even L:

- C_2 to $L = 8000$, C_3 and C_4 to $L = 600$, C_5 to $L = 40$;
- every window shift to $L = 8000$;
- every window scheme, including the complete window of $d = 2$ (Sec. 4).

At every even $L \leq 20$ the two codes agree on C_2 to C_5, r, g and the reference energies E_0 to E_3 to 3×10^{-13}. The window schemes that the production package lacks agree with a brute force, built as explicit operators on the package's own Hamiltonian at $L \leq 14$, to 4.3×10^{-13}. Table 7 gives the production values.

	consistent (vacuum window D_L)			raw		
	$L = 16$	$L = 18$	$L = 20$	$L = 16$	$L = 18$	$L = 20$
C_2	0.1799805130	0.1801641045	0.1803065210	0.1799805130	0.1801641045	0.1803065210
C_3	−0.3238079917	−0.3271370505	−0.3298433574	−0.3238079917	−0.3271370505	−0.3298433574
C_4	0.7154707349	0.7295143865	0.7400078407	0.7082381633	0.7206323697	0.7293066716
C_5	−3.157412985	−3.274129748	−3.376315158	−3.173233312	−3.292870751	−3.398184422
C_6	15.68169942	16.38726028	17.03798989	15.67552153	16.38079206	17.03198660
C_7	−84.42870757	−88.02482107	−91.14377531	−84.17501495	−87.66743530	−90.65688526
C_8	548.2872011	578.4398685	605.8501134	548.2146549	578.3577582	605.7652262

Table 7: Finite-cutoff coefficients $C_n(L)$ in $d = 3$ from the production package `lri_3d_hpc` (10 significant digits; the records hold 20), raw and with the vacuum window D_L. C_2 and C_3 do not depend on the window. The exact sums reproduce C_2 to C_5 of this table to 3×10^{-13}; from C_4 on, neither scheme converges (Fig. 2).

The vacuum window diverges at ϵ^4. $C_2(L)$ and $C_3(L)$ do not depend on the window and converge. From C_4 on, the vacuum window does not suffice (Fig. 2).

- As in $d = 2$, divergent sub-bubbles inside a state see only the level left by the other quanta, while r, g and the vacuum are tuned with the full L.
- In $d = 3$ the mass sub-bubble diverges like $L^{3/2}$ instead of L, so every term is one power worse.
- Uncorrected, $C_4(L)$ falls like $-2.135 \times 10^{-5}\, L^2$.
- With D_L alone it peaks at 0.759 near $L = 32$ and is -0.749 at $L = 600$. The terms are $-7.433 \times 10^{-5}\, L$ and $-\frac{10}{243\,\Gamma(7/2)}\sqrt{L}\ln L$, which the diagonal and zero-mode-number-changing parts of the zero-mode window remove.
- The mass window (χ-attached sunset term), which in $d = 2$ is a finite shift, here removes a further $3.614 \times 10^{-4}\sqrt{L}$.

Complete window. The complete window W of $d = 2$ is carried over unchanged: vacuum window, full zero-mode window, and mass window $-\frac{1}{2}r_2'(L)\{\hat{N}_b, M'\}$, with $\hat{N}_b$ the bath-level operator. With it, $C_4(L)$ converges like $1/L$. At $L = 600$ it is 1.21735. With the limit fixed at $C_4^{(3)}$, a free $\sqrt{L}$ term comes out below 7×10^{-8}. The mass window is used in its leading-order linear form; since r_2 is not linear in L in $d = 3$, the exact form may differ by a finite amount, which the agreement below suggests is small but which is not derived.

At ϵ^5 the nested windows are referred to the outer cutoff (rule W_c of $d = 2$); the external-line window (W_{cs}) changes $C_5(40)$ by -0.003.

Extrapolations. Fit forms (towers of $L^{-k}\ln^j L$ terms, from the $d = 2$ protocol) and windows were chosen after the data were seen, with the continuum values known; the quoted uncertainties are spreads over fits, not standard errors.

- $C_2 = 0.181221785405554$ (15 digits of the closed form; fit to $L \le 8000$).
- $C_3 = -0.353581479(20)$ ($\delta = 1 \times 10^{-7}$): mean and half-range of three fit windows.
- $C_4 = 1.228070(21)$ ($\delta = 5 \times 10^{-6}$): median of one tower over 12 fit windows; the uncertainty is the half-range, including a blind fit to $L \le 400$ that predicts $402 \le L \le 600$ to 1.6×10^{-8}. Towers with one more correction term spread to about 10^{-4}.

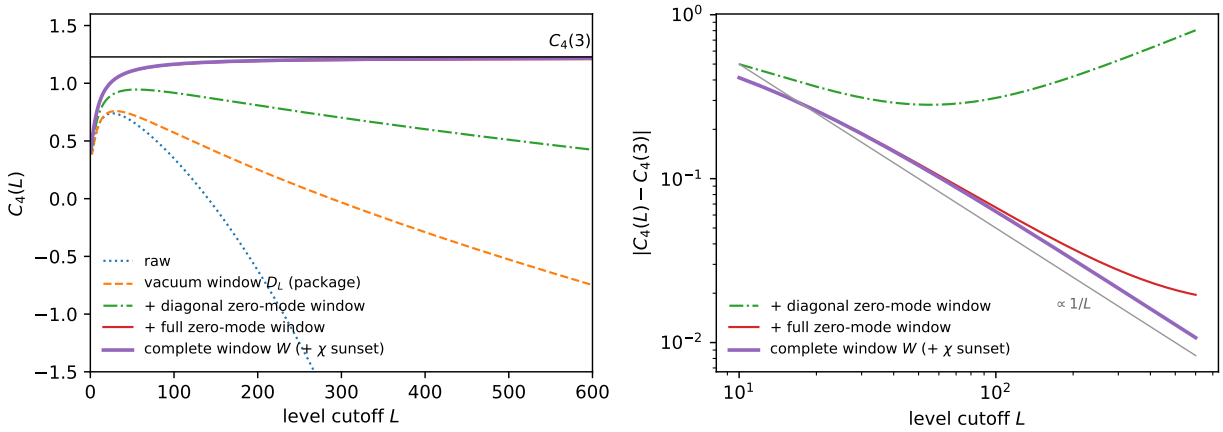


Figure 2: $C_4(L)$ in $d = 3$ from the exact sums, in each window scheme. Left: raw, the vacuum window D_L of the production package, and the zero-mode and mass windows added one by one (the full-zero-mode curve lies under W at this scale). Right: distance from $C_4^{(3)}$ on logarithmic axes. The complete window W converges like $1/L$; without its mass window the distance stops decreasing, has its minimum near $L \approx 1100$ and grows like $3.6 \times 10^{-4}\sqrt{L}$ beyond the plotted range.

- C_5, provisional and calibrated on $C_4^{(3)}$: towers are kept if, on $C_4^W(L)$ at the same $L \le 40$, they reproduce $C_4^{(3)}$ to within 0.036 and pass a blind test. The median of their C_5 fits gives -6.0 ± 0.3 (median -6.005) against $C_5^{(3)} = -5.850351421$. The uncertainty is the worst C_4 miss relative to the remaining C_4 correction at $L = 40$ (0.036 of 0.147), times the remaining C_5 correction at $L = 40$ (1.28, which uses $C_5^{(3)}$ itself).
  - This is a weak test: $C_5^{W_c}(40) = -4.574$ is still 1.28 above the limit.
  - A term in $L^{-1/2}$, which $L \le 40$ cannot exclude, would lower the limit by 0.5 to 1.
  - A window without the mass window also fits at $L \le 40$, although it differs from W_c by a term growing like L^2.
  - Going further needs a new representation of the angular channels: the cost grows like $L^{8.4}$ and the memory like $L^{4.5}$.

Descendant spacings. For the scalar descendant of ϕ (the $n = 1$, $\ell = 0$ quantum), the window removes the ϵ^2 violation of its spacing as $L \to \infty$. At ϵ^3 the external-line window acts in the right direction, but the violation is not shown to vanish at $L \le 16$.

6 Independent checks

One dimension.

- *Window engines.* The complete window through ϵ^5 was computed by two independent engines, one on the solver's kernels and one written from scratch with exact rationals, plus a third engine for a skeptic's check. Four algorithms within the second engine agree to 1.3×10^{-11}; the two engines agree on 3808 numbers to 9.6×10^{-12}; the third reproduces ΔC_5 at $\Lambda \le 28$ to 2.4×10^{-12} (quoted as 10^{-11} in the paper).

- *Eighth-order window.* The complete-window solver reproduces the exact ΔC_4, ΔC_5 at $\Lambda = 46$–72 to 2×10^{-11}.

- *Exact sums against the solver.* The exact sums reproduce the solver's raw C_4 and C_5 at $\Lambda \leq 118$ to 1.4×10^{-12} and 10^{-11}, and the window pieces to 10^{-15}; brute-force Rayleigh–Schrödinger on explicit bases at small cutoffs agrees to 10^{-11}.

- *Eighth-order code.* The optimized eighth-order code reproduces the 38 saved records of its original version to 2.1×10^{-8} and the thermal solver's C_0–C_7 to 6×10^{-14} relative.

- *Blind protocol of the exact sums.* Fit records were frozen and hash-committed before each larger run existed: $\Lambda \leq 1000$ before 2000 for C_4, $\Lambda \leq 320$ before 400 and $\Lambda \leq 400$ before 500 for C_5. Their predictions of the new cutoffs are quoted in Section 3.4.

- *Constrained C_5.* A separately written fit of the constrained extrapolation, made without reading the package's fit code, gives $-731.13(10)$; a separate derivation confirms $\omega_1 = \frac{7}{2}$ as the one-loop exponent of ϕ^6 and the closed form (12).

Two dimensions. Two solvers were written from a first-principles specification of the $d = 2$ model (Sec. 4.1) by agents that saw neither the production code nor each other's work. The first is plain Rayleigh–Schrödinger perturbation theory to ϵ^8 with full wavefunction corrections in exact rational arithmetic, in two codes (`fractions` over the full space, and a reflection-reduced code modulo many primes with rational reconstruction checked on held-out primes), for $L = 0, \ldots, 18$ in all three schemes. The second uses no perturbation theory: the truncated Hamiltonian is built at complex ϵ on a circle with 256-bit arithmetic, r and g are Newton-solved, the eigenvalues are followed from the references, and Cauchy integrals give the coefficients, for $L \leq 8$. Over C_n, r_n, g_n, the reference energies and all sub-cutoff vacua, $n \leq 8$, the production solver agrees with the exact one to 4×10^{-13} (scaled difference $\max |a - b| / \max(1, |b|)$; raw and consistent $L \leq 18$, literal $L \leq 14$), the Cauchy solver with the exact one to 5×10^{-46}, and the production C_n with the Cauchy ones to 1.6×10^{-14}. The large-L sum codes of Sec. 4.2 reproduce these values at small L. The raw scheme was specified from first principles; for the consistent and literal schemes the definitions were taken from the production package, so there the check covers the implementation, not the choice of window. These checks verify that we compute the stated truncation; they do not test its continuum limit.

Three dimensions. These checks show that the codes compute the stated truncation; the windows and fit forms are the assumptions.

- *Two codes:* the exact sums and the production package agree to 3×10^{-13} at $L \leq 20$ (Sec. 5).

- *Original implementation:* the precomputed Hamiltonian of each production cutoff, solved at $L = 14$, agrees with the original implementation shipped in the package to 1.25×10^{-14}.

- *Stored Hamiltonians:* the one of $L = 20$ reproduces those of $L = 16$ and $L = 18$ to 9.8×10^{-15}.

- *Recursion:* every recursion residual is at most 5.1×10^{-19}.

- *Engine regression:* the final ϵ^5 engine reproduces the stored $L \leq 20$ run to 5×10^{-13} and all reduced $L = 40$ sub-sums bit for bit.

7 Caveats

One dimension.

- The windows were derived after their effects had been seen and with the continuum values known, though with no fitted parameter.
- The extrapolation dominates the error, and the exponents (10) are an assumption. Fits that reach the exact C_5 need an effective ω_3 of 380–430, not the gap value 556. A second channel with the leading power $\Lambda^{-1/2}$, from the phase operator of the cutoff kernel, enters at ϵ^4 (cited check `code/lri_d1_fullblock_tail_response`); it does not affect the top logarithm (12) but does affect the lower ones, so tying them to ω_2, ω_3 is not exact.
- The ϵ^4 top-log coefficient fitted from the exact $C_4(\Lambda)$ lies 0.3% from Eq. (12), depending on the basis; whether this is a real violation is undecided. Taking c_5 from the C_2 chain of Eq. (12) instead, a value 0.3% smaller, moves the constrained C_5 by 4×10^{-4} to 1.4×10^{-3} (absolute), far below its quoted uncertainty (`xchk/REPORT_blind.md` of `lri_1d_c5_exact`).
- A test protocol for the raw C_4 and C_5, fixed before most of the large-cutoff data had been seen, had mixed results: its held-out predictions passed; window stability failed for C_5 in two of three bases and for C_4 in one, and tail insensitivity failed for C_5. Every $d = 1$ fit beyond C_3 involves choices made after the data were seen, apart from the blind predictions of the exact sums. The constrained C_5 fit's selection thresholds were set after a run had printed the exact value; the separately written fit is the quoted determination.
- C_8, calibrated on the diagrammatic C_6 and C_7, is not an independent determination.
- We have not constructed an effective Hamiltonian beyond the windows.

Two dimensions. The windows of Secs. 4.4 and 4.5 were derived after their effects had been seen and with the continuum values known, though they contain no fitted parameter. The complete window is verified through ϵ^4; at ϵ^5 the rule W_c and the external-line window are needed as well, and the window is neither derived in full nor proven complete there. The limit $-4/(81\pi)$ of $\delta^{(3)}$ is identified numerically. The quoted limits are spreads over tail models and fit windows, not standard errors. The C_5 value assumes no residual $\ln L$ term; one with $|b| \lesssim 0.01$, a bound judged from the median of the fits rather than a confidence bound, is not excluded at $L \leq 300$ and would move C_5 by up to about 0.1. C_5 does not distinguish W_c from W_{cs}; the external-line window rests on the spacing R_3. The ϵ^3 spacings of the ϕ^2 descendants are not settled at $L \leq 16$, the ϵ^4 window is not known, and C_6 to C_8 are not reproduced. The production tables (raw and consistent schemes) diverge from ϵ^4 on, linearly in L (raw) and like $\ln L$ (consistent), so they are finite-cutoff data of those schemes, not continuum estimates.

Three dimensions. The fit forms and windows were chosen after the data were seen, with the continuum values known. The mass window is used in its leading-order linear form, which is not derived for $d = 3$. The C_4 uncertainty is the spread of one tower over 12 fit windows; towers with one more correction term spread to about 10^{-4}. C_5 is provisional: it is calibrated on $C_4^{(3)}$, the scale of its uncertainty uses $C_5^{(3)}$ itself, a term in $L^{-1/2}$ is not excluded at $L \leq 40$, and at $L \leq 40$ the test cannot separate W_c from the window without the mass window. The ϵ^3 violation of the descendant spacing (scalar descendant of ϕ) is not shown to vanish at $L \leq 16$. The last digit of C_3 is platform-sensitive: one fit window moves by about 10^{-9} between machines, and the replay checks it to a tenth of the stated uncertainty.

8 Reproduction

8.1 Layout

All paths are relative to the folder `HT_Pert/` of the public repository.

- `code/<package>/`: one package per solver or analysis, shipped at the commit its production ran. Each has a `public.mk` with the targets `test` and `replay` (some also `smoke`, `replay-full`, `fetch-data`) and a `public/` folder with the replay scripts.
- `data/HT_<d>d/`: the production records.
- `results/HT_<d>d/`: the expected value of every quoted number (`quoted_numbers.json`) and summaries.

`make test` and `make replay` in `HT_Pert/` run the targets of every package; in a package folder, `make -f public.mk test|replay`. A replay recomputes every quoted number of its package from the shipped records, without a production rerun, prints a PASS/FAIL table and exits non-zero on any failure.

8.2 Packages and records

package (`code/`)	d, role	content, records, cost
`lri_hpc_Hamitonian`	1: solver, analyses	HT solver to ϵ^7; joint tail fits, window engines, identities. Records `data/HT_1d/thermal_expanse` (111 cutoffs) and the analysis outputs
`lri_eh_hpc`	1: window	isolated g^2D window, closed form K
`lri_c8_hpc`	1: solver	eighth order, $\Lambda \leq 92$; records `data/HT_1d/c8`
`lri_d1_full_coefficient_flow`	1: solver	complete window through ϵ^8, $\Lambda \leq 72$; records `data/HT_1d/s10_production`
`lri_1d_dconn_refit`	1: analysis	extrapolations of C_4–C_8 with the complete window
`lri_1d_c4_exact`, `lri_1d_c5_exact`	1: exact sums	C_2–C_4 to $\Lambda = 2000$, C_5 to 500; records `data/HT_1d/exact_c4`, `exact_c5`; blind fit records
`lri_d1_fullblock_tail_response`	1: cited check	second $\Lambda^{-1/2}$ channel
`lri_d2_coeff_hpc`	2: solver	eighth-order production C_0–C_8, raw and consistent windows to $L = 34$ (literal window to $L = 14$); git ref `f847d38`; records `data/HT_2d/prod` (41 self-hashed records, tables, validation receipts); production 2.7 node-hours (symmetry $L \leq 30$: 0.6; Expanse $L = 32, 34$: 2.1, the largest point 48 min and 463 GiB); test 1.2–1.5 min (61 tests; native backend needs x86-64 Linux), smoke 10 s, replay 1 s
`lri_2d_coeff_independent_check`	2: checks, exact sums, windows	clean-room exact RS and Cauchy solvers; exact C_2, C_3 to $L = 600$, C_4 to $L = 2000$ in every scheme and with the complete window, raw C_5 to $L = 300$, $\delta^{(3)}$ to $L = 16000$; the all-order window solver (W_c, W_{cs}, $W_{cs4\chi}$, $L \leq 16$); dated snapshot (no git; derivations in Appendices A to C); records inside the package; production: C_5 to $L = 300$ about 55–60 node-hours with the optimized engine (measured: 130 node-hours for 40% of the tasks with the unmodified engine, 32 for the remaining 60% with the optimized one), C_4 to $L = 2000$ 1.3 node-hours (77 min on one 40-core node), $\delta^{(3)}$ to $L = 16000$ 11 min (18 GB), the exact solvers minutes to an hour per cutoff; test 1.6–1.8 min (with python-flint), replay 3 s, replay-full 15 s

package (`code/`)	d, role	content, records, cost
`lri_3d_c4_exact`	3: exact sums	C_2 at every even $L \le 8000$, C_3, C_4 to $L = 600$, C_5 to $L = 40$, window shifts to $L = 8000$, every window scheme; the paper's $C_2^{(3)}$ to $C_4^{(3)}$ and the C_5 footnote; git ref `d5afa4f`; results in the package (`results/`); test 1–3.5 min, replay about 3 s
`lri_3d_hpc`	3: solver	C_2–C_8 at $L = 16$, 18, 20, raw and vacuum window; the production check of the exact sums and the $L = 14$ gates against the original code; git ref `7c8a67e`; records `data/HT_3d/pert3d`, table `results/HT_3d/SUMMARY.md`; production about 2550 SU on SDSC Expanse; test 1.5–2.1 min, replay about 1 s

8.3 Rules that are part of the results

The fit families and windows, the selection and calibration rules, the rounding of quoted ranges and the status words are part of the results; the `REPRODUCE` file of each package lists them under "Decisions you must not make".

8.4 Result → code, record, command, tolerance

result	value	pkg	records	command	tolerance
$d = 1$ (packages: S = `lri_hpc_Hamitonian`, G = `lri_eh_hpc`, E = `lri_c8_hpc`, W = `lri_d1_full_coefficient_flow`, R = `lri_1d_dconn_refit`, X4 = `lri_1d_c4_exact`, X5 = `lri_1d_c5_exact`, F = `lri_d1_fullblock_tail_response`; records: T = `data/HT_1d/thermal_expanse`, A = the package's `analysis/`, C8 = `data/HT_1d/c8`, S10 = `data/HT_1d/s10_production`, R = the package's summary outputs, X4, X5 = `data/HT_1d/exact_c4`, `exact_c5`, F = the package's `results/`. Each package's integrity checks (code fingerprints, checksum lists, self-hashed records, gates) are merged into one row; `results/HT_1d/quoted_numbers.json` describes every row.)					
integrity of code and records	pass	S	T, A	replay `a1.code`, `a1.records`, `a1.xcheck`, `a1.gates`, `a1.reruns`	exact
largest cutoff of the thermal data	118	S	T, A	replay `a1.lam`	exact
basis size at $\Lambda = 118$	3.5×10^9	S	T, A	replay `a1.states`	as stated
raw $C_4(118)$, % of C_4	67	S	T, A	replay `a1.frac4`	as stated
raw $C_7(118)$, % of C_7	44	S	T, A	replay `a1.frac7`	as stated
tail of $C_2(\Lambda)$, Eq. (9)	$-40/(81\sqrt{\pi})\,\Lambda^{-1/2}$	S	T, A	replay `a1.T2`	bound
ΔC_4 at $\Lambda = 118$	0.080	S	T, A	replay `a1.win4`	as stated
ΔC_5 at $\Lambda = 118$	0.71	S	T, A	replay `a1.win5`	as stated
large-Λ constant of ΔC_4, Eq. (15)	0.0803	S	T, A	replay `a1.win4inf`	as stated
raw C_4 below C_4 (%)	0.19	S	T, A	replay `a1.offset4`	as stated
C_4 from the solver (union): check of its bases A–F part	45.415–45.420	S	T, A	replay `a1.c4AF`	as stated
largest relative C_5 miss, bases A–F	2×10^{-3}	S	T, A	replay `a1.c5pre`	as stated

result	value	pkg	records	command	tolerance
C_5 from the solver (union): check of its freed-tail part	-731.36 to -730.97	S	T, A	replay `a1.c5freed`	as stated
ω_3 from the ϕ^6 gap	556	S	T, A	replay `a1.omega3`	as stated
effective ω_3 of the fits that reach C_5	380–430	S	T, A	replay `a1.omega3eff`	as stated
verdict of the test protocol	unresolved	S	T, A	replay `a1.protocol`	exact
agreement of the window engines through ϵ^5	10^{-11}	S	T, A	replay `a1.engine`	as stated
$[\epsilon^3]v_n$ with D^{conn}, per N_{w}	0.003	S	T, A	replay `a1.ident3`	reading
% of the g^2D violation at ϵ^4 removed	96–100	S	T, A	replay `a1.ident4`	reading
$[\epsilon^5]v_n$ with D^{conn}: zero within $\pm$, per N_{w}	0.1–0.2	S	T, A	replay `a1.ident5`	reading
cutoff placement: raw HT exact through ϵ^3	to 10^{-14}	S	T, A	replay `a1.phase`	bound
cutoff placement at ϵ^4, g^2D alone	$3\eta_4 = 0.0561$ required	S	T, A	replay `a1.phase.g2D`	info
raw $[\epsilon^2]v_n$	$N_{\mathrm{w}}/54$	S	T, A	replay `a1.W54`	bound
C_4, bases A–F, lower cutoffs 40–70	45.415–45.419	S	T, A	replay `a1.c4AF.range`	exact
C_5 misses, bases A–F	-1.53 to -0.51	S	T, A	replay `a1.c5pre.miss`	exact
C_5 misses, leading tail freed	-0.22 to $+0.06$	S	T, A	replay `a1.c5freed.miss`	exact
$[\epsilon^4]v_1(\phi)$, g^2D alone (median)	$+0.0495$	S	T, A	replay `a1.phase.g2D.value`	exact
integrity of code and records	pass	G	A	replay `a1.eh.code`, `a1.eh.receipt`, `a1.eh.bench`	exact
finite part K of the g^2D shift of C_4	0.0616119308426799	G	A	replay `a1.eh.K`	as stated
exact g^2D shift $\Delta C_4(\Lambda)$	$\log\Lambda/17496 + K$	G	A	replay `a1.eh.exact`	bound
largest cutoff of the C_8 data	92	E	C8	replay `c8.reach`	exact
integrity of code and records	pass	E	C8	replay `c8.lanes`, `c8.envelopes`, `c8.exports`, `c8.source`, `c8.manifest`, `c8.gates`, `c8.validation`, `c8.datajson`	exact
$C_8(92)$, raw and g^2D	2883374.25, 2883433.21	E	C8	replay `c8.values`	exact
C_6, C_7, C_8 at every cutoff	table in the package	E	C8	replay `c8.table`	as stated

result	value	pkg	records	command	tolerance
eighth-order code against its original version	2×10^{-8}	E	C8	replay `c8.original`	as stated
raw C_0–C_7 against the thermal solver	6.1×10^{-14}	E	C8	replay `c8.thermal`	as stated
integrity of code and records	pass	W	S10	replay `s10.manifest`, `s10.code`, `s10.records`, `s10.csv`	exact
largest cutoff of D^{conn} through ϵ^8	72	W	S10	replay `s10.winlam`	exact
ΔC_6, ΔC_7, ΔC_8 at $\Lambda = 72$	$+1.1717361$, -1.828185, -8.5641	W	S10	replay `s10.dC678`	exact
complete-window solver against the exact ΔC_4, ΔC_5	2×10^{-11}	W	S10	replay `s10.match`	exact
integrity of code and records	pass	R	R, A	replay `rf.inputs`	exact
blind check of the window continuation (per typical uncertainty)	0.099	R	R, A	replay `rf.blind`	exact
C_4 from the solver (union)	45.415–45.420	R	R, A	replay `rf.C4`	exact
its largest deviation	2×10^{-4}	R	R, A	replay `rf.C4dev`	exact
C_5 from the solver (union)	-731.36 to -730.97	R	R, A	replay `rf.C5`	exact
its largest deviation	3×10^{-4}	R	R, A	replay `rf.C5dev`	exact
C_6	14 320–14 342	R	R, A	replay `rf.C6`	exact
its largest deviation	8×10^{-4}	R	R, A	replay `rf.C6dev`	exact
C_7	$-322\,890$ to $-322\,230$	R	R, A	replay `rf.C7`	exact
its largest deviation	1×10^{-3}	R	R, A	replay `rf.C7dev`	exact
sensitivity of C_7 to the weighting (not quoted)	not quoted	R	R, A	replay `rf.C7w`	info
C_8, calibrated	8.0652×10^6	R	R, A	replay `rf.C8`	exact
its deviation	7×10^{-4}	R	R, A	replay `rf.C8dev`	exact
calibrated range of C_8 ($\times 10^6$)	8.061–8.067	R	R, A	replay `rf.C8cal`	info
uncalibrated range of C_8 ($\times 10^6$)	8.045–8.073	R	R, A	replay `rf.C8uncal`	exact
window effect on C_6, C_7, C_8	C_6: -0.01 to -0.04; C_7: $+4$ to $+19$; C_8: $+17$ to $+32$, -240 to -490 (freed tails)	R	R, A	replay `rf.paired`	exact
noise of ΔC_6, ΔC_7, ΔC_8, $\Lambda = 46$–72	3.95×10^{-10}, 1.34×10^{-8}, 2.32×10^{-7}	R	R, A	replay `rf.noise`	exact
C_2 from the exact sums	0.657131263984(10)	X4	X4	replay `x4.c2`	exact
its δ	4×10^{-12}	X4	X4	replay `x4.c2`	exact

result	value	pkg	records	command	tolerance
$\lvert C_2^{\mathrm{HT}} - C_2 \rvert$	3×10^{-12}	X4	X4	replay `x4.c2`	exact
C_3 from the exact sums	$-3.7198563(10)$	X4	X4	replay `x4.c3`	exact
its δ	2×10^{-8}	X4	X4	replay `x4.c3`	exact
$\lvert C_3^{\mathrm{HT}} - C_3 \rvert$	6×10^{-8}	X4	X4	replay `x4.c3`	exact
largest cutoff of the exact C_2–C_4 sums	2000	X4	X4	replay `x4.lexact`	exact
C_4 from the exact sums	45.4249(9)	X4	X4	replay `x4.c4`	exact
its δ	1×10^{-6}	X4	X4	replay `x4.c4`	exact
$C_4^{\mathrm{conn}}(1000)$	37.8351012026	X4	X4	replay `x4.values`	exact
$C_4^{\mathrm{conn}}(2000)$	39.4111422825	X4	X4	replay `x4.values`	exact
overlap of the runs to 1000 and 2000	3.3×10^{-13}	X4	X4	replay `x4.overlap`	bound
blind prediction of $1000 < \Lambda \leq 2000$	4.0×10^{-10}	X4	X4	replay `x4.blind`	exact
frozen intercept, basis L2, lower cutoff 200	45.4249416926	X4	X4	replay `x4.intercept`	exact
refit intercept, basis L2, lower cutoff 200	45.4249235266	X4	X4	replay `x4.refit`	as stated
C_2, same rule on $\Lambda \leq 118$	0.657131272(35)	X4	X4	replay `x4.c23solver`	exact
C_3, same rule on $\Lambda \leq 118$	$-3.71984(8)$	X4	X4	replay `x4.c23solver`	exact
C_5 from the exact sums	$-731.13(10)$	X5	X5	replay `x5.c5`	exact
its δ	2×10^{-5}	X5	X5	replay `x5.c5`	exact
largest cutoff of the exact C_5 sums	500	X5	X5	replay `x5.records`	exact
$C_5^{\mathrm{conn}}(400)$	-506.619063805112	X5	X5	replay `x5.values`	exact
$C_5^{\mathrm{conn}}(500)$	-519.770442010804	X5	X5	replay `x5.values`	exact
$\Delta C_5(500)$	0.717651334579	X5	X5	replay `x5.values`	exact
overlap of consecutive runs	6×10^{-12}	X5	X5	replay `x5.overlap`	bound
blind prediction, fits frozen at $\Lambda \leq 400$	3×10^{-9}	X5	X5	replay `x5.blind`	exact
blind prediction, fits frozen at $\Lambda \leq 320$	1.5×10^{-8}	X5	X5	replay `x5.blind`	exact
free intercept, basis L2	-730.8460638788	X5	X5	replay `x5.intercept`	exact
free intercept, basis M3	-731.0822822059	X5	X5	replay `x5.intercept`	exact
fitted $\log^3$ coefficient, intercept fixed to C_5	$1.945 - -1.994$	X5	X5	replay `x5.tail`	as stated
c_5 predicted from the C_4 tail	1.9965	X5	X5	replay `x5.constrained`	exact
constrained intercepts	-731.1467, -731.1397, -731.1359	X5	X5	replay `x5.constrained`	exact
separately written constrained fit	$-731.13(10)$	X5	X5	replay `x5.crosscheck`	as stated

result	value	pkg	records	command	tolerance
top-log relation at ϵ^3	5×10^{-6}	X5	X5	replay `x5.toplog`	exact
top-log relation at ϵ^4	0.3%	X5	X5	replay `x5.toplog`	exact
integrity of code and records	pass	F	F	replay `fb.manifest`	exact
leading power of the second channel	$\Lambda^{-1/2}$ at $\epsilon = 0$	F	F	replay `fb.power`	exact
first order of the second channel	from ϵ^4	F	F	replay `fb.onset`	as stated
prefactor of the second channel (not quoted)	$2/(3\sqrt{\pi})$	F	F	replay `fb.amplitude`	as stated
log coefficient of the ϵ^4 phase response (not quoted)	0.4913887735	F	F	replay `fb.B`	as stated
finite-annulus test of the kernel	36 comparisons pass	F	F	replay `fb.annulus`	as stated
what the paper uses from this package	structure only	F	F	replay `fb.caveat`	info
status of the second-channel coefficient	plausible only	F	F	replay `fb.unverified`	info
$d = 2$, perturbative (P = `code/lri_d2_coeff_hpc`, I = `code/lri_2d_coeff_independent_check`; records R = `data/HT_2d/prod`; inside I: RS = I`/rs_exact/results`, CY = I`/cauchy/results`, CT = I`/continuum/exact_C2_C3.csv`, C4 = I`/c4_exact/results/c4_L2000.npz`, WN = I`/residual_log/results/{offdiag_L20000,nm_L8000}.npz`, C5 = I`/residual_log/large_L/results/c5_L300.npz` with I`/residual_log/results/eps5/dC5_L1000_all.npz`, TG = I`/residual_log/large_L/results/tag3_L16000.npz`, S = I`/residual_log/results/legw/legw_L1000.npz`, CE = I`/residual_log/ce_wc/out_*`, CX = I`/complete_exact/results`, TF = I`/residual_log/results/tag`, WD = I`/c4_exact/results/windows_L20000.npz`, CW = I`/residual_log/C4W_by_L.csv`, BP = I`/c4_exact/fits`, RG = I`/residual_log/large_L/results/regression_opt.txt`; SRC = P`/src/lri_coeff`, REF = P`/reference/lri_d2_perturbative_C8/results`)					
record integrity	41 records, self-hashed	P	R	replay `p2.prod.records`	exact
campaign manifests	3, self-hashed	P	R	replay `p2.prod.campaigns`	exact
solver source digest in every record	550e9266…	P	R, SRC	replay `p2.prod.source`	exact
coefficient tables = records	20 rows	P	R	replay `p2.prod.tables`	exact
tables cited by HT	sha256 d86743c1…, ee72158e…	P	R, REF	replay `p2.prod.cited`	exact
production cutoff	$L \le 34$	P	R	replay `p2.prod.lprod`	exact
order	C_0–C_8	P	R	replay `p2.prod.degree`	exact
validation receipts (symmetry, Expanse)	passed	P	R	replay `p2.prod.validation`	receipts
exact RS vs package	4×10^{-13}	I	RS, R	replay `p2.impl_agree`	1 digit
exact RS cutoffs	$L \le 18$	I	RS	replay `p2.rs_lmax`	exact
Cauchy vs exact RS	5×10^{-46}	I	CY, RS	replay `p2.cauchy_vs_rs`	1 digit

result	value	pkg	records	command	tolerance
package vs Cauchy	1.6×10^{-14}	I	CY, R	replay `p2.pkg_vs_cauchy`	2 digits
Cauchy cutoffs	$L \le 8$	I	CY	replay `p2.cauchy_lmax`	exact
exact C_2, C_3 code vs package	4×10^{-14}	I	CT, R	replay `p2.low_agree`	1 digit
large-L C_4 code vs exact RS, package	1.1×10^{-13}	I	C4, RS, R	replay `p2.c4_agree`	bound + record
large-L C_5 code vs package	2.2×10^{-13}	I	C5, R	replay `p2.c5_agree`	bound + record
optimized engine, per task	7.3×10^{-16}	I	RG	replay `p2.c5_opt_regress`	record
$L = 300$ assembly vs $L \le 200$ data	7.6×10^{-13}	I	C5	replay `p2.c5_assembly`	2 digits
$\delta^{(3)}$ chain sums vs rationals	4×10^{-16}	I	TG, TF	replay `p2.delta_rational`	1 digit
$C_2(\infty)$	0.30806541358, 35 digits	I	closed form, CT	replay `p2.c2_digits`	≥ 35 digits
closed-form $C_2(L)$ = HT $C_2(L)$	$\le 10^{-13}$	I	CT	replay `p2.c2_closed_form`	bound
$C_3(\infty)$	-0.8748500(1)	I	CT	replay `p2.c3_limit`	10^{-7}
C_3 deviation	1×10^{-8}	I	CT	replay `p2.c3_dev`	1 digit
$C_3(L)$ cutoffs	$L \le 600$	I	CT	replay `p2.c3_lmax`	exact
C_3 tail, $\ln L/L$ coefficient	0.70734	I	CT	replay `p2.c3_tail`	5 decimals
r_2 slope ($r_2 \simeq 2L/(27\pi)$, Sec. 4.4)	0.0235785	I	C4	replay `p2.r2_slope`	7 decimals
C_2 to C_4	$\delta < 10^{-7}$	I	CT, C4, WN	replay `p2.low_dev`	bound
$C_4(L)$ cutoffs	$L \le 2000$	I	C4	replay `p2.c4_lfour`	exact
κ (slope of consistent − raw)	2.2073×10^{-4}	I	WD	replay `p2.c4_kappa`	4 decimals
$\ln L$ coefficient, consistent	-0.026563(42) (-0.027)	I	C4	replay `p2.c4_log_consistent`	6 decimals
$\ln L$ coefficient, zero-mode	-0.010480(42)	I	C4	replay `p2.c4_log_zm`	6 decimals
$\ln L$ coefficient, raw	-0.026381	I	C4	replay `p2.c4_log_raw`	6 decimals
scheme limits A	5.0026(5), 5.0213(5), 5.0021	I	C4	replay `p2.c4_A_schemes`	4 decimals
blind test, $L = 1200$ to 2000	8.3×10^{-8}, 4.1×10^{-7}	I	C4, BP	replay `p2.c4_blind`	2 digits
C_4, complete window	5.0488682(10)	I	C4, WN	replay `p2.c4W_limit`	7 decimals
C_4 deviation	5×10^{-8}	I	C4, WN	replay `p2.c4_dev`	1 digit

result	value	pkg	records	command	tolerance
C_4^W fit spread	6×10^{-7}	I	C4, WN	replay `p2.c4W_spread`	1 digit
C_4^W free $\ln L$	$\leq 3.5 \times 10^{-6}$	I	C4, WN	replay `p2.c4W_nolog`	2 digits
C_4^W table	= series	I	CW	replay `p2.c4W_table`	10^{-13}
zero-mode + off-diagonal window	5.0386139	I	C4, WN	replay `p2.c4_zmoff`	7 decimals
χ-line mass window	0.0102543	I	WN	replay `p2.c4_chi`	7 decimals
$C_5(L)$ cutoffs	$L \leq 300$	I	C5	replay `p2.c5_lfive`	exact
C_5, W_c + external-line window	-38.835(13) (median -38.83544)	I	C5, TG, S	replay `p2.c5_limit`	3 decimals
C_5 deviation	5×10^{-5}	I	C5, TG, S	replay `p2.c5_dev`	1 digit
residual $\ln L$: shift for $\lvert b\rvert = 0.01$	0.1	I	C5, TG, S	replay `p2.c5_logshift`	1 digit
hold-out test, fit $\leq 200 \to 300$	4.3×10^{-6} vs 9.4×10^{-6}	I	C5, TG, S	replay `p2.c5_blind`	2 digits
free $\ln L$ coefficient by $L_{\max}$	-0.011 $\to$ +0.004	I	C5, TG, S	replay `p2.c5_bdrift`	3 decimals
free $\ln L$, single fits	-0.05 to +0.07	I	C5, TG, S	replay `p2.c5_bspread`	2 decimals
null test on C_4^W	+0.0005	I	C5	replay `p2.c5_null`	1 digit
estimator calibration	bias ≈ -0.004	I	C5, CW	replay `p2.c5_calibration`	4 decimals
tail power $k_1 = 4$ (15 fits)	shift +0.002, half range 0.050	I	C5, TG, S	replay `p2.c5_k1`	3 decimals
bias-corrected median b	+0.008 (15 fits)	I	C5, CW	replay `p2.c5_calibration15`	4 decimals
free $\ln L$, 12-fit medians	$-0.0134 \to +0.0054$	I	C5, TG, S	replay `p2.c5_bdrift_agent`	4 decimals
$C_5^{W_c}(\infty)$	-38.830(13) (median -38.8297)	I	C5	replay `p2.c5_wc`	3 decimals
$C_5^{W_{cs}} - C_5^{W_c}$ limits	≈ -0.005772	I	C5, TG, S	replay `p2.c5_wcs_minus_wc`	10^{-3}
C_5^W linear term	$-0.0019\,L$	I	C5	replay `p2.c5_W_linear`	6 decimals
$\delta^{(3)}(L)$ cutoffs	$L \leq 16000$	I	TG	replay `p2.delta_lmax`	exact
$\delta^{(3)}$ minimum	-0.0159046 at $L = 34$	I	TG	replay `p2.delta_min`	7 decimals
$\delta^{(3)} \to -4/(81\pi)$	within 7×10^{-11}	I	TG	replay `p2.delta_limit`	bound
$\delta^{(3)}$, $\ln L/L$ term	$-1/(54\pi)$	I	TG	replay `p2.delta_lnL`	5×10^{-7}
$S(\infty)$	0.367187	I	S	replay `p2.S_limit`	6 decimals
$\delta_\infty S_\infty$	-0.005772	I	S	replay `p2.delta_S`	6 decimals

result	value	pkg	records	command	tolerance
window shifts to	$L = 1000$	I	S, C5	replay `p2.shift_lmax`	exact
R_2, window W	restored, $R_2 \approx 0.0114/L$	I	CX	replay `p2.ident2_W`	4 decimals
R_2, consistent window	0.0243	I	CX	replay `p2.ident2_cons`	4 decimals
R_3, single quantum	$\geq 95\%$ at $L = 14$, 16	I	CE	replay `p2.ident3_chi`	≥ 0.95
$[\epsilon^3]$ residual in W_c	$0.063\, a\, w_{aa}$	I	CE	replay `p2.ident3_wc`	10^{-3}
$R_3(\chi_{11})$ at $L = 16$	-1.6×10^{-4}; W_c: 0.0156	I	CE	replay `p2.ident3_L16`	2 digits
R_3, ϕ^2 descendants	-0.057, -0.059: not settled	I	CE	replay `p2.ident3_phi2`	3 decimals
$R_4(\chi_{11})$ local slope	0.028 per unit of $\ln L$	I	CE	replay `p2.eps4_R4slope`	3 decimals
ϵ^4 χ-line window	gaps 95%, 90%	I	CE, TF	replay `p2.eps4_chiwindow`	per cent
C_6 shift identity (App. C.5)	$\delta^{(3)}S_1 = +0.0018$ to $+0.0025$	I	CE, S	replay `p2.c6_identity`	10^{-12}
$C_6(16)$ (not reproduced)	165.4 (+7.5 per step)	I	CE	replay `p2.c6_L16`	range
$d = 3$, *perturbative* (E = `code/lri_3d_c4_exact`, results E`/results`; P = `code/lri_3d_hpc`, records R = `data/HT_3d/pert3d`, table T = `results/HT_3d/SUMMARY.md`; ref = P`/results/reference_coefficients_L14.json`)					
$C_2^{(3)}$	0.181221785405554 (15 digits)	E	E`/results/windows_L8000.npz`	replay `C2-3D`	15 digits
$C_3^{(3)}$	-0.353581479(20), $\delta = 1 \times 10^{-7}$	E	E`/results/ c4_L600.npz`	replay `C3-3D`	PDG; central to 1/10 of the uncertainty
$C_4^{(3)}$, complete window	1.228070(21), $\delta = 5 \times 10^{-6}$	E	E`/results/` (c4, offdiag, dchi, windows)	replay `C4-3D`	PDG
blind fit, no $\sqrt{L}$ term	1.6×10^{-8}; $\lvert b\rvert < 7 \times 10^{-8}$	E	same	replay `C4-BLIND`, `C4-NODIV`	10^{-7}
$C_5^{(3)}$ (provisional)	-6.0 ± 0.3	E	E`/results/c5raw_all_L40.npz`, `dC5_L600.npz`	replay `C5-3D`, `C5-LHALF`	1 decimal
cutoffs	600, 40, 8000	E	E`/results/`	replay `CUTOFFS`	exact
fit summaries	regenerated	E	E`/results/`	replay `FITS-REGEN`, `FITS-QUOTED`	quoted lines
exact sums vs production, $L \leq 20$	3×10^{-13}	E, P	ref, R`/results/`	replay `VAL-PKG-C4`, `VAL-PKG-C5`	10^{-12}
window schemes vs brute force, $L \leq 14$	4.3×10^{-13}	E, P	E`/results/brute/`	replay `VAL-PKG-C5`	10^{-12}
sums recomputed at small L	11 stages	E, P	E`/results/`	test (all stages)	10^{-12} to 10^{-9}

result	value	pkg	records	command	tolerance
production $C_n(L)$, $L = 16$, 18, 20 (Table 7, T)	6 rows	P	R/`results/L*/`	replay `TABLE`	stored strings
$C_4(20)/C_4^{(3)}$, read by `HT/`	60.26%	P	T	replay `C4-RATIO`	2 decimals
original implementation, $L = 14$	1.25×10^{-14}	P	R/`gates/`, ref	replay `GATE-L14`	5×10^{-13}
stored Hamiltonians, recursion	9.8×10^{-15}, 5.1×10^{-19}	P	R/`results/`	replay `CROSS-STORE`, `RESIDUAL`	5×10^{-13}, 10^{-16}
records and code identity	194 files	P	`data/HT_3d/MANIFEST.sha256`	replay `REC-MANIFEST`, `REC-SRC`, `REC-CHAIN`	sha256

A Two dimensions: operators, exact fourth order and the complete window

Appendices A to C give the derivations and the supporting data behind Section 4. This appendix covers the operators and schemes at low order, the solver checks, the exact $C_4(L)$, the cause of its $\ln L$ divergence class by class, and the complete window at ϵ^4. Appendix B treats order ϵ^5: the window at all orders, the double counting that rule W_c removes, and the tail of C_5. Appendix C treats the descendant spacings and the external-line windows W_{cs} and $W_{cs4\chi}$, and ends with the status of each window component. Paths are relative to the package `code/lri_2d_coeff_independent_check/` (package I of Section 8), and each subsection names its records. Numbers that Section 4 also quotes have reproduction rows in Section 8. The other numbers of these appendices are read from the records named in each subsection, or evaluated from them where the text says so; the reproduction scripts do not recompute them. "Exact" data are rationals, or float64 evaluations of exact finite sums with a measured rounding error of at most 1.4×10^{-11}. Limits $L \to \infty$ are fits; their uncertainties are spreads over tail models and fit windows, not standard deviations.

A.1 Operators and schemes at low order

Records: `SPEC.md`, `rs_exact/README.md`, `zeromode_exact/README.md`, `cauchy/README.md`.

Factorized vertices. With $Z = a_0 + a_0^\dagger$, $a_0 = A_{00}$, and $\chi = \phi - Z$, normal ordering factorizes:

$$V = \sum_{m=0}^{4} \frac{1}{12}\binom{4}{m} :Z^{4-m}: \otimes \langle :\chi^m: \rangle_\theta, \qquad M = \sum_{m=0}^{2} \binom{2}{m} :Z^{2-m}: \otimes \langle :\chi^m: \rangle_\theta. \tag{30}$$

At $\epsilon = 0$ the weights factorize holomorphically, $w_{ab} = v(a)v(b)$ with $v(a) = (\frac{1}{2})_a/a!$. The squared V-vertex coefficient times the Wick count of a bundle of m χ lines is $\gamma_m = m!\left[\binom{4}{m}/12\right]^2$: $\gamma_1 = \frac{1}{9}$, $\gamma_2 = \frac{1}{2}$, $\gamma_3 = \frac{2}{3}$, $\gamma_4 = \frac{1}{6}$.

Space and particle cap. The truncated space holds the occupation states with spin zero, even under $(a, b) \to (b, a)$, and $\ell \le L$; spin zero forces ℓ even, and the field parity $(-1)^P$ is conserved. Truncated operators are the projections of the full ones; nothing else is truncated. The level cutoff does not bound P, because zero-mode quanta cost no level. A path argument shows that

a contribution to the order-ϵ^n energy of $|p\rangle$ reaches at most $P = p + 4\lfloor n/2 \rfloor$, which gives the cap $P \le p + 16$ through ϵ^8. It was also tested: caps $p + 18$ and $p + 20$ reproduce the $p + 16$ rationals identically, while $p + 14$ changes C_8 and nothing below. The Cauchy code confirms it with absolute caps: $P \le 21$ and $P \le 23$ reproduce $P \le 19$, and $P \le 17$, 18 change only C_8.

References and the projection Π. Each reference $|p\rangle$ is non-degenerate in its sector, so $E_p(\epsilon)$ is unique. Because $w_{00} = 1$ and every window has vanishing diagonal elements on level-0 states, $\langle p|M|p\rangle = 2p$ and $\langle p|V|p\rangle = \frac{1}{2}p(p-1)$ at every ϵ. With $\mathcal{E}_p^{(n)}$ the order-n energy without the r_n, g_n terms, Eq. (7) reads $C_n = \Pi[\mathcal{E}^{(n)}]$ with

$$\Pi f = f(2) - f(1) - \tfrac{1}{3}\big[f(3) - f(0)\big] = -\tfrac{1}{3}\Delta^3 f(0), \qquad \Pi[p^3] = -2, \qquad \Pi[p^4] = -12, \tag{31}$$

and Π annihilates every polynomial of degree at most two in p. Only the part of an energy shift that is at least cubic in p reaches C_n.

Where the schemes differ. The diagonal windows (consistent, literal, zero-mode) are of order ϵ^2 and vanish on $|p\rangle$, so they first enter through $\langle p|H_1 G_p D_2 G_p H_1|p\rangle$ at ϵ^4, with $H_1 = F_1 + g_1 V$ and the resolvent $G_p = Q_p/(\frac{p}{2} - H_0)$, Q_p the projection off $|p\rangle$. Raw and consistent therefore differ first at C_4. Raw and consistent vacua agree through ϵ^3, so the consistent and literal windows differ first at C_6; the exact data confirm both statements for every $L > 0$. At ϵ^4 only the ϵ^2 parts of the windows matter:

$$D_2^{\mathrm{cons}}(\ell) = E_v^{(2)}(L) - E_v^{(2)}(L-\ell), \qquad E_v^{(2)}(\lambda) = g_1^2 \langle 0|V G_0 V|0\rangle_\lambda = -\frac{(\lambda+2)(\lambda+4)}{864}, \tag{32}$$

where $\langle\cdot\rangle_\lambda$ restricts the intermediate states to level $\le \lambda$, and the zero-mode window is $D_2^{\mathrm{zm}}(z,\ell) = h_{zz}^{(2)}(L) - h_{zz}^{(2)}(L-\ell)$, the diagonal of Eq. (20). The r_2 and g_2 terms of the sub-cutoff energies cancel in these differences, and for $z = 0$ the two windows coincide.

Degenerate partners. For $z \ge 5$ the zero-mode channel $|z\rangle$ has degenerate partners with level $\ell = 2, 4, \cdots \le \frac{z-1}{2}$ and $P = z - 2\ell$. First order lifts the degeneracy (smallest separation $\frac{7}{3}$, at $z = 5$, $\ell = 2$), so the branch through $|z\rangle$ is unique; the references are never degenerate. The window objects project the partners out (Appendix B.1).

A.2 Solver checks in detail

Records: `compare_with_package.txt`, `compare_three_way.txt`, `rs_exact/README.md`, `cauchy/README.md`, the production records `../../data/HT_2d/prod/`, `continuum/exact_C2_C3.csv`, `continuum/asymptotics_C2_C3.txt`; code `continuum/exact_low_orders.py`.

The three computations. Section 4.2 and the independent-checks paragraph of Section 6 give their scope and agreement. The production solver propagates Taylor jets of the wave functions in float64, with scalar reductions in extended long double; its consistent run at $L = 34$ took 2893 s on 64 threads with a peak memory of 463 GiB. The exact solver works in a similarity frame in which every matrix element is rational; its modular code uses 72 primes with Chinese remaindering, and every rational reconstruction is verified on eight held-out primes. The Cauchy solver evaluates Δ_ε at $\epsilon_k = \rho\, e^{2\pi \mathrm{i} k/K}$ with $\rho = \frac{1}{64}$, $K = 128$ and 256-bit arithmetic, and $C_n = K^{-1} \sum_k \Delta_\varepsilon(\epsilon_k)\epsilon_k^{-n}$. It solves the consistent vacua exactly. On states with $\ell \ge 2$ the window is $E_v(\lambda) - E_v(\lambda - \ell)$, so the self-consistent condition $(H_\lambda + D_\lambda)\psi = E_v(\lambda)\psi$ becomes the linear generalized eigenproblem $A_\lambda \psi = E_v(\lambda)\, \mathcal{P}_0 \psi$, with $A_\lambda = H_\lambda - E_v(\lambda - \hat{\ell})(1 - \mathcal{P}_0)$, $\mathcal{P}_0$ the projection on level-0 states, and the sub-cutoff vacua $E_v(\lambda - \ell)$ known from smaller λ.

Internal checks of the exact codes.

- $C_0 = 1$, $C_1 = -\frac{1}{6}$, $r_1 = 0$, $g_1 = \frac{1}{3}$ for every L and scheme, and the tuning conditions hold exactly at every order. The fraction and modular codes agree identically for every number at $L \le 12$.
- A third, Hermitian construction of H was diagonalized at 110 digits at $\epsilon = 10^{-3}$, 10^{-5}, 10^{-8}, with the exact series for r, g and the vacua inserted ($L \le 6$, every scheme and channel). The residual divided by ϵ^9 tends to a constant, which confirms every coefficient through ϵ^8 to about 10^{-8}.
- The Cauchy results do not change with the circle radius, the number of points, the precision, the continuation path or the particle cap; the quoted errors (about 10^{-45} on C_8) are set by the most aliased variant. All 882 rational identifications have denominators of the form $2^a 3^b 5^c 7^d$.

C_2 and C_3. The closed form (19) gives $C_2(0) = \frac{17}{81}$, $C_2(2) = \frac{19}{72}$, $C_2(4) = \frac{179}{640}$ and $C_2(6) = \frac{833719}{2903040}$, equal to the exact rationals (evaluated against `rs_exact/results/raw_L{0,2,4,6}.json`). Its approach to the limit is $C_2(L) - C_2 = -\frac{40}{81\pi} L^{-1} + 0.235785\, L^{-2} + \dots$. An independent diagrammatic code, which contracts the factorized vertices (30) with the level cutoff on every intermediate state, gives $C_3(L)$ in float64 to $L = 600$, where $C_3(600) = -0.865794525554$. Its $\ln L/L$ coefficient is 0.70734, which is $4.4999 \times \frac{40}{81\pi}$; the factor $\frac{9}{2}$ is identified numerically from this fit. It is not the one-loop ϕ^6 value $\omega_1 = \frac{7}{2}$ of Eq. (10), and which operator sets the $d = 2$ tail is not derived. With the tower 1, $\ln L/L$, $1/L$, $\ln^k L/L^2$ ($k \le 2$), $\ln L/L^3$, $1/L^3$ the limit is -0.8748500372, -0.8748500106 and -0.8748500065 for $L_{\min} = 32$, 64 and 128, which gives $C_3 = -0.8748500(1)$.

A.3 Exact $C_4(L)$ to $L = 2000$

Records: `c4_exact/README.md`, `c4_exact/C4_by_L.csv`, `c4_exact/results/{c4_L1000,c4_L2000,windows_L20000}.npz`, `c4_exact/fits/fit_summary.txt`, `c4_exact/fits/blind_predictions_L1200_2000.json`, `c4_exact/validate_{rs,pkg,corrected}.txt`, `zeromode_exact/README.md`, `continuum/{fits_summary,zm_corrected_fits}.txt`; code `c4_exact/{c4lib,c4_compute,c4_run,windows,fitlib,fits_c4}.py`.

At $L \le 30$ the production data could not separate a slow power tail from a logarithm. Fits of the consistent C_4 with towers that contain $\ln L/L$ landed 2.4 to 5% below the continuum value, and the same fits with the zero-mode window added gave 4.90 to 5.04 (`continuum/zm_corrected_fits.txt`). The exact $C_4(L)$ settles the question.

Fourth-order perturbation theory. Write $H = H_0 + \sum_{k\ge1} \epsilon^k H_k$ with

$$H_1 = F_1 + g_1 V, \qquad H_2 = r_2 M + g_2 V + g_1 \dot{V} + D_2, \qquad H_3 = r_3 M + g_3 V + g_2 \dot{V} + \tfrac{1}{2} g_1 \ddot{V} + r_2 \dot{M} + D_3, \tag{33}$$

where $\dot{V} = \partial_\epsilon V$ and $\dot{M} = \partial_\epsilon M$ act through $w_{ab}(\epsilon)$. With $G = G_p$, $\langle\cdot\rangle = \langle p|\cdot|p\rangle$ and $E^{(1)} = \langle H_1\rangle$,

$$\begin{aligned} E^{(4)} = {} & \langle H_4\rangle + 2\langle H_1 G H_3\rangle + \langle H_2 G H_2\rangle + 2\langle H_1 G (H_1 - E^{(1)}) G H_2\rangle + \langle H_1 G (H_2 - \langle H_2\rangle) G H_1\rangle \\ & + \langle H_1 G (H_1 - E^{(1)}) G (H_1 - E^{(1)}) G H_1\rangle - \langle H_1 G H_1\rangle \langle H_1 G^2 H_1\rangle . \end{aligned} \tag{34}$$

A diagonal window enters C_4 only through $g_1^2 \langle V G D_2 G V\rangle$; the mass window D^M enters through more terms (Appendix A.5). The order-3 couplings r_3, g_3 need the complete $\mathcal{E}_p^{(3)}$ for $p = 0, 1, 3$, not only its part that survives Π.

Diagram evaluation. With Eq. (30), a K-vertex chain is a sum over χ-line patterns n_{ij} (lines between vertices $i<j$) of a Wick count, the vertex coefficients, an exact rational zero-mode path sum and a χ level sum. A time slice s with n_s χ lines and z_s zero-mode quanta has the energy denominator $\frac{1}{2}(z_s-p)+\frac{1}{2}n_s+$(level of the slice). The χ sums use the holomorphic factorization: with $v_k(y)=(1-y)^{-k/2}$, $\sum_{ab}w_{ab}y^a\bar{y}^b=v_1(y)v_1(\bar{y})$, and a bundle of n χ lines has the generating function $[v_1(y)v_1(\bar{y})-1]^n=\sum_k\binom{n}{k}(-1)^{n-k}v_k(y)v_k(\bar{y})$. Angular averaging at every vertex forces spin zero in every slice, so the holomorphic and antiholomorphic levels of a slice are equal, j_s, and its level is $2j_s$. The density of a pattern is then a sum of squares of one-variable coefficients, with no angular-momentum sum left. ϵ-derivative vertices differentiate the line weights.

One run for all cutoffs. The densities do not depend on L; the cutoff only imposes $\max_s 2j_s\le L$. Accumulating by $\max_s j_s$ and summing cumulatively gives every even $L\le L_{\max}$ in one run. The window terms depend on L explicitly and are evaluated separately.

Pruning and symmetry. The zero-mode amplitude of a four-V chain with n χ lines is a polynomial of degree $8-n$ in p. Since Π kills degree ≤ 2, only $n\le 5$ reaches C_4: 384 of the 641 four-vertex patterns. Time reversal maps them onto each other in 168 pairs with identical contributions, so 216 patterns are computed. The full set reproduces every individual $E_p^{(4)}$ of the exact solver.

Window shifts without cancellation. Evaluated naively, the window $h_{zz}^{(2)}(L)-h_{zz}^{(2)}(L-\ell)$ is built from terms of size $h_{zz}^{(2)}\sim L^2$, while the part that survives Π is of order $1/L$: a relative cancellation of about L^3. The script `windows.py` applies Π to the rational zero-mode polynomials first and only then multiplies by single-signed level sums. It runs to $L=20000$ and agrees with the full computation to 6×10^{-13} for $L\le1000$.

Cost and precision. On one 40-core node the run to $L_{\max}=600$ takes 51 s, to 1000 four minutes and to 2000 about 77 minutes (two Slurm jobs on the symmetry cluster, 59 minutes to the time limit and 17 minutes from the checkpoint; the package README's 80 to 90 minutes is the estimate made during the run). Everything is float64. Overlapping runs agree to 6.7×10^{-12} (600 against 1000, from the README) and 1.4×10^{-11} (1000 against 2000, evaluated from `c4_L1000.npz` and `c4_L2000.npz`), and the cancellation in the assembly is about 4×10^3 at $L=1000$.

Validation. Table 10 lists the checks. The zero-mode scheme itself was solved exactly through ϵ^8 for $L=0$ to 18 (`zeromode_exact/`), by exact perturbation theory with the degenerate partners treated by a projected equation. Two implementations (modular and fraction) are identical for $L\le10$. A non-perturbative Cauchy check covers every coefficient and every sub-cutoff energy $E_z^{(k)}(\lambda)$ at $L\le6$, and closure tests show that this scheme needs only the zero-mode channels $z\le14$ through ϵ^8. The zero-mode C_4 minus the consistent C_4 equals an exact closed form in L (`zeromode_exact/closed_form_dC4.py`) at every L; for example $C_4^{\rm zm}(2)=\frac{6786263}{3732480}$.

Exact window shifts. Fits of the cancellation-free shifts to $L=20000$ (residuals $\le5\times10^{-14}$) give

$$\Delta C_4^{\rm cons}(L)\equiv C_4^{\rm cons}-C_4^{\rm raw}=\kappa L-\frac{5}{8748\pi}\ln L+0.000446687+o(1),$$
$$\Delta C_4^{\rm zm}(L)\equiv C_4^{\rm zm}-C_4^{\rm cons}=\Big(\frac{4}{81\pi}+\frac{5}{4374\pi}\Big)\ln L+0.0187584+o(1), \tag{35}$$

check	range	worst deviation
r_2, g_2, r_3, g_3, C_2, C_3 against the exact rationals	$L = 0$ to 18	3.4×10^{-15} relative (C_3: 1.3×10^{-14})
raw and consistent C_4 against the exact rationals	$L = 0$ to 18	1.1×10^{-13} relative
every $E_p^{(4)} - 2pr_4 - \frac{1}{2}p(p-1)g_4$, $p = 0$ to 3, raw, all 641 patterns	$L = 0$ to 18	2.1×10^{-14} relative
$E_v^{(2)}(\lambda)$ of Eq. (32)	$\lambda \le 18$	2.2×10^{-16}; exact for $\lambda \le 12$
C_2, C_3, raw and consistent C_4, r_3, g_3 against the production solver	$L = 2$ to 34	1.1×10^{-13} relative
zero-mode C_4 against brute-force fraction arithmetic	$L = 0$ to 12	1.1×10^{-13} relative
zero-mode C_4 against the exact zero-mode solver	$L = 0$ to 18	1.1×10^{-13} relative
pruned ($n \le 5$) against the full pattern set	$L \le 18$	10^{-13}
time-reversal pairs	168 pairs, $L = 40$	5.5×10^{-16}

Table 10: Validation of the large-L C_4 code (`c4_exact/README.md`, `c4_exact/validate_*.txt`).

with $\kappa = 2.2073 \times 10^{-4}$; the analytic evaluation in `continuum/asymptotics_C2_C3.txt` gives $\kappa = 2.20729299645 \times 10^{-4}$. The $\ln L$ coefficients are those of Table 12. Split by the number of outer χ lines, $\Delta C_4^{\rm zm}$ has logarithms only for two lines ($\frac{4}{81\pi}$) and four lines (3.6387×10^{-4}). Evaluated from `C4_by_L.csv`, it has a minimum of 0.0982 near $L = 36$ and then grows logarithmically: 0.1028 at $L = 100$, 0.1313 at 1000 and 0.1418 at 2000.

Fits. The tail models (`fitlib.py`) are towers T_{jk}: a constant, $\ln^i L/L$ for $i \le j$ and $\ln^i L/L^2$ for $i \le k$; T_{jkm} adds $\ln^i L/L^3$ for $i \le m$. A model may add a free $\ln L$ and single terms such as $\ln^3 L/L$. The table of Section 4.4 uses T_{22} and $T_{22} + \ln^3 L/L$, each with a free $\ln L$, the $\ln^2 L/L$ coefficient free or fixed to $-5/\pi$ (end of Appendix A.5), and $L_{\rm min} \in \{200, 300, 400\}$, $L \le 2000$: twelve fits, with the median and half the full spread quoted. At $L = 2000$ the data are $C_4 = 4.29126$ (raw), 4.73178 (consistent) and 4.87359 (zero-mode). The raw fit uses $C_4^{\rm raw} = C_4^{\rm cons} - \Delta C_4^{\rm cons}$ with the exact shift. The difference of the two fitted β, 0.016082, equals the $\ln L$ coefficient of $\Delta C_4^{\rm zm}$. Subtracting the fitted $1/L$ and $1/L^2$ terms from $C_4(L)$ leaves straight lines in $\ln L$: $5.0026 - 0.02656 \ln L$ (consistent) and $5.0213 - 0.01048 \ln L$ (zero-mode). Without a $\ln L$ term every tower drifts with the fit window, and the limits, 4.881 to 4.896 (zero-mode) and 4.648 to 4.684 (consistent), are far from 5.0489 and from each other. The fitted β of the consistent scheme alone would not exclude the nearby simple value $-1/(12\pi) = -0.0265258$; the class decomposition of Appendix A.4 fixes it at $-\frac{365}{4374\pi}$.

Out-of-sample and blind tests. Fitted on $[200, 600]$ and used to predict $(600, 2000]$, the zero-mode model T_{22} with $\ln L$ misses by at most 9.0×10^{-8}, and models without $\ln L$ miss by 5.0×10^{-6} to 2.5×10^{-5}. Fitted on $[200, 1000]$, the models with $\ln L$ miss by 8.5×10^{-9} (T_{22}) and 5.5×10^{-8} (T_{232}), those without by 4.3×10^{-7} to 6.7×10^{-6}. Before the $L = 2000$ run finished, predictions for $L = 1200$ to 2000 were recorded from fits on $L \le 1000$. At $L = 2000$, prediction minus data is -8.6×10^{-9} (T_{22}) and -6.4×10^{-8} (T_{232}) for the models with $\ln L$ in the zero-mode and consistent schemes (-7.5×10^{-9} and -8.3×10^{-8} raw). For the models without $\ln L$ it is $+4.3 \times 10^{-7}$ to $+1.8 \times 10^{-6}$ (zero-mode), $+1.2 \times 10^{-6}$ to $+4.6 \times 10^{-6}$ (consistent) and -4.1×10^{-7} to -8.6×10^{-7} (raw). The data are accurate to about 10^{-11}, so the blind test favors the logarithm, by factors of 5 to 500 (the ratios of the deviations above).

A.4 Mechanism and the $\ln L$ coefficients by class

Records: `residual_log/results/log_coeffs.txt`; code `residual_log/{log_coeffs,offdiag_window, decompose}.py`.

The budget defect. The cutoff constrains the total level of each intermediate state. View a fourth-order process as an outer structure with a sub-bubble nested in one of its intermediate states. The sub-bubble may use only the level $L - \ell_{\text{other}}$, where ℓ_{other} is the level of everything else in that time slice, while the counterterm that cancels its divergence was tuned on level-0 references, where it has the full L. The mismatch is the density of the divergence at the cutoff times the level taken by the other quanta. Table 11 lists the three divergent sub-bubbles of the ϕ^4 vertex.

sub-bubble	divergence	deficit for spectators of level ℓ	restored by
vacuum (four χ lines)	$E_v^{(2)}(\lambda) = -\frac{(\lambda+2)(\lambda+4)}{864}$, quadratic	$\sim \ell L$	consistent window
sunset (three χ lines; mass)	$r_2(\lambda) = \frac{2\lambda}{27\pi} + \ldots$, linear	$\frac{2}{27\pi}\ell$ per mass insertion; does not decay	in part by the diagonal zero-mode window; fully by D^{zm} and D^M
coupling (two χ lines)	$g_2 \sim \ln \lambda$	$\ln \frac{L}{L-\ell} \sim \frac{\ell}{L}$	not needed for soft spectators; finite effects from hard ones

Table 11: Divergent sub-bubbles and their level-budget deficits in $d = 2$ ($\Delta_\sigma = \frac{1}{2}$ at $\epsilon = 0$).

The sunset slope is $dr_2/dL = \frac{2}{27\pi} = 0.0235785$ (the exact $[r_2(2000) - r_2(1998)]/2$ is 0.0235785). Because the divergence is linear, the deficit is proportional to the spectator level and independent of L, and it hits every insertion of the mass operator M, not only the diagonal of its zero-mode part.

Linked clusters with zero-mode transfer. Take two components that share only the zero mode: an outer bubble O with m χ lines and an inner bubble B with m'. Let E_O, E_B be their slice energies and $q = \frac{1}{2}\Delta z_B$ the zero-mode transfer, with Δz_B the change of z caused by B. The four vertices have six time orderings: two sequential ones ($OOBB$, $BBOO$), in which no slice holds lines of both components, and four overlapping ones ($OBBO$, $BOOB$, $OBOB$, $BOBO$), in which some slice holds both. Since $H_0 = \ell + \frac{P}{2}$ is additive, the six orderings and the unlinked terms of perturbation theory cancel in the continuum. At leading degree in p the zero-mode amplitude does not depend on the ordering, and the cancellation reduces to the identity

$$\frac{1}{E_O E_B (E_O + q)} + \frac{1}{E_O E_B (E_B - q)} - \frac{1}{q\, E_O (E_B - q)} + \frac{1}{q\, E_B (E_O + q)} = 0 \tag{36}$$

for every q, including the limit $q \to 0$; its first two terms come from the four overlapping orderings, the last two from the sequential orderings and the unlinked terms. With outer level $2a$ and inner level $2b$, the overlapping orderings need $a + b \le \frac{L}{2}$, and the others do not. The truncation therefore errs by

$$-\sum_{a,b \le L/2 < a+b} \Big[\frac{1}{E_O E_B (E_O + q)} + \frac{1}{E_O E_B (E_B - q)}\Big] \times (\text{zero-mode amplitude}). \tag{37}$$

This is the nested insertion, into every outer state of level ℓ, of the window part of the second-order zero-mode effective interaction,

$$D_\lambda(z \to z') = g_1^2 \sum_{k:\, \lambda - \ell < \ell_k \le \lambda} \frac{\langle z'|V|k\rangle\langle k|V|z\rangle}{E_z^0 - E_k^0} = h^{(2)}_{z'z}(\lambda) - h^{(2)}_{z'z}(\lambda - \ell), \tag{38}$$

in its ket-referenced (Bloch) form and including $z' \neq z$: the window $D^{\rm zm}$ of Eq. (20). The diagonal zero-mode scheme keeps only $q = 0$. At subleading degree in p the orderings differ by commutator terms. These are confined to the window region and are $\mathcal{O}(\ln L/L)$ for the numbers $(m, m') = (2, 2)$, $(1, 3)$ and $(3, 1)$ of outer and inner χ lines.

The ratio $\frac{2}{3}$. Take the outer two-χ bubble, with $:Z^2:$ at each outer vertex, and the inner zero-mode sunset, with $:Z:$ at each inner vertex. Per unit amplitude the window of the nested sunset at outer level $2a$ is $-\frac{2}{\pi}a$, and the outer weight (pattern density over the squared denominator) tends to $\frac{1}{4a^2}$. The window sum is therefore

$$\sum_{a\le L/2} \frac{1}{4a^2}\Big(-\frac{2a}{\pi}\Big) = -\frac{1}{2\pi}\ln L + \mathcal{O}(1). \tag{39}$$

At leading degree in p the zero-mode operators commute and act as $a_0 \to \sqrt{p}\,e^{\mathrm{i}\theta}$, $a_0^\dagger \to \sqrt{p}\,e^{-\mathrm{i}\theta}$, so $:Z^2: \to p\,(e^{2\mathrm{i}\theta} + 2 + e^{-2\mathrm{i}\theta})$ and $:Z: \to \sqrt{p}\,(2\cos\theta)$. The closed-path amplitude is the angular average $p^3\langle(2\cos\theta)^6\rangle_\theta = 20\,p^3$. Split it by the zero-mode transfer of the inner sunset:

- $\Delta z_B = 0$ (inner vertices a_0, $a_0^\dagger$ in either order, two ways) times the phase-free part of the outer factor $(e^{2\mathrm{i}\theta} + 2 + e^{-2\mathrm{i}\theta})^2$, which is $1 + 4 + 1 = 6$: in total 12;
- $\Delta z_B = +2$ (inner $a_0^\dagger a_0^\dagger$) times the $e^{2\mathrm{i}\theta}$ coefficient of the outer factor, $2 \cdot 2 = 4$, and likewise $\Delta z_B = -2$: in total 8.

So $20 = 12 + 8$. The coupling factor is $g_1^4\gamma_2\gamma_3 = \frac{1}{81}\cdot\frac{1}{2}\cdot\frac{2}{3} = \frac{1}{243}$. With $\Pi[p^3] = -2$ the window logarithm restored in C_4 is

$$(-2)\cdot 20\cdot\Big(-\frac{1}{2\pi}\Big)\cdot\frac{1}{243} = \frac{20}{243\pi} = \underbrace{\frac{12}{243\pi}}_{\Delta z=0\ (=\frac{4}{81\pi})} + \underbrace{\frac{8}{243\pi}}_{\Delta z=\pm 2}. \tag{40}$$

The part that changes the zero-mode number is $\frac{8}{12} = \frac{2}{3}$ of the diagonal part, and a diagonal window captures $\frac{12}{20} = \frac{3}{5}$ of this class. The split $20 = 12 + 8$ and the γ_m follow from the algebra above; the factor $\frac{1}{243}$ is also in `log_coeffs.py`.

The coefficients by class. Table 12 lists every class. The two other logarithmic classes follow the same pattern. For four χ lines times a one-line bubble, the outer weight tends to $\frac{1}{4}$, and when the outer state has level $L - 2j$ the inner one-line bubble keeps the budget $2j$; its window is then $-\frac{1}{2\pi}(\frac{1}{j} - \frac{2}{L})$, which sums to $-\frac{1}{8\pi}\ln L$; with $g_1^4\gamma_4\gamma_1 = \frac{1}{4374}$ and the amplitude $20p^3$ this gives $+\frac{5}{4374\pi}$. For one χ line times the vacuum, Eq. (32) gives $E_v(L) - E_v(L-2a) = -[2a(2L+6) - 4a^2]/864$: the term linear in L gives κL, and the $4a^2$ term with weight $1/(4\pi a^3)$ gives $(864\pi)^{-1}\ln L$, which the factor $-\frac{40}{81}$ of the one-line outer state (couplings, zero-mode amplitude and Π; `log_coeffs.py`) turns into $-\frac{40}{81\cdot 864\pi} = -\frac{5}{8748\pi}$ in C_4. The $\ln L$ coefficient of each scheme is minus the sum of the logarithms it misses:

$$\beta_{\rm raw} = -\Big(-\frac{5}{8748\pi} + \frac{20}{243\pi} + \frac{5}{4374\pi}\Big) = -\frac{725}{8748\pi}, \qquad \beta_{\rm cons} = -\Big(\frac{20}{243\pi} + \frac{5}{4374\pi}\Big) = -\frac{365}{4374\pi},$$
$$\beta_{\rm zm} = -\frac{8}{243\pi}, \qquad \beta_W = 0, \tag{41}$$

in agreement with the fitted values of Section 4.4 to 10^{-6} or better. The χ-attached class has no net logarithm, but its two parts do: $+0.0369938\ln L$ directly at ϵ^4 and $-0.0369938\ln L$ through

class (outer state × sub-bubble)	analytic	exact fit	raw	cons	zm	W
one χ line × vacuum bubble (also κL)	$-\frac{5}{8748\pi} = -1.81933 \times 10^{-4}$	-1.81933×10^{-4}	m	r	r	r
two χ lines × zero-mode sunset, $\Delta z = 0$	$\frac{4}{81\pi} = 0.0157190067$	0.015719006698	m	m	r	r
four χ lines × one-line zero-mode bubble	$\frac{5}{4374\pi} = 3.63866 \times 10^{-4}$	$3.63865887 \times 10^{-4}$	m	m	r	r
two χ lines × zero-mode sunset, $\Delta z = \pm 2$	$\frac{8}{243\pi} = 0.0104793378$	0.010479337749	m	m	m	r
all other zero-mode classes (diagonal and off-diagonal)	0	$\lvert\cdot\rvert < 2 \times 10^{-10}$				
χ-attached sunsets (D^M)	0	-2.9×10^{-9}; -2.6×10^{-7}				

Table 12: $\ln L$ coefficients of the C_4 shift that each window class restores. "m": missing in the scheme; "r": restored. The fits use the exact shifts to $L = 20000$ (the χ-attached class to $L = 8000$) with $L_{\min} = 2000$ and 4000 (`residual_log/results/log_coeffs.txt`).

the retuned g_3. This is a local vertex renormalization, a sunset with one hard external χ line, which the tuning of g absorbs. The decomposition must be made by window class, not by diagram topology: single four-vertex topologies carry L, $\ln^2 L$ and $\ln L$ terms that cancel against counterterm chains (`decompose.py`), and the window classes are the combinations that are separately finite in the continuum.

Why the tuning cannot absorb the defect. The two tuning conditions fix two state-independent numbers, r_n and g_n. A shift of r or g changes the reference energies by $2p\,\delta r + \frac{1}{2}p(p-1)\,\delta g$, which Π removes. The budget defect is different in kind: it is an operator proportional to the level of the spectator quanta, $\sim\{\hat{\ell}, M\}$, and through the outer amplitude $20p^3$ it is cubic in p. No choice of tuning conditions can represent it.

Contrast with one dimension. In $d = 1$ the cutoff is on the free energy $E^0 = \ell + \frac{P}{4}$ (Eq. (5)). Every four-particle vacuum shell then carries the same energy, so the vacuum window is of order one per missing shell and does not decay, and D^{conn} repairs it (Section 2.7). The references have $E^0 = \frac{p}{4}$, so their own windows hold vacuum shells as well, and how many depends on where the cutoff sits relative to them (the cutoff placement of Section 3.3). By power counting the sunset density falls like level$^{-1/2}$ ($r_2 \sim \Lambda^{1/2}$), so its budget deficit $\sim\ell/\Lambda^{1/2}$ vanishes: this is why the vacuum correction suffices in $d = 1$ and does not in $d = 2$. The connected ("folded") ϵ^4 term that D^{conn} does not restore also decays; its contribution to the ϵ^4 spacing of $\partial\phi$ is -6.8×10^{-4} at $\Lambda = 16.25$ and -3.8×10^{-4} at $\Lambda = 64.25$ (`lri_hpc_Hamitonian`, record `analysis/identity_checks/challenge-theory/folded_ph0.25.json`). With the vacuum window alone the ϵ^2 descendant spacings vanish (Table 4). In $d = 2$ every structure is one power worse: the vacuum window grows like L (the κL of the raw scheme), the sunset window is of order one, and the coupling window is $\sim\ell/L$. The references have level 0, so their own windows are empty and nothing depends on where the cutoff sits relative to them; the zero-mode dependence that $d = 1$ carries through $E^0 = \ell + \frac{P}{4}$ appears in $d = 2$ as the separate zero-mode window.

A.5 The complete window at ϵ^4

Records: `residual_log/WINDOW_SPEC.md`, `residual_log/results/brute_vs_codes.txt`, `residual_log/C4W_by_L.csv`, `residual_log/results/{offdiag_L20000,nm_L8000}.npz`, `residual_log/fits/fits_`

`final.txt`; code `residual_log/{nm_window,nm_window_exact,brute_full,brute_nm,nm_zz_check, offdiag_hermitian_check,fits_final}.py`.

Principle. Section 4.4 defines the window, $D = D^{\rm zm} + D^M$ with Eqs. (20) and (21), and states its principle. For a mass insertion between states of levels ℓ_s and $\ell_{s'}$ the budget is the average $L - \frac{1}{2}(\ell_s + \ell_{s'})$, which comes from averaging the internal orderings of the sunset: nested and exchange orderings for a χ hop $A^\dagger A$, and the two assignments for pair creation $A^\dagger A^\dagger$ and pair annihilation AA. $h^{(2)}$ contains all inner bubble types with one to four χ lines; the zero-line part and the r_2, g_2 terms do not depend on the level and cancel. Bra-referenced and Hermitian versions of $h^{(2)}$ give the same C_4 to 10^{-16}, because the product $\langle VG\, D^{\rm zm} GV\rangle$ through which it enters is symmetric.

How the window enters C_4. Since $D^{\rm zm}$ vanishes on level-0 states, its only contribution is $g_1^2\langle VG\, D^{\rm zm} GV\rangle$. D^M is symmetric, but it does not vanish on $|p\rangle$, because M' creates two quanta. It therefore enters already at ϵ^3, through $2g_1\langle VGD^M\rangle$, which changes r_3 and g_3 but not C_3. At ϵ^4

$$\begin{aligned}\delta\mathcal{E}_p^{(4)} &= g_1^2\langle VGD^MGV\rangle + 2g_1\big[g_1\langle VGVGD^M\rangle - g_1 V_{pp}\langle VG^2D^M\rangle - \tfrac{1}{4}\langle VG(P-p)GD^M\rangle\big],\\ \delta C_4^M &= \Pi\big[\delta\mathcal{E}^{(4)}\big] + 2g_1\,\delta g_3\,\Pi\big[\langle VGV\rangle\big];\end{aligned} \tag{42}$$

r_3 drops out of C_4, and Π kills $\langle H_2GD^M\rangle$ and $\langle D^MGD^M\rangle$. The complete coefficient is

$$C_4^W(L) = C_4^{\rm zm}(L) + \delta C_4^{\rm off}(L) + \delta C_4^M(L), \tag{43}$$

where $\delta C_4^{\rm off}$ is the shift from the elements $z' \neq z$ of $D^{\rm zm}$. All three pieces are exact for every even $L \leq 2000$, and the shifts are known further, to $L = 20000$ (off-diagonal) and $L = 8000$ (D^M).

Derivation status. $D^{\rm zm}$ follows from the linked-cluster argument of Appendix A.4; it is exact at leading degree in p, and the differences at subleading degree vanish as $L \to \infty$. D^M is derived at leading order in $1/L$. Its slope can be replaced by the exact sum $t_M(j)$ of the sunset contributions that the cutoff removes for a spectator of level $2j$, with both zero-mode orderings averaged; $t_M(j) \to \frac{2}{\pi}j$ per unit amplitude (Appendix A.4). This changes δC_4^M by -4.5×10^{-4} at $L = 100$ and -2.4×10^{-4} at $L = 200$ (`nm_window_exact.py`), that is by $\mathcal{O}(1/L)$, with the same limit. The Newton-coefficient form of Appendix B.1, another refinement, changes C_4 much less (4.9×10^{-7} at $L = 100$), also with the same limit; why the two refinements differ this much at finite L was not analyzed.

check	L	agreement
`offdiag_window.py`, diagonal part, against $C_4^{\rm zm} - C_4^{\rm raw}$ of `c4_exact`	≤ 2000	4×10^{-11}
`nm_window.py` against `brute_nm.py` (exact perturbation theory with $\{\hat{\ell}, M'\}$ as an operator): δC_4 and δg_3	2, 4, 6	10^{-14}
`brute_full.py` (complete scheme as operators; off-diagonal window from auxiliary $\|z\rangle$ channels) against $C_4^{\rm zm}$, $C_4^{\rm zm} + \delta C_4^{\rm off}$ and C_4^W	2 to 12	$\leq 6.3\times10^{-14}$
`nm_window.py` on a workstation ($L \leq 400$) against the symmetry cluster ($L \leq 2000$, 8000)	overlap	6×10^{-14}

Table 13: Checks of the window codes against brute-force exact perturbation theory in fraction arithmetic, with D added as an explicit operator (`residual_log/results/brute_vs_codes.txt`).

Verification at finite L. Table 13 lists the checks. The brute-force values of the complete scheme are $C_4^W(2) = 1.966884466461$, $C_4^W(4) = 2.290042077448$, $C_4^W(6) = 2.546923269518$, $C_4^W(8) = 2.756245037529$, $C_4^W(10) = 2.928832263364$ and $C_4^W(12) = 3.073608283249$; the comparison is limited by the float64 precision of the large-L code, about 6×10^{-14}.

What does not work.

- Any diagonal window (consistent, zero-mode): it misses the transitions $\Delta z \neq 0$ and the χ parts by construction.
- Replacing D^{zm} by its leading form $-\frac{g_1^2}{3\pi}\{\hat{\ell}, :Z^2:\}$ (`nm_zz_check.py`). The difference tends to $0.042838 + \frac{5}{4374\pi}\ln L$: the inner coupling and one-line bubbles carry finite parts from hard outer states and the four-line $\times$ one-line logarithm, so the exact zero-mode window is required. Only the χ part may use the leading form.
- A modified tuning prescription (Appendix A.4).

scheme	$C_4(2000)$	large L	limit
raw	4.29126	$-\kappa L - \frac{725}{8748\pi}\ln L$	diverges
consistent (= literal at this order)	4.73178	$-\frac{365}{4374\pi}\ln L$	diverges
zero-mode (diagonal)	4.87359	$-\frac{8}{243\pi}\ln L$	diverges
+ off-diagonal zero-mode window	4.97161	no $\ln L$	5.0386139, 0.0102545 short
+ mass window D^M (= W)	4.98172	no $\ln L$	5.0488682
continuum			5.0488684314

Table 14: C_4 at $L = 2000$ and its large-L form, window by window (`residual_log/C4W_by_L.csv`, `residual_log/fits/fits_final.txt`).

Large L. Table 14 builds the complete window up piece by piece. The off-diagonal shift is $\delta C_4^{\mathrm{off}}(L) = \frac{8}{243\pi}\ln L + 0.0172932 + \mathcal{O}(\ln^2 L/L)$, from exact values to $L = 20000$. It removes the logarithm of the zero-mode scheme but leaves a finite miss of 0.0102545. The mass-window shift converges, $\delta C_4^M(\infty) = 0.0102543394$ with a spread of 1.5×10^{-8} over twelve fits of the exact values to $L = 8000$, and supplies the missing finite part: the separately extrapolated pieces give $5.0386139 + 0.0102543 = 5.0488682$. The approach is slow, like $\ln^2 L/L$: at $L = 2000$, $C_4^W - C_4^{(2)} = -0.0672$ (Fig. 1c).

Fits of C_4^W. The data are accurate to 10^{-10} or better, so the uncertainty is entirely the extrapolation.

- The towers T_{22}, $T_{22} + \ln^3 L/L$ and T_{23} without $\ln L$, with $L_{\min} \in \{200, 300, 400\}$ (nine fits), give 5.04886823 with a half spread of 5.6×10^{-7}, 2.0×10^{-7} below the continuum value; this is the value quoted in Section 4.4.
- No-log fits outside this set (towers $T_{22} + \ln^3 L/L$ and T_{23} at $L_{\min} = 600$, T_{33} at $L_{\min} = 400$) move the limit by up to 1.8×10^{-5} from the continuum value; T_{33} at $L_{\min} = 600$ is over-fitted ($\ln^2 L/L$ coefficient -25, limit 2.8×10^{-4} low).
- With a free $\ln L$ term the stable towers at $L_{\min} = 200$ to 300 give its coefficient between -1.8×10^{-6} and $+3.5 \times 10^{-6}$, and the limit moves by -4.3×10^{-5} to $+2.2 \times 10^{-5}$ (the tower T_{33}, outside this set, gives a coefficient of -8.1×10^{-6} at $L_{\min} = 200$). Richer towers at larger $L_{\min}$ become unstable, for example with a free $\ln L$ coefficient up to 5×10^{-4} and a $\ln^2 L/L$ coefficient of $+16$ to $+30$; these fits have more freedom than the data constrain.

Hence the quoted uncertainty: the half spread of the nine stable fits, rounded up to 10^{-6}. The fits outside the stable set show that the model dependence can reach 2×10^{-5}, and 4×10^{-5} with a free $\ln L$ term, which the complete window excludes by construction (Table 12). The fitted tail is about $-1.40 \ln^2 L/L$ and comes entirely from C_4^{zm}. Fixing it at $-5/\pi$, the value used as a prior in the fit scripts, makes the fits 500 times worse, so that prior is not supported. In the fits of the raw, consistent and zero-mode schemes (Section 4.4) the variants with the coefficient fixed only widen the quoted spread.

B Two dimensions: order ϵ^5

This appendix supports Section 4.5. It gives the window at all orders, shows that it diverges linearly at ϵ^5 and why, derives the rule W_c that removes the divergence, and gives the details of the C_5 extrapolation.

B.1 The window at all orders

Records: `residual_log/WINDOW_SPEC.md` (status table and §§2.2 to 2.10), `complete_exact/README.md`.

For every even $\lambda \le L$ the Hamiltonian at sub-cutoff λ is the truncation to $\ell \le \lambda$ of $H_0 + \epsilon F_1 + rM + gV$ plus $D_\lambda = D_\lambda^{\mathrm{zm}} + D_\lambda^M$, with one $r(\epsilon)$, $g(\epsilon)$ fixed at $\lambda = L$. Scheme W is this window with self-consistent auxiliary objects (last item below): through ϵ^4 it is the complete window of Section 4.4; at ϵ^5 it diverges (Appendix B.3), and rule W_c repairs it. The ingredients are the following.

- *Auxiliary channels.* $E_z(\lambda)$ and $\psi_z(\lambda)$ are the eigenvalue and the right eigenvector of this Hamiltonian on the branch through $|z\rangle$, normalized by $\langle z|\psi_z\rangle = 1$.
- *Bloch zero-mode window.* With $U_{z'z}(\lambda) = \langle z'|\psi_z(\lambda)\rangle$ and $h(\lambda) = U\,\mathrm{diag}(E_z(\lambda))\,U^{-1}$,
$$D^{\mathrm{zm}}|z,\chi\rangle = \sum_{z'} \big[h(\lambda) - h(\lambda - \ell_\chi)\big]_{z'z}|z',\chi\rangle \qquad (\ell_\chi \ge 2).$$
Here h is the full Bloch matrix; at ϵ^2 its level-dependent part is $h^{(2)}$ of Eq. (20), so the window reduces to that equation.
- *Mass window.* $D^M = \frac{1}{4}\{e_1(\lambda) - e_1(\lambda - \hat{\ell}), M'\}$ with the Newton coefficient $e_1 = E_1 - E_0$, the combination that the mass counterterm renormalizes: $E_z = \sum_k e_k \binom{z}{k}$ and $\langle z|M|z\rangle = 2z$ (§2.4 of `WINDOW_SPEC.md`). At ϵ^2, $e_1^{(2)}(\lambda) - e_1^{(2)}(\lambda - \ell) = -\frac{4}{27\pi}\ell + \mathcal{O}(\ell/\lambda)$, so $D^{M,(2)}$ is Eq. (21) up to $\mathcal{O}(1/\lambda)$. This form changes C_4 relative to the slope form by 1.8×10^{-6}, 4.9×10^{-7}, 1.2×10^{-7} and 2.9×10^{-8} at $L = 50$, 100, 200 and 400, with the same limit, and it makes the scheme an exact rational at every L.
- *Coupling window* (scheme $W{+}V$ only). $D^V = \frac{1}{2}\{e_2(\lambda) - e_2(\lambda - \hat{\ell}), V'\}$ with $e_2 = E_2 - 2E_1 + E_0$ and $V' = V - \frac{1}{12}{:}Z^4{:}$.
- *Partners.* The degenerate partners of $|z\rangle$, $z \ge 5$ (Appendix A.1), are projected out of every window object. A reading that keeps them through a projected equation in the auxiliary channels differs by 2.8×10^{-5} in C_4 and -1.6×10^{-4} in C_5 at $L = 16$.
- *Self-consistency* (scheme W). $h(\lambda - \ell)$ and $e_1(\lambda - \ell)$ belong to the Hamiltonian at sub-cutoff $\lambda - \ell$ with its own window. Rule W_c (Appendix B.4) replaces this.

C_5 needs $D^{\mathrm{zm},(2)}$, $D^{\mathrm{zm},(3)}$, $D^{M,(2)}$ and $D^{M,(3)}$; $D^{M,(4)}$ does not enter. C_4 sees only the ϵ^2 windows, so C_4^W is the coefficient of Appendix A.5 up to the difference between the e_1 form and the slope form given above.

The coupling window is invalid. In the exact small-L solver $C_4(W+V) - C_4(W) = 0.484$, 0.491 and 0.492 at $L = 10$, 12 and 14. This is a finite shift, although power counting (§3 of `WINDOW_SPEC.md`) suggests that the effect of D^V vanishes. D^V also makes the ϵ^3 spacing grow (Table 22). It is not used below.

B.2 Exact ϵ^5 data

Records: `residual_log/results/eps5/{validation_c5,validation_vs_complete_exact}.txt`, `residual_log/results/eps5/{c5_L160,c5_L200,dC5_L1000_all}.npz`, `complete_exact/README.md`; code `residual_log/eps5/`, `residual_log/brute_W5.py`.

Two independent exact computations exist.

1. *Large-L diagram sums* (`residual_log/eps5/`). The C_4 machinery of Appendix A.3 is extended to five-vertex chains, ϵ-Taylor orders at every vertex, open zero-mode paths and window insertions; the perturbation-theory expression for $E^{(5)}$ is unrolled symbolically into 201 products of vertices, windows and resolvents. The code gives $C_n(L)$, $n \le 5$, for every even $L \le 160$ in the raw scheme and eleven window schemes (among them those of Table 16), raw C_5 to $L = 200$ (extended to $L = 300$ in Appendix B.6), and the scheme shifts $\delta C_n(L) = C_n(\text{scheme}) - C_n(\text{raw})$ for every even $L \le 1000$. The shifts need no five-vertex chain. The cost was about 25 node-hours on the symmetry cluster.

2. *Exact small-L solver* (`complete_exact/`). Scheme W and its variants as explicit operators in exact perturbation theory through ϵ^8, $L = 0$ to 16, modulo 208 primes ($L \le 12$) or 128 primes ($L = 14$, 16), every value verified on eight held-out primes, with descendant channels in every scheme. An independent fraction implementation agrees identically at $L = 2$ to 10; closure tests with random entries outside the claimed window sets change nothing; and the raw, consistent and zero-mode schemes reproduce the exact solvers of Appendix A.2 exactly to $L = 16$. Examples, with D^M in the e_1 form: $C_4^W(2) = \frac{11011547}{5598720}$ (the slope form of Appendix A.5 gives 1.966884466461) and $C_5^W(2) = -\frac{879593527}{80621568}$.

quantity	reference	worst deviation
raw, consistent C_2 to C_5	exact solver, $L \le 18$	1.0×10^{-13} relative
raw, consistent C_2 to C_5	production solver, $L \le 34$	2.2×10^{-13} relative
zero-mode C_2 to C_5	exact zero-mode solver, $L \le 18$	9×10^{-14} relative
C_5 of W and three diagnostic variants	explicit-operator fraction arithmetic (`brute_W5.py`), $L \le 8$	1.7×10^{-14} relative
C_3 to C_5 of W, W_2, raw, consistent, zero-mode	`complete_exact`, $L \le 16$	1.0×10^{-15} relative
full runs against raw plus shift	all schemes, $L \le 160$	1.7×10^{-11} absolute
raw C_5, run to $L = 200$ against run to $L = 160$	overlap	1.3×10^{-11} absolute

Table 15: Validation of the exact ϵ^5 data; scheme names as in Appendix B.3 (`residual_log/results/eps5/validation_c5.txt`, `validation_vs_complete_exact.txt`).

Table 15 lists the checks. Raw C_5 is -31.898 at $L = 160$ and -32.597 at $L = 200$, against the continuum value -38.834. The last row of the table is evaluated from `c5_L160.npz` and `c5_L200.npz`.

B.3 Scheme W diverges linearly

Records: `residual_log/results/eps5/classes_c5.txt`, `residual_log/fits/fits_c5.txt`, `residual_log/results/eps5/dC5_L1000_all.npz`, `residual_log/WINDOW_SPEC.md` (status after the exact ϵ^5 test,

item 1); code `residual_log/eps5/{classes_c5,fits_c5}.py`.

The exact shifts have the form $aL + b_2 \ln^2 L + b_1 \ln L + c + \mathcal{O}(\ln^k L/L)$, with no $L \ln L$ term, and their linear coefficients are fixed to better than 2×10^{-6}. Raw C_5 to $L = 160$ fixes its own linear coefficient only roughly. A free fit gives $+0.0004 \pm 0.002$ (half range over tail towers and $L_{\min} = 30$, 40, 60; step 2 of `fits_c5.txt`). A profile over a fixed raw coefficient $a_{\rm raw}$ (fits of $C_5^{\rm raw}(L) - a_{\rm raw} L$ with logarithmic towers, $L \le 160$) is sharper: the intercept is most stable, and needs no logarithms, at $a_{\rm raw} = +0.0004(4)$; the (4) is the range of $a_{\rm raw}$ over which the intercept stays most stable, a judgement read off the profile (step 4 of `fits_c5.txt`), not a statistic. These fits were made with the data then available and were not repeated with the data to $L = 300$. Table 16 gives the coefficients. The scheme names are: subscript 2 for windows of order ϵ^2 only; consistent and zero-mode with their ϵ^3 windows; Bloch for the Bloch $D^{\rm zm}$ at ϵ^2 and ϵ^3 without D^M; W_{3e} for W with the eigenvalue-diagonal ϵ^3 zero-mode window $E_z^{(3)}(L) - E_z^{(3)}(L - \ell)$ in place of the Bloch one.

	consistent$_2$	consistent	zero-mode$_2$	zero-mode	Bloch$_2$	Bloch	W_2	W_{3e}	W
shift	-0.001429	-0.000392	$+0.003119$	-0.001381	$+0.000991$	-0.000364	-0.000974	-0.005474	-0.002329
total	-0.0010	$+0.0000$	$+0.0035$	-0.0010	$+0.0014$	$+0.0000$	-0.0006	-0.0051	-0.0019

Table 16: Linear coefficient of $C_5^S(L)$ by scheme S. "Shift": the coefficient of $\delta C_5^S(L) = C_5^S - C_5^{\rm raw}$, from the exact shifts to $L = 1000$ (`residual_log/results/eps5/classes_c5.txt`). "Total": shift plus the raw coefficient $+0.0004(4)$ from the profile fit of raw $C_5(L)$, $L \le 160$ (`residual_log/WINDOW_SPEC.md`); every total carries its uncertainty ± 0.0004.

So $C_5^W(L)$ carries about $-0.0019\, L$. The Bloch zero-mode windows alone (scheme Bloch) leave no linear term; the eigenvalue-diagonal ϵ^3 form leaves $-0.0051\, L$ and is excluded. The consistent scheme has no linear term but keeps about $+0.2 \ln L$.

Location. Differences of schemes isolate the window classes (Table 17). The whole linear defect of W is the Z–χ part $2\langle Z\chi\rangle_\theta$ of $D^{M,(2)}$, the only part of M' that connects a pure zero-mode state to a one-χ state. That part cannot simply be dropped: at ϵ^4 its $\ln L$ cancels that of the $\chi\chi$ parts, and together they give the finite $+0.0102$ that C_4 needs. A variant that evaluates the Z–χ window at the smaller of the two levels keeps the linear term of C_5 and adds $+0.00295 \ln L$ to C_4.

B.4 Double counting and the rule W_c

Records: `residual_log/WINDOW_SPEC.md` (§2.2a); code `residual_log/{brute_W5,brute_desc}.py`, `residual_log/ce_wc/`.

Diagnosis. A Z–χ sunset ($Z \to 3\chi \to \chi_{aa}$) followed by a vertex that absorbs χ_{aa} into the zero mode is a connected process from a pure zero-mode state to a pure zero-mode state, that is, a term of the Bloch matrix $h^{(3)}$. Inside an outer state with spectator level ℓ its sunset has only the budget $L - \ell$. The outer $D^{M,(2)}$, acting on the transition $z \to (z \mp 1, \chi_{aa})$, restores it once. The zero-mode window $D^{{\rm zm},(3)} = h^{(3)}(L) - h^{(3)}(L - \ell)$ restores it a second time: in scheme W the auxiliary channel at $L - \ell$ carries its own mass window, so $h(L - \ell)$ contains this sunset restored only to $L - \ell$, while $h(L)$ contains it restored to L, and their difference adds the restoration from $L - \ell$ to L once more. In general, with self-consistent auxiliary objects (h, e_1, E_0, ...), every window nested inside them is counted twice. At ϵ^5 this affects only $D^{M,(2)}$ inside $h^{(3)}$ and $e_1^{(3)}$, which is why the defect is purely Z–χ and why C_4 is unaffected. From ϵ^6 on it affects $D^{\rm zm}$, the hop and pair parts of D^M inside $h^{(4)}$, and the vacuum window inside $E_0^{(4)}$ of the consistent scheme.

class	C_5: coefficient of L	C_4: finite part and $\ln L$
ϵ^2 vacuum window	-0.001429	the linear $+0.000221\,L$
ϵ^3 vacuum window	$+0.001037$	none
ϵ^2 diagonal z-dependent window	$+0.004548$	$+0.016083\ln L$
ϵ^2 off-diagonal Bloch window	-0.002129	$+0.010479\ln L$
ϵ^3 Bloch window	-0.001355	none
ϵ^3 eigenvalue-diagonal window (instead of Bloch)	-0.004500	none
$D^{M,(2)}$	-0.001965	$+0.010246$, no $\ln L$
$D^{M,(3)}$	0.000000	none
partner terms in $E_z^{(3)}$	0	0
$D^{M,(2)}$, Z–χ part	-0.001965	$+0.012293 - 0.002947\ln L$
$D^{M,(2)}$, χ hop (χ in, χ out)	0	$-0.000894 + 0.001474\ln L$
$D^{M,(2)}$, pair creation	0	$-0.000579 + 0.000737\ln L$
$D^{M,(2)}$, pair annihilation	0	$-0.000579 + 0.000737\ln L$

Table 17: Window classes at ϵ^5 and their ϵ^4 content (`residual_log/results/eps5/classes_c5.txt`). The lower block splits $D^{M,(2)}$ by the leg type of its M' vertex; the parts add up to $D^{M,(2)}$ within 8×10^{-11}. The C_4 column gives fits of the shifts to $L\le1000$: for the ϵ^2 vacuum window it lists only the linear term (its $\ln L$ is in Table 12), and the $D^{M,(2)}$ entry is a finite-L estimate of the limit 0.0102543 of Appendix A.5.

The rule. An auxiliary channel at sub-cutoff $\lambda<L$ represents the zero-mode cluster embedded in a state of the cutoff-L theory whose spectators have level $L-\lambda$. The windows inside it are therefore the cutoff-L windows of that embedded state:

$$D^{\rm zm}_{\rm aux}|z,\chi\rangle=\sum_{z'}\big[h(L)-h(\lambda-\ell_\chi)\big]_{z'z}|z',\chi\rangle,\qquad D^{M}_{\rm aux}=\tfrac14\big\{e_1(L)-e_1(\lambda-\hat\ell),M'\big\}. \tag{44}$$

$D^{\rm zm}_{\rm aux}$ is absent on auxiliary level-0 states, where $D^{M}_{\rm aux}$ uses $e_1(L)-e_1(\lambda)$. The outer windows keep their form, with h and e_1 now computed under this rule. At the order that enters C_5,

$$\begin{aligned}h^{(3)}(\lambda;L)&=h^{(3)}(\lambda)+\tfrac12\big[e_1^{(2)}(L)-e_1^{(2)}(\lambda)\big]H_3^{M'}(\lambda),\\ e_1^{(3)}(\lambda;L)&=e_1^{(3)}(\lambda)+g_1\big[e_1^{(2)}(L)-e_1^{(2)}(\lambda)\big]\big[A_1(\lambda)-A_0(\lambda)\big],\qquad A_z(\lambda)=\langle z|M_0'G_zV_0|z\rangle_\lambda,\end{aligned} \tag{45}$$

where $H_3^{M'}$ is the response of $h^{(3)}$ to M_0' inserted with unit coefficient at ϵ^2, the subscript 0 marks the ϵ^0 part of an operator, and $G_z=Q_z/(\frac z2-H_0)$. The added terms complete the inner sunsets to the outer cutoff L, so that the outer difference no longer restores them a second time. The rule has no fitted parameter; its status is provisional (Appendix C.6). Three exact implementations agree: explicit-operator fraction arithmetic (`brute_W5.py`, `brute_desc.py`), the large-L diagram code, and `ce_wc/`, a copy of the small-L solver with the auxiliary windows referred to L at all orders. They give identical C_5 at $L\le8$, C_3 to C_5 within 10^{-15} at $L\le16$, and the same ϵ^3 spacing of χ_{11} as rationals at $L=8$ to 14.

B.5 Tests T1 to T3

Records: `residual_log/results/eps5/{classes_c5,ce_wc_tables}.txt`, `residual_log/fits/fits_c5.txt`, `residual_log/results/eps5/dC5_L1000_all.npz`.

We test each window in three ways: T1, C_4 stays unchanged and convergent; T2, C_5 converges to the continuum value; T3, the ϵ^3 spacing R_3 of the single-quantum descendants vanishes (Appendix C adds T4, the ϵ^4 spacing).

T1 passes. Windows of order ϵ^3 and higher do not enter C_4, and $|C_4^{W_c} - C_4^W| \le 6 \times 10^{-13}$ for $L \le 1000$. So $C_4^{W_c}$ has the limit 5.0488682(10) of Section 4.4.

T2 passes, with the caveat of Section 4.5. The W_c shift has the linear coefficient -0.0003922 (half range 2×10^{-6} over tail towers and fit windows; step 1 of `fits_c5.txt`), equal to the consistent scheme's -0.0003923, so with the raw $+0.0004(4)$ the scheme has no linear term. Its limit and the residual-logarithm question are those of Appendix B.6. Table 18 compares the schemes.

scheme	windows	linear	limit	limit − continuum
raw	none	+0.0004	needs L or $\ln L$ terms	
consistent	vacuum, ϵ^2 and ϵ^3	+0.0000	needs $\ln L$ ($+0.2 \ln L$)	
zero-mode	diagonal, ϵ^2 and ϵ^3	−0.0010	diverges	
Bloch	Bloch D^{zm}, ϵ^2 and ϵ^3	+0.0000	≈ -38.5	$\approx +0.3$
W	$D^{\mathrm{zm}} + D^M$	−0.0019	diverges	
W_c	W with rule W_c	+0.0000	−38.830(13)	+0.004

Table 18: Large-L behavior and extrapolated C_5 by scheme. "Linear" is the total linear coefficient of $C_5^S(L)$ of Table 16, with uncertainty ± 0.0004. The W_c limit uses the protocol of Appendix B.6 (towers with the $\ln^3 L/L$ term, $L \le 300$); for Bloch, whose tail also carries logarithms, the limit and the miss are only approximate. Raw: fits without L, $\ln^2 L$ and $\ln L$ terms miss the continuum value by 1.9 (step 3 of `fits_c5.txt`, $L \le 160$). Continuum value $C_5^{(2)} = -38.8336459$.

The scheme Bloch misses by about 0.3: the finite part of D^M is needed, and W_c keeps it.

T3 fails in W_c. W_c removes the slow growth of the ϵ^3 spacing that W has, but leaves a constant (Table 22). A repair of this defect shifts C_5 by less than the extrapolation spread: the external-line window of Appendix C, which identifies the cause and derives the repair, shifts it by -0.005772, against a spread of ± 0.013. So T2 cannot decide it, and T3 is the decisive test.

B.6 The tail of C_5 and the residual logarithm

Records: `residual_log/large_L/fits/{c5_fits300,legw_logs}.txt`, `residual_log/large_L/results/{regression_opt,compare_c5_L300_vs_L200}.txt`, `residual_log/large_L/results/c5_L300.npz`, `residual_log/fits/parent_check_largeL/refit_c5_300.txt`; code `residual_log/large_L/{c5_fits300,c5_fits}.py`, `residual_log/large_L/lib_opt/`, `residual_log/fits/parent_check_largeL/refit_c5_300.py`.

$C_5^{W_{cs}}$ (Section 4.5, Appendix C.3) differs from $C_5^{W_c}$ by $\delta^{(3)}(L)S(L)$, which carries no $\ln L$. Towers without the $\ln^3 L/L$ term that C_5 needs bias the limit, because a free $\ln L$ then stands in for the missing tail.

Tail power. The tail of $C_n(L)$ carries $\ln^{n-2} L/L$: $C_2 \sim 1/L$, $C_3 \sim \ln L/L$, $C_4 \sim \ln^2 L/L$ and $C_5 \sim \ln^3 L/L$ (Appendices A.2 and A.5). Hold-out tests, made after the data to $L = 300$ existed, support this power (Table 19): fit $[L_{\min}, 240]$ and predict $L = 242$ to 300 with towers without $\ln L$ whose $1/L$ part has the powers $\ln^i L$, $i \le k_1$. The prediction error falls by one to two orders of magnitude for each added power up to $k_1 = n - 2$ ($k_1 = 2$ for C_4^W, 3 for $C_5^{W_{cs}}$) and changes by at most a factor of two beyond it. The test thus finds the smallest adequate tail, but it cannot exclude one more power: for C_4^W, $k_1 = 3$ is slightly better than 2.

	$k_1 = 0$	1	2	3	4
C_4^W: hold-out error	6.1×10^{-4}	9.7×10^{-6}	1.1×10^{-7}	$\mathbf{5.2 \times 10^{-8}}$	6.5×10^{-8}
$C_5^{W_{cs}}$: hold-out error	1.2×10^{-2}	3.5×10^{-4}	5.3×10^{-6}	$\mathbf{3.7 \times 10^{-7}}$	5.7×10^{-7}
$C_5^{W_{cs}}$: limit − continuum	+0.81	+0.16	+0.015	−0.003	+0.012

Table 19: Selection of the tail power k_1 (largest i in $\ln^i L/L$) by hold-out prediction of $L = 242$ to 300 from fits to $L \le 240$, towers without $\ln L$ ($1/L^2$ with $\ln^{0,1} L$ or $\ln^{0,1,2} L$; $L_{\min} = 40$, 60, 80; median of the largest errors), and the limit of the same towers fitted to $L \le 300$ (`residual_log/large_L/fits/c5_fits300.txt`, §1). Bold: smallest error.

With the 15 fits of Eq. (25) ($L_{\min} = 30$ to 100, $k_2 = 0$ to 2, $L \le 300$), $k_1 = 4$ gives a median of −38.8334, shifted by +0.002 from the $k_1 = 3$ value, with a half range of 0.050 instead of 0.013 (replay check `p2.c5_k1`); with the towers of Table 19 the limit moves by +0.015.

Raw C_5 to $L = 300$. The sum is split into 10012 independent tasks (groups of chain patterns and level ranges). The large-L engine as first run (`residual_log/large_L/lib/`, not the production solver) computed 3987 of them in about 130 node-hours. An optimized engine (`residual_log/large_L/lib_opt/engine.py`) finished the remaining 6025 tasks in 32 node-hours; only the density without ϵ-derivatives changes, through a memoized tree over the inclusion–exclusion terms, a parallel diagonal convolution and batched axis convolutions. Per task it agrees with the first engine to 7.3×10^{-16} (norm-relative; 1448 tasks, covering every kind of density kernel at $L = 300$ and all tasks with a bundle of six lines at $L = 200$). The $L = 300$ assembly, 40% old and 60% new tasks, reproduces the $L \le 200$ data to 7.6×10^{-13} (C_5, relative) and 5.1×10^{-13} (C_4). Raw $C_5(300) = -33.6509669991$.

Hold-out tests with $L \le 300$. Table 20 compares fits without and with a free $\ln L$ at fixed k_1.

	fit $L \le 200$, predict 202 to 300		fit $L \le 240$, predict 242 to 300	
k_1	no $\ln L$	free $\ln L$	no $\ln L$	free $\ln L$
2	4.2×10^{-5}	7.1×10^{-6}	1.6×10^{-5}	2.0×10^{-6}
3	4.3×10^{-6}	9.4×10^{-6}	6.0×10^{-7}	5.7×10^{-7}
4	6.5×10^{-6}	1.4×10^{-5}	6.3×10^{-7}	7.7×10^{-7}

Table 20: $C_5^{W_{cs}}$: hold-out prediction errors, median over the 15 fits of Eq. (25) (`residual_log/fits/parent_check_largeL/refit_c5_300.txt`). The medians over twelve fits with $L_{\min} = 30$ to 80 (`c5_fits300.txt`) for the fit to $L \le 200$ with $k_1 = 3$ are 4.7×10^{-6} and 9.6×10^{-6}.

With the misspecified $k_1 = 2$ a free logarithm helps. With $k_1 = 3$ and the fit ending at $L = 200$, the model without $\ln L$ predicts $L = 202$ to 300 better by a factor of two. With the fit ending at $L = 240$ the two are equal, so at that level the hold-out test no longer separates a logarithm from none; for C_4^W, which has no logarithm, a free one also predicts slightly better there.

Drift of the free $\ln L$ coefficient. With $k_1 = 3$ towers, $L_{\min} = 30$ to 80 (medians of twelve fits, §2 of `c5_fits300.txt`):

$L_{\max}$	200	220	240	260	280	300
free $\ln L$ coefficient b	-0.0134	-0.0080	-0.0005	$+0.0030$	$+0.0047$	$+0.0054$
calibration, injected 0	-0.0007	-0.0021	-0.0028	-0.0033	-0.0035	-0.0036
calibration, injected -0.03	-0.0307	-0.0321	-0.0328	-0.0333	-0.0335	-0.0336
$C_5^{W_{cs}}$: limit without $\ln L$ − continuum	-0.0049	-0.0051	-0.0043	-0.0032	-0.0024	-0.0017

The calibration applies the same estimator to the synthetic series $7.7\,C_4^W + 2\ln^3 L/L + b\ln L$ with a known, injected b: C_4^W has no logarithm and a known limit, the factor $7.7 \approx |C_5/C_4|$ gives it the size of C_5, and the $\ln^3 L/L$ term gives it the top logarithm of C_5. The estimator settles by $L_{\max} \approx 250$ and then reads about 0.004 low. An independent refit (`refit_c5_300.py`), with the 15 fits of Eq. (25), gives the same drift (-0.011, -0.004, $+0.003$ and $+0.004$ at $L_{\max} = 200$, 240, 280 and 300), and on C_4^W, which has no logarithm, the same estimator returns $+0.0005$. Section 4.5 draws the conclusion.

The limit. The 15 fits of Eq. (25) to $L \le 300$ give Eq. (26); their range is -38.8430 to -38.8174. $C_5^{W_c}$ lies higher by about $-\delta_\infty S_\infty = +0.0058$, as the external-line window requires. As $L_{\max}$ grows, the W_c median moves away from the continuum value and the W_{cs} median toward it. The same protocol applied to C_4^W gives 5.04868 ± 0.00082 (half range of the 15 fits) against 5.0488684.

Classes excluded as sources of a logarithm. The ϵ^3 external-line class gives no $\ln L$ for any per-unit coefficients, because the $\ln L$ coefficients of $S(L)$ cancel within the hop sector and within the pair sector separately (6×10^{-7} and -1.7×10^{-6}; `legw_logs.txt`). The spectator window $D^{M,(3)}$ has no L, $\ln^2 L$ or $\ln L$ term within the fit uncertainties. The other classes that carry logarithms cannot be separated by fits of the exact shifts alone.

B.7 Consequences for C_6 to C_8

Records: `residual_log/results/eps5/ce_wc_tables.txt`, `complete_exact/README.md` (§9).

The double counting recurs, and grows, at every order with nested windows. Rule W_c removes it by construction and is implemented at all orders in `residual_log/ce_wc/`. At $L = 14$ the schemes W_c and W already differ by about 1%: $C_6 = 157.83$ against 159.46, $C_7 = -1439.9$ against -1458.3, and $C_8 = 14710$ against 14962. The ϵ^3 remainder of T3 is a constant defect attached to χ lines; it enters C_5 (through the shift $\delta^{(3)}S$, Appendix C.4) and every higher coefficient. Its repair (Appendix C) changes C_6 to C_8 by only about 0.01%, but the same defect recurs at ϵ^4 and ϵ^5. These differences between schemes are much smaller than the distance of C_6 to C_8 at $L \le 16$ from the continuum values, and C_6 to C_8 are not reproduced in any scheme (Section 4.7).

C Two dimensions: descendant spacings and the external-line windows

This appendix supports Section 4.6 and the external-line window of Section 4.5. The spacings R_n of Eq. (27) test a window at a fixed order without any continuum input; they exposed the defects that remain beyond the complete ϵ^4 window, and a tagged external line located the ϵ^3 defect exactly.

C.1 Descendant channels

Records: `complete_exact/README.md` (§§4, 8 and 9), `complete_exact/results/{descendant_summary.md,descendant_fits.txt}`, `complete_exact/results/descendant_<scheme>_L<L>.json`,

`residual_log/results/descendant_e2.txt`, `residual_log/results/eps5/{descendant_eps3,ce_wc_tables}.txt`, `residual_log/fits/parent_check_wcs/verify.txt`; code `residual_log/descendant_e2.py`.

Two kinds of descendants are used: $\chi_{aa} = \partial^a\bar{\partial}^a\phi$ (one quantum (a,a), level $2a$, primary $|1\rangle$) and $(\partial\bar{\partial})^a\phi^2$ (primary $|2\rangle$). The reference of the latter at $\epsilon = 0$ lies in a degenerate block and is fixed as the eigenvector of H_1 with eigenvalue $E_2^{(1)}$. The small-L solver `complete_exact/` and its W_c copy `residual_log/ce_wc/` compute these channels exactly in every scheme for $L \le 16$, and $R_0 = R_1 = 0$ exactly everywhere. A spacing counts as restored if it decays like $1/L$.

Order ϵ^2. Since $z = 0$ for χ_{aa}, the consistent and zero-mode windows coincide on it. $R_2(\chi_{aa})$ is a sum over two kinds of intermediate states. In the nested ones the external quantum is absorbed and re-emitted. In the exchange ones the first vertex emits the new χ_{aa} together with a bubble, and the second absorbs the old χ_{aa} with the bubble. The exchange states hold two (a,a) quanta, so the level cutoff leaves the bubble only $L-4a$ instead of L. For the sunset this is a deficit of $\frac{2}{\pi}(2a) \times g_1^2 \cdot \frac{2}{3}w_{aa}$, and all other finite parts cancel, so

$$\lim_{L\to\infty} R_2(\chi_{aa}) = \frac{8a\, w_{aa}}{27\pi} \neq 0 \tag{46}$$

in the consistent, literal and zero-mode schemes; the raw truncation also lacks the vacuum window, which by Eq. (32) grows linearly in L. The mass window repairs it. On the one-quantum state $\langle\chi_{aa}|M'|\chi_{aa}\rangle = 2w_{aa}$ and $\ell = 2a$, while $\langle 1|D^M|1\rangle = 0$, so at first order D^M shifts R_2 by exactly $-\frac{1}{27\pi}\cdot 4a \cdot 2w_{aa} = -\frac{8a\,w_{aa}}{27\pi}$.

a	w_{aa}	$\frac{8a\,w_{aa}}{27\pi}$	$R_2(2000)$	$R_2(20000)$	with D^M, $L=2000$	with D^M, $L=20000$
1	0.25	0.023578510	0.023584405	0.023579100	5.90×10^{-6}	5.90×10^{-7}
2	0.140625	0.026525824	0.026532463	0.026526487	6.64×10^{-6}	6.63×10^{-7}
3	0.0976563	0.027631067	0.027637989	0.027631757	6.92×10^{-6}	6.91×10^{-7}
5	0.0605621	0.028559298	0.028566467	0.028560012	7.17×10^{-6}	7.14×10^{-7}
10	0.0310454	0.029280172	0.029287560	0.029280905	7.39×10^{-6}	7.33×10^{-7}

Table 21: The ϵ^2 spacing of χ_{aa} in the consistent window at large L, and the residual $R_2 - \frac{8a\,w_{aa}}{27\pi}$ left after adding D^M (`residual_log/results/descendant_e2.txt`).

Table 21 gives the numbers: with D^M the residual decays like $1/L$, with coefficient 1.2×10^{-2} ($a=1$) to 1.5×10^{-2} ($a=10$). The exact small-L solver confirms this for the full schemes at $L \le 16$. In the raw scheme R_2 grows (χ_{11}: 0.1077 at $L=16$). In the consistent and zero-mode schemes $R_2(\chi_{11})$ is constant (0.0243 at $L=16$), and the ϕ^2 descendant decreases only slowly (0.1459 at $L=16$, effective exponent -0.27). In W, and in W_c, which is identical at this order, R_2 is restored: $0.0114/L$ for χ_{11} (0.000714 at $L=16$), $0.0146/L$ for χ_{22} and $0.634/L$ for the ϕ^2 descendant.

Order ϵ^3. None of the schemes W, W_c and their variants restores the ϵ^3 spacing. The external-line window W_{cs} (Appendix C.3) reduces it for χ_{aa} by at least 95% at $L=14$ and 16, but a $1/L$ decay is not established. Table 22 collects $R_3(\chi_{11})$; the two sources agree where they overlap (scheme W to all printed digits at $L=6$ to 14).

For the other descendants at $L=16$: χ_{22} has $R_3 = 0.04048$ (consistent), 0.03447 (W, growing) and 0.01755 (W_c, flat). The ϕ^2 descendant of level 2 has 0.09469 (consistent, growing), 0.03864 (W, growing after a minimum of 0.0074 near $L=6$ to 8) and -0.0337 (W_c; -0.0306, -0.0413 and -0.0337 at $L=6$, 10 and 16, not settled).

scheme	$L=4$	$L=8$	$L=12$	$L=14$	$L=16$	behavior
raw	0.1617	0.2479	0.3355	0.3793	0.4232	grows linearly
consistent (= zero-mode)	0.04007	0.03941	0.04002	0.04037	0.04071	constant
ϵ^2 windows only	0.1874	0.2750	0.3643	0.4089		grows linearly
W without D^M	0.05598	0.05519	0.05577	0.05611		constant
W with $D^{M,(2)}$ only	0.05812	0.06237	0.06598	0.06749		grows slowly
W	0.02316	0.02686	0.02962	0.03074	0.03173	grows, $\sim 0.009 \ln L$
$W+V$	0.08723	0.1138	0.1362	0.1469	0.1573	grows
W_c	0.01785	0.01618	0.01579	0.01570	0.01565	constant
W_{cs}	0.004225	0.000896	0.000121	−0.000049	−0.000157	falls (Appendix C.4)

Table 22: R_3 of $\chi_{11} = \partial\bar{\partial}\phi$ by scheme, exact (`complete_exact/README.md`, `residual_log/results/eps5/{descendant_eps3,ce_wc_tables}.txt`, `residual_log/fits/parent_check_wcs/verify.txt`). "ϵ^2 windows only" is the window of Appendix A.5 with no ϵ^3 part; "without D^M" and "with $D^{M,(2)}$ only" are W without the mass window and with its ϵ^2 part only (their W_c versions are identical at this order).

Structure of the ϵ^3 defect. For $a \ll L/2$ the ϵ^3 spacing of χ_{aa} is proportional to $a\,w_{aa} = \frac{1}{8}\langle\chi_{aa}|\{\hat{\ell}, M'\}|\chi_{aa}\rangle$ in every scheme. At $L=12$, $R_3/(a\,w_{aa}) = 0.16$, 0.15, 0.15 for $a=1$, 2, 3 in the consistent scheme, 0.22, 0.21, 0.21 in W without D^M, 0.119, 0.114, 0.110 in W, and 0.063 to 0.068 in W_c. This is the operator structure of a mass window attached to χ lines, as at ϵ^2. The diagnostics locate it:

- D^{zm} is not the problem at ϵ^3: W without D^M leaves a constant (0.056 for χ_{11}).
- In W, $D^{M,(2)}$ acting at third order produces the slow growth, and $D^{M,(3)}$ shifts the spacing by an almost L-independent -0.036. The growth is the double counting of Appendix B.4, which W_c removes.
- W_c leaves a near constant: at $L=16$, $R_3/(a\,w_{aa}) = 0.06259$, 0.06242 and 0.06337 for $a=1$, 2 and 3, changing slowly with L. Section 4.5 compares it with the first-order shift of the external-line window, $4a\,w_{aa}\,\delta^{(3)}(L) \to -\frac{16}{81\pi} a\,w_{aa}$, with $\frac{16}{81\pi} = \frac{2}{3}\cdot\frac{8}{27\pi} = 0.06288$.

The factor $\frac{2}{3}$ equals the $\Delta z = \pm 2$ share of the angular average of Appendix A.4, which suggested a missing channel of the ϵ^3 exchange sunset that changes the zero-mode number. The tagged-line test of Appendix C.2 shows instead that the defect is the level budget of the external legs, charged in each slice only where the legs are present. Whether the $\frac{2}{3}$ has a simple diagrammatic origin is not derived.

Order ϵ^4 and higher. The defect propagates. In W_c, $R_4(\chi_{11}) = 0.019$, 0.037, 0.049, 0.057, 0.063, 0.068, 0.073 and 0.076 for $L = 2, 4, \ldots, 16$; $R_4(\chi_{22})$ rises from 0.034 ($L=6$) to 0.070 ($L=16$), and R_4 of the ϕ^2 descendant from 0.14 to 0.40. In W, $R_4(\chi_{11}) = 0.097$ at $L=16$, and orders 4 to 8 are violated by larger, growing amounts. Appendices C.4 and C.5 give the ϵ^4 results of the external-line windows.

C.2 The tagged-line test

Records: `residual_log/results/tag/tag_L<L>_a<a>_*.json`; code `residual_log/brute_desc_tag.py` (exact fraction arithmetic).

In the χ_{aa} channel the external line is tagged. The initial quantum is a separate copy τ_{in} of the mode (a, a) that can only be absorbed, and the final quantum a copy τ_{out} that can only be created, once. The three one-quantum states $|\tau_{\text{in}}\rangle$, $|\tau_{\text{out}}\rangle$ and the regular $|\chi_{aa}\rangle$ all stand for the real descendant. All three are projected out of every resolvent, as the reference state is in ordinary perturbation theory. The order-n energy is the sum of the components on $|\tau_{\text{in}}\rangle$ and $|\tau_{\text{out}}\rangle$ of the order-n right-hand side of the Rayleigh–Schrödinger recursion (the vector whose overlap with the reference gives $E^{(n)}$). Histories that end on the regular $|\chi_{aa}\rangle$ would count those ending on $|\tau_{\text{out}}\rangle$ twice and are dropped. The tags count toward the level budget, and toward ℓ in every window, with charges $(\nu_{\text{in}}, \nu_{\text{out}})$.

- With charges $(2a, 2a)$ the construction is identical to the real channel; it reproduces W_c exactly ($L = 2$ to 14 for $a = 1$, $L = 6$ to 12 for $a = 2$).
- With charges $(0, 0)$ the external line is invisible to the cutoff and to the windows (budget-free). Its self-energy then has exactly the budget of the ϕ self-energy, on which r is tuned.

L	charged ($= W_c$): R_2	R_3	budget-free: R_2	R_3	in only: R_3	out only: R_3
2	0.004630	0.021917	0.017361	0.028412	0.024564	0.026879
6	0.001808	0.016678	0.008545	0.012127	0.012487	0.013542
10	0.001122	0.015932	0.005728	0.007330	0.009882	0.010568
14	0.000813	0.015702	0.004317	0.005055	0.008861	0.009368

Table 23: R_2 and R_3 of χ_{11} with a tagged external line, W_c windows (`residual_log/results/tag/`). "In only" and "out only" charge one leg by 2.

With the legs budget-free, $R_2 \approx 0.06/L$ and $R_3 \approx 0.07/L$ both go to zero (Table 23); charged, R_3 tends to 0.0156. For $a = 2$ the budget-free R_3 is 0.0313, 0.0227, 0.0177 and 0.0144 at $L = 6$ to 12, against 0.0221, 0.0195, 0.0185 and 0.0180 in W_c. So the whole ϵ^3 remainder of W_c is the level budget of the external legs; spectators, nested windows and the auxiliary objects are already treated correctly.

Why average-level charging is exact at ϵ^2 and not at ϵ^3. A mass-type sub-process on a χ line, one of the sub-bubbles that rM' cancels, has two external legs, the legs of the M' term. The truncation charges a leg only in the slices of the sub-process where it is present: an absorbed leg before its absorption, an emitted leg after its emission. D^M instead charges both legs with their average level $\frac{1}{2}(\ell_s + \ell_{s'})$ in every slice.

- At ϵ^2 the sub-process is the sunset. It has two orderings, normal (no leg present) and exchange (both legs present), whose weights are equal at large bubble level, so the average is exact and $R_2 \to 0$.
- At ϵ^3 there are also mixed orderings. In the vertex-corrected sunset (vertices v_1, v_2, v_3; lines v_1v_2, twice v_1v_3 and twice v_3v_2; the legs at v_1 and v_2) the six orderings have $(0, 0)$, $(1, 0)$, $(0, 1)$, $(2, 1)$, $(1, 2)$ and $(2, 2)$ external legs present in their two internal slices. Their weights differ, so the response to the level of a leg is not half the response to a spectator.
- The tadpole with a sunset on its loop has both legs at one vertex and charges exactly one leg in every slice, so it is unaffected.

C.3 The ϵ^3 external-line window W_{cs}

Records: `residual_log/WINDOW_SPEC.md` (§2.5a), `residual_log/large_L/fits/delta3_fits.txt`, `residual_log/large_L/results/{validation_small.txt,tag3_L16000.npz,tag3nu_L4000.npz}`, `residual_log/ce_wc/out_wcs/`; code `residual_log/{brute_desc_tag,delta3}.py`, `residual_log/large_L/{tagchan,delta3_fits}.py`, `residual_log/ce_wc/{ce_core,ce_solver,ce_run,tables_wcs}.py`.

Definition. Scheme W_{cs} is W_c with one more term at ϵ^3. The full ϵ^3 window is

$$D^{(3)} = \tfrac{1}{4}\{\Delta e_1^{(3)}(L,\hat{\ell}), M_0'\} + \tfrac{1}{4}\{\Delta e_1^{(2)}(L,\hat{\ell}), M_1'\} + \frac{\delta^{(3)}(L)}{2}\,X_0, \qquad \Delta e_1(L,\ell) = e_1(L) - e_1(L-\ell), \tag{47}$$

with $\delta^{(3)}$ of Eq. (23) and M_k' the ϵ^k part of M'.

- X is M' with every term multiplied by η times the sum of the levels of its χ legs. The crossing sign is $\eta = +1$ for terms with one creator and one annihilator (χ hops and $Z \leftrightarrow \chi$ hops) and $\eta = -1$ for pair creation and pair annihilation ($\chi\chi$ or $Z\chi$). Zero-mode legs have level 0.
- $e_1^{\rm tag}(\lambda;\nu_{\rm in},\nu_{\rm out})$ is $E - E_0$ of the ϕ channel $|1\rangle$ at sub-cutoff λ whose external zero-mode line is tagged as in Appendix C.2. The tags have level 0 in H_0 but add $\nu_{\rm in}$, $\nu_{\rm out}$ to the budget level; r and g are those of the references at L. With zero charges the tagged channel equals the ϕ channel exactly through ϵ^3. $e_1^{(3)}$ and $e_1^{\rm tag,(3)}$ are taken with r_3 removed.
- $\delta^{(3)}(L)$ is one number per outer cutoff, used in every channel, including the auxiliary ones (rule W_c). Everything else ($D^{(2)}$, $D^{\rm zm}$ at all orders, $D^{M,(k)}$ for $k \geq 4$, rule W_c) is unchanged.

Derivation of the operator form. For a hop $s \to s'$ with spectator level $\ell_{\rm spec}$ and χ legs of levels $\nu_{\rm in}$, $\nu_{\rm out}$, the slice-resolved window is $\frac{1}{2}[e_1(L) - e_1^{\rm tag}(L-\ell_{\rm spec};\nu_{\rm in},\nu_{\rm out})]M'_{s's}$. To first order in the charges this is $\frac{1}{2}[e_1'\ell_{\rm spec} + \mu(\nu_{\rm in}+\nu_{\rm out})]M'_{s's}$, with $e_1' = de_1/d\lambda$ and $\mu = \frac{1}{2}e_1' + \delta$ the response to one unit of leg level; δ is its departure from the average-level rule of D^M. As $L \to \infty$ in and out legs respond equally, by time reversal; at finite L they differ by $\mathcal{O}(1/L)$ (below). Crossing a leg exchanges the slices in which it is present, so a pair leg responds with $e_1' - \mu = \frac{1}{2}e_1' - \delta$. With $\frac{1}{2}(\ell_s + \ell_{s'}) = \ell_{\rm spec} + \frac{1}{2}(\nu_{\rm in}+\nu_{\rm out})$ this is D^M plus $\frac{\delta}{2}X$. δ is calibrated on the smallest charges, $(2,2)$: with $\delta_\nu^{(3)}(L) = \frac{1}{2\nu}\big[e_1^{(3)}(L-\nu) - e_1^{\rm tag,(3)}(L;\nu,\nu)\big]$ for symmetric charges (ν,ν), $\delta^{(3)} = \delta_2^{(3)}$. For χ_{11} the window is therefore exactly the slice-resolved one; for larger a it is its linearization, which becomes exact as $L \to \infty$ (below).

What is fixed and what is assumed. Nothing is fitted: $\delta^{(3)}(L)$ is an exact rational computed from the tagged ϕ channel at each L. The descendant tests do not probe the sign of the pair terms, because R_3 of any descendant involves only diagonal hop terms of X. The pair sign rests on the crossing argument and on the separate cancellation of $\ln L$ in the Z–χ and $\chi\chi$ parts of the C_5 response $S(L)$ (Appendix C.4).

$\delta^{(3)}(L)$ at large L. A chain-sum version of the tagged ϕ channel (`residual_log/large_L/tagchan.py`) evaluates $\delta^{(3)}(L)$ for every even $L \leq 16000$ in one run of 11 minutes. It reproduces the exact rationals for $L = 2$ to 16 to 4×10^{-16} (relative 2.6×10^{-14}), including the values the W_{cs} runs used, and runs with different $L_{\max}$ agree to 10^{-11}. Table 24 gives selected values; Fig. 1b shows the curve.

L	$\delta^{(3)}(L)$	$\delta^{(3)}(L)+\frac{4}{81\pi}$	$\delta_4^{(3)}(L)$
6	-0.0148018	$+9.2\times10^{-4}$	-0.014450
10	-0.0155308	$+1.9\times10^{-4}$	-0.016733
16	-0.0158036	-8.5×10^{-5}	-0.016729
34 (minimum)	-0.0159046	-1.9×10^{-4}	
100	-0.0158494	-1.3×10^{-4}	-0.015993
400	-0.0157725	-5.3×10^{-5}	-0.015806
1000	-0.0157458	-2.7×10^{-5}	-0.015759
4000	-0.0157278	-8.8×10^{-6}	-0.015731
16000	-0.0157217	-2.7×10^{-6}	

Table 24: $\delta^{(3)}(L)$: exact rationals for $L\le16$, chain sums beyond (`residual_log/large_L/fits/delta3_fits.txt`). Third column: distance from $-\frac{4}{81\pi}=-0.0157190$. Last column: the coefficient $\delta_4^{(3)}$ calibrated on charges $(4,4)$ (the exact form for χ_{22}).

$\delta^{(3)}(L)$ falls to its minimum at $L=34$ and then rises toward the limit of Eq. (24); the next term is $+0.013856/L$, followed by $\mathcal{O}(\ln L/L^2)$. Free-limit fits with a $\ln L/L$ term ($L_{\min}=200$ to 2000, data to $L=16000$) give $\delta_\infty+\frac{4}{81\pi}$ between -3×10^{-11} and $+7\times10^{-11}$; without $\ln L/L$ they miss by -5×10^{-7} to -1.5×10^{-6}. With the limit fixed, the $\ln L/L$ coefficient is -0.0058946, equal to $-\frac{1}{54\pi}$. A fit to $L\le4000$ predicts $L=16000$ to 4×10^{-10} with the $\ln L/L$ term (-2×10^{-6} without). Also $\delta^{(3)}(2)=-0.0093879$, $\delta^{(3)}(4)=-0.0136276$, and $\delta^{(3)}(16)$ is the rational $-17345700448126303825/1097581272385718321152$. Because $L\le16$ lies on the falling branch before the minimum, extrapolations in $1/L^2$ from $L\le16$, which do not see the $\ln L/L$ term, miss the limit: they give -0.0158 to -0.0165 (`residual_log/fits/parent_check_wcs/fits.txt`). With the limit identified numerically, $4|\delta_\infty|/\frac{8}{27\pi}=\frac{2}{3}$: the leg coefficient is two thirds of the ϵ^2 exchange-sunset defect.

The nonlinearities are finite-L effects. Define the single-leg coefficients $d_{\rm in}$ and $d_{\rm out}$ as $\delta_2^{(3)}$ with only one leg charged (by 2); with the $(2,2)$ cross term they give $\delta_2^{(3)}=\frac{1}{2}(d_{\rm in}+d_{\rm out})+\frac{1}{4}(\text{cross term})$. At $L\le16$, $d_{\rm in}$ and $d_{\rm out}$ are smaller in magnitude than $\delta^{(3)}$, and the coefficients for larger charges are larger: at $L=16$, $\delta_4^{(3)}$ and $\delta_6^{(3)}$ (the exact χ_{22} and χ_{33} forms) exceed $\delta_2^{(3)}$ in magnitude by 6% and 13% (evaluated from the $\nu=4$ and $\nu=6$ lines of `delta3_fits.txt`). At large L both effects disappear (`delta3_fits.txt`):

- $d_{\rm in}$ and $d_{\rm out}$ both tend to $-\frac{4}{81\pi}$. Their difference falls like $1/L$ (1.6×10^{-5} at $L=1000$, 3.9×10^{-6} at 4000), so time reversal is restored, and the $(2,2)$ cross term vanishes (-4.0×10^{-4} at $L=1000$, -1.2×10^{-4} at 4000).
- $\delta_\nu^{(3)}$ for $\nu=4$, 6, 8 and 16 tends to the same limit, with differences of about $-(\nu-2)\times0.0062/L$ from $\delta_2^{(3)}$ (last column of Table 24).

So the linearized window (47), calibrated on the charges $(2,2)$, is asymptotically exact for every a.

C.4 Tests of W_{cs}

Records: `residual_log/fits/parent_check_wcs/{verify,fits,r3zero,blind16}.txt`, `residual_log/results/legw/legw_L1000.npz`, `residual_log/large_L/fits/legw_logs.txt`, `residual_log/ce_wc/{out_wcs,out_local,out_wcs4}/`, `residual_log/large_L/results/tag3_L16000.npz`; code `residual_log/{nm_legw,brute_legw}.py`.

T1 passes. C_0 to C_4, r_1 to r_3 and g_1 to g_3 of W_{cs} are identical to those of W_c as exact rationals for $L = 2$ to 16 (r_4 differs).

T2 passes, with the same caveat. $C_5^{W_{cs}}(L) - C_5^{W_c}(L) = \delta^{(3)}(L)\, S(L)$ holds to 1.2×10^{-16} or better for $L = 2$ to 16, with $S(L)$ the C_5 response to $\frac{1}{2}X$ inserted at ϵ^3 (`nm_legw.py`, checked by brute force at $L = 2$, 4, 6). $S(L)$ is known to $L = 1000$ and converges with no $\ln L$:

L	10	20	50	100	200	400	1000	∞
$S(L)$	0.286252	0.319024	0.344641	0.354908	0.360610	0.363699	0.365693	0.367187

A free $\ln L$ coefficient comes out at 7×10^{-7} or less (`fits.txt`, $L_{\min} = 100$ to 400; other tail towers give at most 4×10^{-6}, `legw_logs.txt`). By type of M' term the $\ln L$ coefficients of S are $+\frac{1}{16}$ (Z–χ hop), $-\frac{1}{16}$ (Z–χ pair), $-\frac{1}{16}$ (χ hop), $+\frac{1}{16} + \frac{1}{27}$ (pair creation) and $-\frac{1}{27}$ (pair annihilation). They cancel within the hop sector and within the pair sector separately, so a leg window with one coefficient per unit leg level for hops and another for pairs gives no $\ln L$ in C_5, whatever the two coefficients are; the large-L check confirms this to 10^{-6}. This holds for either sign of the pair terms. With the crossing sign the coefficients also cancel within the Z–χ sector and within the $\chi\chi$ sector; with an unsigned leg weight these two sectors would carry $\pm\frac{1}{8}$ each. Only the signed weight therefore keeps C_5 free of $\ln L$ if the Z–χ and $\chi\chi$ parts of the window have independent coefficients, and this is the support for the sign. With the exact $\delta^{(3)}(L)$, $C_5^{W_{cs}} - C_5^{W_c} = -0.005478$, -0.005625 and -0.005700 at $L = 50$, 100 and 200 (evaluated from `tag3_L16000.npz` and `legw_L1000.npz`), tending to $\delta_\infty S_\infty = -0.005772$ with no $\ln L$. The window shift is small compared with the effect of a residual $\ln L$ with $|b| \approx 0.01$, about ± 0.1 (Section 4.5).

T3 for χ_{aa}: passes at $L \le 16$, with a possible small remainder. Table 25 gives R_3 of χ_{11}, χ_{22} and χ_{33} in W_{cs}.

L	χ_{11}	χ_{22}	χ_{33}
6	+0.001876 [0.016678]	+0.005487 [0.022139]	+0.029714 [0.047060]
8	+0.000896 [0.016183]	+0.002279 [0.019477]	+0.012071 [0.029985]
10	+0.000401 [0.015932]	+0.000987 [0.018459]	+0.003591 [0.021791]
12	+0.000121 [0.015789]	+0.000345 [0.017971]	+0.001503 [0.019864]
14	−0.000049 [0.015702]	−0.000012 [0.017707]	+0.000554 [0.019012]
16	−0.000157 [0.015647]	−0.000225 [0.017554]	+0.000046 [0.018566]

Table 25: R_3 in W_{cs}, exact, with the W_c value in brackets (`residual_log/fits/parent_check_wcs/verify.txt`; $L = 16$ from `residual_log/ce_wc/out_wcs/wcs_L16.json`).

The table is the basis of the statement in Section 4.6 (at most 3% of the W_c value left at $L = 14$ and 16 for $a = 1$ to 3). In W_{cs}, $R_3(\chi_{11})$ is 0.0019 at $L = 6$ and below 2×10^{-4} from $L = 12$ on, against about 0.0157 in W_c; it crosses zero between $L = 12$ and 14. Whether it tends to zero was tested by a hold-out test (`blind16.txt`, made when the $L = 16$ value already existed): fits to $L \le 14$ predict $L = 16$.

- Free-limit models ($a + b/L^2$, $a + b/L^2 + c/L^3$, $a + b/L + c/L^2$; $L_{\min} = 6$, 8) give a χ_{11} limit of -0.0005 to -0.0007 and miss $L = 16$ by 5×10^{-6} to 2.8×10^{-5}.
- The zero-limit model $b/L + c/L^2 + d/L^3$ misses by 9×10^{-6} ($L_{\min} = 8$) and 2.1×10^{-5} ($L_{\min} = 6$), equally well.

So for χ_{11} a zero limit is not excluded, and a small negative remainder of 3 to 5% of the W_c value is slightly favored. For χ_{22} the same free-limit fits give -0.0014 to $+0.0001$, up to about 8% of its W_c value (0.0176); for χ_{33} they scatter between -0.008 and $+0.022$ and decide nothing. The exact statement behind the window is the budget-free test of Appendix C.2, where R_3 falls roughly like $1/L$ at $L \le 14$ ($R_3 \approx 0.07/L$). The residual at $L \le 16$ is consistent with finite-L effects (the linearization and the slow convergence of R_3 in W_c), but R_3 itself has not been computed beyond $L = 16$. The exact slice-resolved forms for χ_{22} and χ_{33} use $e_1^{\mathrm{tag}}(L; 2a, 2a)$ instead of the linearization. They give -0.0004, -0.0009, -0.0012 (χ_{22}) and $+0.0019$, -0.0013, -0.0022 (χ_{33}) at $L = 10$, 12 and 14, drifting slowly negative. The linearized window is closer to zero at these L, and both forms share the same large-L limit.

T3 for the ϕ^2 descendants: not settled. The level-2 descendant is shifted by exactly $\frac{3}{2}\delta^{(3)}(L)$ (ratio 1.5000 at every L), the first-order expectation value of $\frac{\delta}{2}X$ in that state. W_{cs} is right for this state only if its W_c spacing tends to $-\frac{3}{2}\delta_\infty = \frac{6}{81\pi} = +0.0236$. Fits of the W_c spacing at $L \le 16$ give $+0.005$ to $+0.039$ (`fits.txt`), so they neither confirm nor exclude it. At $L = 16$ the spacings are -0.057 (level 2; W_c: -0.0337) and -0.059 (level 4; W_c: -0.0257), and the level-4 one is still moving away from zero. Two-particle descendants have large finite-L terms: their R_2 is still 0.04 at $L = 16$, against 0.0007 for χ_{11}.

T4 (ϵ^4 spacing) is not restored by W_{cs}.

- W_{cs} leaves R_4 of every χ_{aa} exactly unchanged. The direct effect of the $\delta^{(3)}$ term on R_4 ($+0.002115$ at $L = 6$) is cancelled exactly by the change it makes in $e_1^{(4)}$ of the auxiliary channels, hence in $D^{M,(4)}$.
- R_4 still grows: $R_4(\chi_{11}) = 0.0489$, 0.0570, 0.0632, 0.0682, 0.0726 and 0.0765 at $L = 6$ to 16, a local slope of about 0.028 per unit of $\ln L$ (least-squares slope over $L = 8$ to 16).
- With budget-free legs (tagged, all windows of W_{cs} through ϵ^3), $R_4(\chi_{11}) = -0.0235$, -0.0164, -0.0118, -0.0086, -0.0062 and -0.0043 at $L = 4$ to 14, going to zero. So the growth is the same leg defect at ϵ^4.
- The direct transcription to ϵ^4 (diagnostic scheme W_{cs4}: $D^{(4)} \mathrel{+}= \frac{1}{2}[\delta^{(4)}X_0 + \delta^{(3)}X_1]$, with $\delta^{(4)}$ from the tagged ϕ channel) removes the growth but does not track the budget-free values: $R_4(\chi_{11}) = -0.0165$, -0.0133, -0.0115, -0.0108 and -0.0107 at $L = 6$ to 14, flattening at -0.0107. The reason is that the zero-mode proxy fails at ϵ^4 (Appendix C.5): in the tagged ϕ channel D^{zm} acts on the regular zero modes only, so the tag is a spectator of the zero-mode clusters, whereas the real external zero mode joins them. At zero charge the proxy already differs from $e_1^{(4)}$, and $\delta^{(4)}(L)$ grows roughly like $\ln L$ (-0.0802 at $L = 6$ to -0.0991 at $L = 14$; Table 28).

Higher coefficients. W_{cs} changes C_6 to C_8 by about 0.01% at $L \le 16$ (Table 26), much less than the differences of about 1% between W_c and W. Every auxiliary object still follows rule W_c.

C.5 The ϵ^4 χ-line window $W_{cs4\chi}$

Records: `residual_log/WINDOW_SPEC.md` (§2.5b), `residual_log/fits/parent_check_p5/verify_p5.txt`, `residual_log/ce_wc/{out_wcs,out_wcs4,out_wcs4c,out_wcs4x,x4}/`, `residual_log/results/legw/legw_L1000.npz`, `residual_log/results/tag/{tag4bf_*,tag4_*,spec4_*}.json`; code `residual_log/`

L		C_5	C_6	C_7	C_8
14	W_{cs}	-20.25163	157.8451	-1440.041	14711.12
14	W_c	-20.24684	157.8302	-1439.928	14710.24
14	relative change	-2.4×10^{-4}	$+9.4 \times 10^{-5}$	-7.8×10^{-5}	$+5.9 \times 10^{-5}$
16	W_{cs}	-21.06383	165.3880	-1521.286	15664.25
16	W_c	-21.05893	165.3722	-1521.165	15663.31
16	relative change	-2.3×10^{-4}	$+9.6 \times 10^{-5}$	-7.9×10^{-5}	$+6.0 \times 10^{-5}$

Table 26: C_5 to C_8 in W_{cs} and W_c at $L = 14$ and 16, exact rationals rounded (`residual_log/ce_wc/out_wcs/`, `residual_log/ce_wc/out_local/`).

`{brute_desc_tag4,brute_desc_chi4,delta4chi,p5_prepare_x4}.py`, `residual_log/ce_wc/tables_p5.py`, `residual_log/fits/parent_check_p5/verify_p5.py`.

Why the zero-mode proxy fails at ϵ^4. Through ϵ^3 a χ line and the zero-mode line have the same self-energy function up to $\mathcal{O}(1/L)$. That is why D^M can use e_1 and why the leg window can be calibrated on the tagged ϕ channel. At ϵ^4 they differ: in the ϕ channel the external zero mode belongs to the zero-mode sector and joins the zero-mode clusters that D^{zm} windows, and a χ line cannot. Both parts of the χ mass window are then off.

- *Leg part.* The transcription W_{cs4} flattens at $R_4(\chi_{11}) = -0.0107$, while the budget-free χ_{11} line goes to zero.
- *Uniform (spectator) part.* Define the χ-line function $e_1^{\chi}(\lambda) = \Sigma_\chi(\lambda)/w_{11}(\epsilon)$, with $\Sigma_\chi(\lambda) = E(\chi_{11}\text{ line};\lambda) - E_0(\lambda)$ from the budget-free tagged χ_{11} channel at sub-cutoff λ under rule W_c, without a window on its own line, and the spectator function $Y_n(L,\ell) = \Delta e_1^{\chi,(n)}(L,\ell) - \Delta e_1^{(n)}(L,\ell)$ (Table 27). At fixed ℓ, Y_2 and Y_3 vanish like $1/L$, so D^M with e_1 is right for χ-line spectators through ϵ^3. $Y_4(L,2)$ does not vanish: it is 0.0829, 0.0824 and 0.0817 at $L = 12$, 14 and 16, about $+0.082$. $Y_4(L,4)$ levels off at 0.1426 at $L = 14$ and 16, $Y_4(L,6)$ is still growing there, and for hard spectators (ℓ near L) every Y_n is large.

L	$Y_2(L,2)$	$Y_3(L,2)$	$Y_4(L,2)$	$Y_4(L,4)$	$Y_4(L,6)$	$Y_4(L,L)$
6	-0.0114	-0.0599	$+0.0624$	-0.0567	-2.4772	-2.477
8	-0.0068	-0.0346	$+0.0789$	$+0.0971$	-0.1060	-2.925
10	-0.0045	-0.0225	$+0.0826$	$+0.1312$	$+0.1085$	-3.328
12	-0.0032	-0.0158	$+0.0829$	$+0.1405$	$+0.1600$	-3.706
14	-0.0024	-0.0117	$+0.0824$	$+0.1426$	$+0.1756$	-4.067
16	-0.0019	-0.0090	$+0.0817$	$+0.1426$	$+0.1803$	-4.421

Table 27: The spectator function $Y_n(L,\ell)$, exact rationals rounded (`residual_log/ce_wc/x4/`).

Definition. Scheme $W_{cs4\chi}$ is W_{cs} plus, at ϵ^4,

$$D^{(4)} \mathrel{+}= \tfrac{1}{4}\{Y_4(L,\hat{\ell}), M'_{cc,0}\} + \tfrac{1}{2}\big[\delta_{4\chi}(L)\, X_0 + \delta^{(3)}(L)\, X_1\big], \tag{48}$$
$$\delta_{4\chi}(L) = R_4^{\mathrm{bf}}(L) - R_4^{W_{cs}}(\chi_{11};L) + \delta^{(3)}(L) - \tfrac{1}{4}Y_4(L,2).$$

- M'_{cc} is the $\chi\chi$ part of M' (hops and pair terms without a zero-mode leg). The anticommutator uses the same per-state rule as D^M, and in auxiliary channels Y_4 is evaluated at the outer reference (rule W_c).
- $R_4^{\rm bf}(L)$ is the ϵ^4 spacing of the budget-free χ_{11} line at cutoff L, and X_1 is the ϵ^1 part of the leg operator X.
- The second term is a calibration on χ_{11}. On that state the window adds $w_0 Y_4(L,2) + 4[w_0\delta_{4\chi} + w_1\delta^{(3)}]$, with $w_{11} = w_0 + w_1\epsilon + \ldots$, $w_0 = \frac{1}{4}$, $w_1 = -\frac{1}{4}$; this is fixed so that the charged line has the ϵ^4 self-energy of the budget-free one. Hence $R_4(\chi_{11}) = R_4^{\rm bf}(L)$ by construction, and χ_{11} is not a test.
- Nothing is fitted: $R_4^{\rm bf}$ and Y_4 are exact brute-force rationals at each $L \le 16$.

Two variants serve as comparisons. Variant "all" applies Y_4 also to the Z–χ terms of M'; whether a $Z \leftrightarrow \chi$ sub-process uses e_1, e_1^χ or a mixture is not derived, so the $\chi\chi$ form is the definition and "all" brackets the choice. The leg-only variant keeps the uniform part on e_1: $D^{(4)}$ += $\frac{1}{2}[(\delta_{4\chi} + \frac{1}{4}Y_4(L,2))X_0 + \delta^{(3)}X_1]$.

L	$\delta^{(3)}$	$\delta_{4\chi,\rm leg}$	$Y_4(L,2)$	$\delta_{4\chi}$	proxy $\delta^{(4)}$
6	−0.01480	−0.08012	+0.06243	−0.09572	−0.08023
8	−0.01529	−0.08409	+0.07888	−0.10381	−0.08554
10	−0.01553	−0.08729	+0.08258	−0.10794	−0.09023
12	−0.01567	−0.09007	+0.08293	−0.11080	−0.09469
14	−0.01575	−0.09264	+0.08240	−0.11324	−0.09908
16	−0.01580	−0.09509	+0.08172	−0.11552	

Table 28: Coefficients of the ϵ^4 windows (`residual_log/ce_wc/x4/`, `residual_log/results/tag/`). $\delta_{4\chi,\rm leg} = \delta_{4\chi} + \frac{1}{4}Y_4(L,2)$ is the coefficient of the leg-only variant; the proxy $\delta^{(4)}$ is that of W_{cs4}.

Exact invariances. For the leg-only variant ($L = 2$ to 16) and for $W_{cs4\chi}$ and "all" ($L = 4$ to 16), the following are identical to W_{cs} as exact rationals (`verify_p5.txt`): C_0 to C_5; r_1 to r_4 and g_1 to g_4; E_p through ϵ^4; R_2 and R_3 of every descendant. r_5, g_5 and C_6 change (at $L = 16$, $r_5 = 10.4473$ against 10.2495 and $g_5 = -2.2890$ against -2.2681). R_5 of χ_{aa} is exactly unchanged, as R_4 was under the ϵ^3 window, and it grows: $R_5(\chi_{11}) = 0.0737$, 0.0839, 0.0934, 0.1042, 0.1158, 0.1285 and 0.1420 at $L = 4$ to 16. That is the ϵ^5 instance of the same leg defect.

T4: the ϵ^4 spacing. χ_{11} matches the budget-free line by construction (-0.0043 at $L = 14$, -0.0028 at 16, going to zero). The tests are χ_{22} and χ_{33} against their own budget-free lines (Table 29).

- χ_{22}: $W_{cs4\chi}$ closes 95% of the gap between W_{cs} and the budget-free line at $L = 14$ (-0.0204 against -0.0248) and agrees to 10^{-5} at $L = 8$ (-0.04044 against -0.04045). The remaining gap, +0.0026, +0.0037 and +0.0044 at $L = 10$, 12 and 14, is not shrinking: L times the gap grows (0.026, 0.044, 0.062). A zero limit for $a \ge 2$ is therefore not established. The gap reflects the dependence of the ϵ^4 leg response on the charge, a few percent of the leg coefficient (about 3.5% at $L = 14$), which the calibration on χ_{11}, with charges $(2,2)$, cannot see. $R_4(\chi_{22})$ itself falls ($LR_4 = -0.31$, -0.30, -0.29, -0.26 at $L = 10$ to 16); the budget-free χ_{22} line was not computed at $L = 16$.

	scheme	$L=6$	8	10	12	14	16
χ_{22}	W_{cs} $(=W_c)$	+0.0341	+0.0398	+0.0485	+0.0564	+0.0634	+0.0697
	proxy W_{cs4}	−0.0340	−0.0335	−0.0297	−0.0267	−0.0245	
	leg only	−0.0338	−0.0319	−0.0264	−0.0215	−0.0172	−0.0136
	$W_{cs4\chi}$	−0.0594	−0.0404	−0.0312	−0.0250	−0.0204	−0.0165
	budget-free	−0.0497	−0.0405	−0.0338	−0.0288	−0.0248	
χ_{33}	W_{cs} $(=W_c)$	+0.0894	+0.0715	+0.0448	+0.0474	+0.0544	+0.0619
	proxy W_{cs4}	+0.0220	−0.0013	−0.0331	−0.0354	−0.0334	
	leg only	+0.0221	+0.0004	−0.0296	−0.0300	−0.0259	−0.0212
	$W_{cs4\chi}$	−0.2381	−0.0331	−0.0432	−0.0387	−0.0329	−0.0275
	budget-free		−0.0723	−0.0569	−0.0490	−0.0429	

Table 29: R_4 of χ_{22} and χ_{33} by scheme, exact (`residual_log/fits/parent_check_p5/verify_p5.txt`). The $\chi\chi$ form ($W_{cs4\chi}$) and the variant "all" give identical R_4 for every descendant.

- χ_{33}: 90% of the gap is closed at $L=14$ (-0.0329 against -0.0429); the gap is $+0.0137$, $+0.0103$ and $+0.0101$ at $L=10$, 12 and 14.
- The leg-only variant misses χ_{22} by 0.0073 to 0.0076 and χ_{33} by 0.017 to 0.019 at $L=12$ and 14, so the uniform part Y_4 is needed. The proxy W_{cs4} happens to land near χ_{22} at $L=12$ and 14 but misses χ_{11} (-0.0064 at $L=14$) and χ_{33} ($+0.0095$).
- The ϕ^2 descendant of level 2 still grows in every scheme: $R_4=0.0404$, 0.1472, 0.2080, 0.2459, 0.2710 and 0.2885 in $W_{cs4\chi}$ at $L=6$ to 16 (W_{cs}: 0.1301 to 0.3949). It is not settled, as at ϵ^3.

L	W_c	W_{cs}	leg only	$W_{cs4\chi}$	"all"
12	149.30050	149.31434	149.29008	149.28512	149.28594
14	157.83017	157.84508	157.81947	157.81354	157.81431
16	165.37223	165.38804	165.36121	165.35460	165.35533

Table 30: C_6 by scheme, exact rationals rounded (`residual_log/fits/parent_check_p5/verify_p5.txt`); the continuum value is 362.366(4).

C_6. The leg-only variant and the proxy W_{cs4} add the same operator, $\frac{1}{2}[\delta X_0+\delta^{(3)}X_1]$, with different coefficients δ of X_0. Their C_6 values therefore fix the response of C_6 to $\frac{1}{2}X_0$ inserted at ϵ^4: it equals the C_5 response $S(L)$ of Appendix C.4 to 10^{-14} at $L=6$ to 14 (replay check `p2.c6_identity`). Relative to W_{cs}, the leg-only variant thus shifts C_6 by $\delta_{4\chi,\mathrm{leg}}(L)S(L)+\delta^{(3)}(L)S_1(L)$, where S_1 is the C_6 response to $\frac{1}{2}X_1$; the second term is $+0.0018$ to $+0.0025$ at $L=6$ to 14 ($S_1=-0.12$ to -0.16). The shift is -0.0256 at $L=14$ and -0.0268 at $L=16$, and the uniform part of $W_{cs4\chi}$ adds -0.0035, -0.0050, -0.0059 and -0.0066 at $L=10$ to 16. The total $W_{cs4\chi}$ shift relative to W_{cs} is -0.032 at $L=14$ and -0.033 at $L=16$ (Table 30).

The window coefficient grows with $\ln L$. $\delta_{4\chi,\mathrm{leg}}=-0.0801$, -0.0841, -0.0873, -0.0901, -0.0926 and -0.0951 at $L=6$ to 16, a local slope of -0.014 to -0.018 per unit of $\ln L$ (-0.0183 between 14 and 16), and $\delta_{4\chi}$ runs from -0.0957 to -0.1155. This is forced: in W_{cs}, R_4 grows with a local slope of about 0.028 per unit of $\ln L$, while the budget-free line goes to zero, so the slope

should approach -0.028 as the budget-free spacing flattens. Since the C_6 response to the ϵ^4 leg term is $S(L)$, the leg window's C_6 shift then tends to about $-0.028\, S_\infty \ln L \approx -0.010 \ln L$, plus a uniform part of order 0.01.

A structural prediction, not a measurement. If this window is complete at ϵ^4, C_6 in W_c and W_{cs} carries the opposite divergence, about $+0.010 \ln L$, from the missing χ-line budget, and $W_{cs4\chi}$ removes it. This parallels ϵ^3, where the missing constant leg window left $C_5^{W_c}$ about 0.006 above $C_5^{W_{cs}}$. It cannot be tested at $L \le 16$: there $C_6 \approx 165.4$ in every scheme of Table 30, against the continuum value 362.366(4), $C_6(L)$ still rises by about 7.5 per step of 2 in L, and the external-line windows shift C_6 by about 2×10^{-4} of its value. No statement about $C_6(\infty)$ is possible from $L \le 16$ in any scheme.

Cost. The χ-line objects are exact brute force. The budget-free line at $L = 16$ took 6.1 hours and 76 GB, the χ-line function 7.3 hours and 110 GB; $L = 18$ would need about 20 hours and more than 200 GB per process, more than one node holds. A large-L test needs an ϵ^6 version of the shift machinery and a diagrammatic tagged χ channel for $\delta_{4\chi}(L)$ and $Y_4(L, \ell)$; neither is built.

C.6 Status of the window components and outlook

Records: `residual_log/WINDOW_SPEC.md` (status table, §§2.5a and 2.5b), `zeromode_exact/README.md` (§§9 and 10), `complete_exact/README.md` (§§1 and 5), `continuum/fits_summary.txt`.

Structure at higher orders. Since $C_n = -\frac{1}{3}\Delta_p^3 \mathcal{E}_p^{(n)}$ at every order, only degree three or more in p survives, and at order n the number of surviving χ lines is at most $\frac{1}{2}(4n-6)$: five at ϵ^4, seven at ϵ^5, nine at ϵ^6. The following is conjectured from power counting.

- Nested χ sunsets with χ spectators start at ϵ^5. $\{\hat{\ell}, M'\}$ contains them, because $\hat{\ell}$ counts every quantum present.
- Coupling bubbles with hard χ spectators give windows of order one from ϵ^5 on. Their window behaves like $\ln \frac{L}{L-\ell}$, so the leading form is not enough; D^V as specified is not a correct form, and a correct one is open.
- The zero-mode effective Hamiltonian is needed at all orders as a matrix $h^{(k)}_{z'z}(\lambda)$, not only its eigenvalues; the slope of the mass window acquires a $\ln L$ at ϵ^3 ($r_3 \sim L \ln L$); and divergent three-vertex sub-bubbles with spectators appear.
- The double counting of Appendix B.4 recurs at every order with nested windows; rule W_c removes it by construction.

Next steps.

1. The residual logarithm of C_5: bounding $|b|$ well below 0.01 needs exact $C_5(L)$ to $L \gtrsim 400$.
2. Large-L tests of the ϵ^4 window: the prediction $C_6^{W_c} \supset +0.010 \ln L$ and $C_6(\infty)$ need the ϵ^6 shift machinery and a diagrammatic tagged χ channel for $\delta_{4\chi}(L)$ and $Y_4(L, \ell)$, the analogue of `residual_log/large_L/tagchan.py` with an external χ line. The χ_{22} gap would then show whether the a-dependence of the leg response is a finite-L effect, as it was at ϵ^3.
3. The ϵ^5 leg window: $R_5(\chi_{aa})$ grows, and the same construction one order higher is needed before C_7.

component	status
D^{zm} at ϵ^2, Eq. (20)	derived (linked clusters); exact agreement with brute force at $L \leq 12$; with D^M, the fitted limit of C_4^W agrees with the continuum (Section 4.4)
D^M at ϵ^2, Eq. (21), or in the e_1 form	derived at leading order in $1/L$; verified at ϵ^4
Bloch D^{zm} at ϵ^3	supported at ϵ^5: no linear term (scheme Bloch); the eigenvalue-diagonal form is excluded ($-0.0051\,L$)
self-consistent auxiliary windows (scheme W)	double counting; C_5^W diverges like $-0.0019\,L$
rule W_c (auxiliary windows referred to L)	provisional, not derived in full; T1 and T2 pass, T3 fails
coupling window D^V	invalid (finite C_4 shift of about 0.49; growing ϵ^3 spacing)
ϵ^3 external-line window (W_{cs}, Eq. (47))	provisional, not derived in full; $\delta_\infty = -\frac{4}{81\pi}$ identified numerically; linearization asymptotically exact; T1 and T2 pass; T3 for χ_{aa}: at least 95% removed at $L = 14$ and 16 ($a = 1$ to 3); extrapolated remainder 3 to 5% for χ_{11} (zero not excluded), up to about 8% for χ_{22}, undetermined for χ_{33}; cancellation as $L \to \infty$ expected but not established; ϕ^2 descendants not settled
C_5 residual logarithm	no $\ln L$ term preferred by the fits with the $\ln^3 L/L$ tail; $\lvert b\rvert \lesssim 0.01$ judged from the median of the fits to $L \leq 300$ (Section 4.5); bounding it well below 0.01 would need $L \gtrsim 400$
ϵ^4 χ-line window ($W_{cs4\chi}$, Eq. (48))	provisional; exact invariances through C_5 and R_3; χ_{11} by construction; χ_{22}, χ_{33} gaps 95%, 90% closed at $L = 14$ (χ_{22} gap not shrinking); ϕ^2 not settled; grows like $\ln L$
ϵ^5 leg window	open: $R_5(\chi_{aa})$ grows
coupling window with χ legs; higher-order windows; completeness	conjectured from power counting; not computed
C_6 to C_8 with a complete window	not available

Table 31: Status of the window components in $d = 2$. Every component was derived after its effect had been seen and with the continuum values known (Section 7); none has a fitted parameter.

4. The ϕ^2 descendants at ϵ^3 and ϵ^4 need larger L or a treatment of two-particle legs beyond the linearization.

Consequences for the production solver.

- The production solver computes its stated truncation correctly through ϵ^8 (Section 4.2); no implementation change is needed for what it claims to compute.
- The windows it implements (consistent, literal) are incomplete in $d = 2$ from ϵ^4 on. Its consistent C_4 follows $5.0026(5) - \frac{365}{4374\pi} \ln L + \ldots$, and its consistent C_5 needs about $+0.2 \ln L$ (`residual_log/WINDOW_SPEC.md`): fits without logarithms miss the continuum value by 1.6 to 1.7 (steps 3 and 4 of `residual_log/fits/fits_c5.txt`, $L \leq 160$). Fits of C_4 to C_8 from its data at $L \leq 30$ (the data available when they were made) fall short of the diagrammatic values in magnitude, by 6 to 13% for the medians over the tail towers (`continuum/fits_summary.txt`); they are no evidence against those values.
- The windows matter much more at high order. The diagonal zero-mode window alone changes C_4 by about $+0.10$ but C_8 by $+67\%$ at $L = 2$, $+23\%$ at $L = 14$ and $+18.5\%$ at $L = 18$ (§9 of `zeromode_exact/README.md`).
- A production implementation of the window must drop the assumption that D is diagonal

in the occupation basis. It needs the Bloch matrices $h_{z'z}(\lambda)$ for every $\lambda \le L$ (in scheme W, zero-mode channels $z \le 20$ for the references through ϵ^8, $z \le 21$ with descendant channels), the Newton coefficients $e_1(\lambda)$, rule W_c inside every auxiliary channel, and the partners projected out of the window objects. It needs the non-diagonal D^M, which does not vanish on the references and enters r_3 and g_3. The ϵ^3 external-line window needs the tagged ϕ channel that gives $\delta^{(3)}(L)$ and the reweighted operator X, and the ϵ^4 window the budget-free and spectator χ-line channels.

- The exact Fock basis grows fast (463 GiB at $L = 34$). A reliable extrapolation needs exact large-L sums of the kind used here, or at least the exact window shifts, to fix the $\ln L$ structure.

D Provenance and reviews

Each package folder carries `PROVENANCE` and `REVIEWS` files with commits, dates, authors and review rounds; the review logs and agent transcripts themselves are not published. Codex below is the Codex CLI with model gpt-6-astra at reasoning effort xhigh.

One dimension.

- *Solver* (`lri_hpc_Hamitonian`). Based on a supplied package written by OpenAI GPT-6 astra (via ChatGPT); the Expanse layer, fixes and the analyses (joint tail fits, window engines, identity checks) were written by Claude Code sessions in September and October 2026. Codex reviewed the plan (five rounds, approved) and inspected the code twice; the second round's fixes and all later analyses were not inspected by Codex. The analyses were checked by internal multi-agent workflows with independent engines and skeptic lenses.

- *Eighth-order code* (`lri_c8_hpc`). A GPT-6 astra rewrite of its own original eighth-order code; the port to the symmetry cluster by a Claude Code session; Codex inspection rounds 1–8, approved.

- *Complete-window solver* (`lri_d1_full_coefficient_flow`). Supplied by the author as a GPT-6 astra package; the extension on the symmetry cluster by Claude Code, Codex rounds 1–4, approved. The refits that use it (`lri_1d_dconn_refit`) were made by a Claude Code workflow with an independent verifier stage and had no Codex review.

- *Exact sums* (`lri_1d_c4_exact`, `lri_1d_c5_exact`). Written by the Claude Code session that also wrote this note, in October 2026, ported from the $d = 2$ exact-sum code. Codex approved the plans and inspected each package twice; the last fix round of each is not re-inspected. The constrained C_5 fit was checked by two separate Claude Code subagents (a blind re-implementation and a theory check).

- *Second channel* (`lri_d1_fullblock_tail_response`). A GPT-6 astra research package, shipped verbatim; reviewed by a Claude Code workflow, which rated the sign and normalization of its J term plausible but did not re-derive them.

Two dimensions, Appendices A to C. These appendices merge the method write-up of `lri_2d_coeff_independent_check`, which earlier shipped as a separate note in that package, in its public edition of 14 October 2026 (file paths made relative; an appendix on inconsistencies between internal sources and the values that later data superseded removed). The write-up and the computations

behind it were made by Claude Code sessions (Claude Opus 5.5) between 8 and 14 October 2026: the coordinating session HT2dPert and the agents it launched. These were two clean-room solver agents that saw only the model specification, a continuum agent, the exact-C_4 agent, the zero-mode agent, the complete-window agent, the window-derivation agent, the large-L agent and a write-up agent. The parent checks were made by the coordinating session. HT2dPert merged the write-up into this note on 16 October 2026, leaving out material that the main text already states and values that later data superseded. The merged text was then read by a fresh agent with no project context and reviewed against the write-up by the reviewing session; their findings were checked against the records named in each subsection. None of this material had an external code review.

Two and three dimensions. See the provenance files of the packages named in Sections 4 and 5.

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